UNIVERSITY OF CAPE COAST

ISRAEL SERIES

ALGEBRA AND TRIGONOMETRY

(MAT101)

PAST QUESTIONS

WITH

SOLUTIONS

FOR TERTIARY STUDENTS

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A.T

2017/2018

Section A

1. The average of 5 consecutive integers starting with m as the first integer is n. What is the average of 9 consecutive integers that start with (m+2)?

A. m+4 B. n+6 C. m+5
D. n+4

2. The curve $y = x^2 + bx + 256$ cuts the $x-axis$ at $(h, 0)$ and $(k, 0)$. If h and k are integers, find the least value of b.

A. -32 B. -255 C. -256 D. -257

3. If the ratio of the sum of the first six terms of a G.P. to the sum of the first three terms of the G.P. is 9, find the common ratio of the sequence.

A. 3 B. $\frac{1}{3}$ C. 2
D. $\frac{1}{2}$

4. Find all the real solutions to the equation $2|-2x-2|-3=13$.

A. -5 B. 3 C. -5, 3 D. 0

5. The sum of the first n terms of an A.P. is 2n and the sum of the first 2n terms is 3n. Find the sum of the first 3n terms.

A. n B. 5n C. 4n
D. None

6. A square is drawn by joining the midpoints of the sides of a given square. A third square is drawn inside the second square and this process is continued indefinitely. If a side of the first square is 4cm, determine the sum of the areas of all the squares?

A. 18 B. 32 C. 36
D. 64

7. If in sets A, B, C, the set $B \cap C$ consists of 8 elements, set $A \cap B$ consists of 7 elements and $C \cap A$ consists of 7 elements. Find the minimum element in the set $A \cup B \cup C$

A. 8 B. 14 C. 22 D. 15

8. The sixth term of the expansion $(x-2)^n$ is $-8064x^5$. Find the value of n.

A. 5 B. 6 C. 10 D. 11

9. The roots of the equation $3kx^2 - 18tx + (2k+3) = 0$ are α and $\frac{1}{\alpha}$, where t and k are constants. Find the value of k.

A. $-\frac{5}{4}$ B. 2 C. 3 D. $\frac{21}{5}$

10. In the expansion of $(a+b)^{15}$, find the numerical coefficient of the term containing a^2b^{13}.

A. 105 B. 78 C. 70 D. 20

11. Form the polynomial equation whose zeros are $2+\frac{1}{\sqrt{2}}$, $2-\frac{1}{\sqrt{2}}$

A. x^2-3x-1 B. $2x^2+x-2$
C. $2x^2-8x+7$ D. x^2+5x+1

12. Find the zeros of the polynomial x^3+x^2-7x-3

A. $-3,\ 2+\sqrt{2},\ 1+\sqrt{2}$ B. $3,\ 1+\sqrt{2},\ 1-\sqrt{2}$
C. $-3,\ 1+\sqrt{2},\ 1-\sqrt{2}$ D. 1, 3, 7

Use this information to answer Questions **13** and **14**

The polynomial $P(x) = a_3x^3 + a_2x^2 + a_1x + a_0$ has α, β and γ

13. Find a relation between the sum of the zeros and the coefficients of $P(x)$.

A. $\alpha+\beta+\gamma = \frac{a_1}{a_3}$ B. $\alpha+\beta+\gamma = -\frac{a_0}{a_3}$
C. $\alpha+\beta+\gamma = -\frac{a_2}{a_3}$ D. $\alpha+\beta+\gamma = \frac{a_2}{a_3}$

14. Which of the following relationships is correct?

A. $\alpha+\alpha\beta+\alpha\gamma = \frac{a_1}{a_3}$ B. $\alpha\beta+\beta\gamma+\alpha\gamma = \frac{a_1}{a_3}$
C. $\alpha\beta+\beta+\beta\gamma = \frac{a_1}{a_3}$ D. $\alpha\beta\gamma = \frac{a_0}{a_3}$

15. If $P(x) = x^3 + 3x^2 - x - 3$ and -1 is a zero of $P(x)$, find the other zeros.

A. (1,3) B. (-3,-1)

C. (3,1) D. (-3,1)

16. Consider an infinite geometric series with first term 'a' and common ratio r. If its sum is 4 and the second term is $\frac{3}{4}$, find the values of a and r.

A. $\frac{4}{7}, \frac{3}{7}$ B. $2, \frac{3}{8}$ C. $\frac{3}{2}, \frac{1}{2}$ D. $3, \frac{1}{4}$

17. Find the sum of the infinite series $\sum_{k=0}^{\infty} 5\left(\frac{2}{3}\right)^k$

A. 0 B. $\frac{10}{3}$ C. 10 D. 15

18. State the position of the term containing x^{10} in the expansion $(2x^3 + 4)^{12}$

A. 3rd B. 6th C. 9th D. 11th

19. If one of the roots of the quadratic equation $x^2 + mx + 24 = 0$ is 1.5, find the value of m.

A. -22.5 B. -17.5 C. -10.5 D. -8.5

20. The sum of the 4th *and* 12th terms of an A.P. is 20. Find the sum of the first 15 terms of the A.P.

A. 300 B. 120 C. 150 D. 170

21. Given that $\log_a x = \log_{a^k} x^k ; x > 0, k \in N$. If $a = k = 2$, find the value of x.

A. 4 B. 8 C. 16 D. 32

Use the equation $M^2 - 3M + 5 = 0$ to answer Questions 22 and 23

22. Which of the following matrices satisfies the equation?

A. $M = \begin{pmatrix} -1 & -3 \\ -1 & 2 \end{pmatrix}$ B. $M = \begin{pmatrix} -1 & 3 \\ 1 & -2 \end{pmatrix}$

C. $M = \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}$ D. $M = \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}$

23. Find the inverse of the matrix M.

A. $M = \begin{pmatrix} 0.4 & 0.6 \\ -0.2 & 0.2 \end{pmatrix}$ B. $M = \begin{pmatrix} -0.4 & -0.6 \\ -0.2 & 0.2 \end{pmatrix}$

C. $M = \begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}$ D. $M = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix}$

Use the two equations $\log_y x - 4\log_x y = 0$ *and* $\log_{10} x + \log_{10} y = 1$ to answer Questions 24 and 25

24. Find the value of x.

A. 2 B. $\sqrt{5}$ C. $\sqrt{10}$ D. -10

25. Find the value of y.

A. 2 B. $\sqrt{5}$ C. $\sqrt{10}$ D. -10

Use the function $f(x) = \frac{x-2}{x^3-5x}$ to answer Questions 26 to 28.

26. Find the domain of the function

A. $(-\infty, -2.2] \cup [-2.2, 2.2] \cup [2.2, \infty)$

B. $(-\infty, -2.2) \cup (-2.2, 0) \cup (0, 2.2) \cup (2.2, \infty)$

C. . $(-\infty, -2.2] \cup [-2.2, 0] \cup [0, 2.2] \cup [2.2, \infty)$

D. $(-\infty, -2.2) \cup (-2.2, 2.2) \cup (2.2, \infty)$

27. Find the vertical asymptotes.

A. $x = \pm 2.2$ B. $x = 0, 2.2, 3.2$

C. $x = 0, \pm 2.2$ D. $x = \pm 2.2, 3.2$

28. Find the horizontal asymptote.

A. $y = 0$ B. $y = 1$ C. $y = 2$ D. $y = \infty$

29. Let X, Y *and* Z be sets such that $Y \subseteq X$. Which of the following is true?

A. $X \cup Z = Y \cup Z$ B. $Y' \cap Z' \subseteq X' \cap Z'$

C. $X \cap Z \subseteq Y \cap Z$ D. $X \cap (Y \cup Z) = Y \cup (X \cap Z)$

30. Find the Cartesian product of $(Z \times N) \cap (N \times Z)$, where N,Z are natural natural numbers and integers respectively.

A. $Z \times N$ B. $N \times Z$ C. $N \times N$ D. $Z \times Z$

SECTION B

1. (a) Sketch the graph of $f(x) = \sqrt{1-x^2}$.

Deduce the sketch of the graph of

$h(x) = -2 + \sqrt{-(x+2)^2+1}$ in the same plane.

(b) Show that if x is small enough to neglect its cube and higher powers

$$\sqrt{\frac{1-x}{1+x}} = 1 - x + \frac{x^2}{2}.$$

(c) Find the truth set of $|x+2| \geq |x+1| + |x-3|$.

2. Consider the matrix

$$P = \begin{vmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix}$$

(a) Find the characteristic equation of the matrix

(b) By determining a suitable expression, find the matrix P^3

(c) Hence, find the inverse matrix P^{-1}

(d) Use your results to solve the linear equations

$$-y + z = 2$$

$$2x - y + 2z = 5$$

$$x + z = 4$$

3. (a) Sketch the parabola $y = x^2 - 4x + 5$.

(b) If the parabola is translated 3 units left and 2 units upwards, find,

(i) the equation of the translated curve

(ii) the points of the intersection of both curves

(iii) the roots of the translated curve.

(c) Express the function

$$f(x) = \frac{x+1}{x^2 - x - 6}$$

in its partial fractions.

4. A sequence is defined by the sttatements

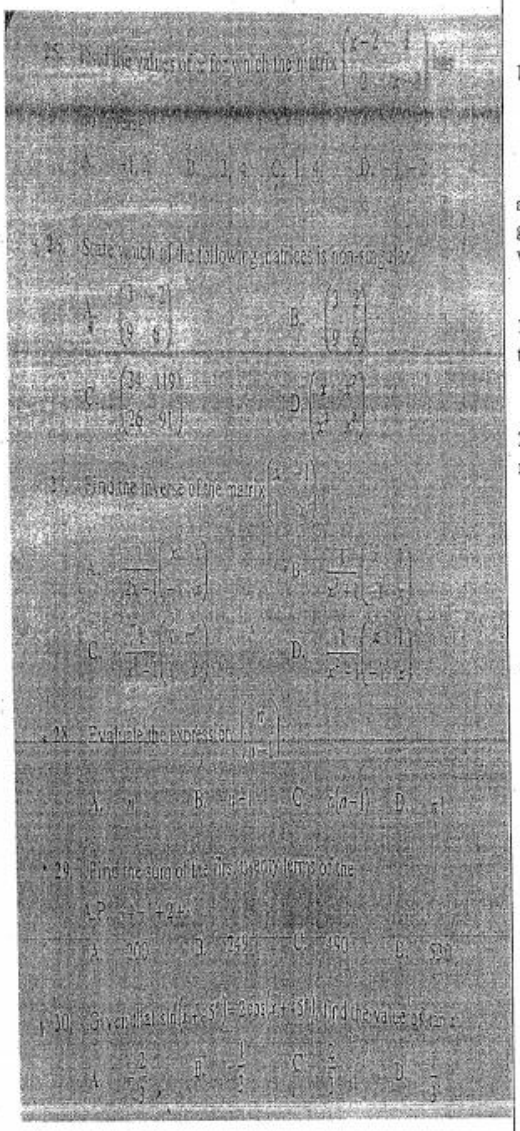
$u_1 = 1, \quad u_{n+1} = u_n + \left(\frac{1}{4}\right)^n, \quad \forall n \geq 1$

(a) Write down the values of u_3 *and* u_4

(b) Find an expression for u_n in terms of n only

(c) Evaluate $\lim_{n\to\infty} u_n$

2015/16
SECTION A



2015/2016

SECTION B

1. (a) A function $f(x)$ is given by

$$f(x) = \frac{3x - 9}{x^2 - x - 2}$$

Examine the behavior of the function by finding all possible asymptotes and hence, sketch its graph. Show also that $f(x)$ cannot lie between two certain values.

(b) If α, β *and* γ are roots of $x^3 - 4x^2 - 5x - 1 = 0$, without solving the equation, find the equation whose roots are α^2, β^2 *and* γ^2

2. (a) Find the characteristic equation of the matrix

$$P = \begin{vmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix}$$

Hence, find the inverse of P

(b) Use matrix method to solve the linear equations

$$-y + z = 2$$

$$2x - y + 2z = 5$$

$$x + z = 4$$

3. (a) Express $g(x) = \sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$ *where* $R > 0$.

(i) Determine the maximum and minimum values of g and t00000000001the corresponding values θ.

(ii) Solve the equation for $4\sin\theta\cos\theta = \sqrt{3}\sin\theta - \cos\theta$, $0^0 \leq \theta \leq 360^0$

(b) Show that if x is small enough to neglect its cube and higher powers

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$$\sqrt{\frac{1-x}{1+x}} = 1 - x + \frac{x^2}{2}.$$

4. (a) A sequence is defined by the statements

$u_1 = 1, \quad u_{n+1} = u_n + \left(\frac{1}{4}\right)^n, \ \forall n \geq 1$

(i) Write down the values of u_3 *and* u_4

(ii) Find an expression for u_n in terms of n only

(iii) Evaluate $\lim_{n\to\infty} u_n$

(b) Solve the simultaneous equation

$\log_y x - 4\log_x y = 0$

$\log_{10} x + \log_{10} y = 1$

2014/2015

SECTION A

1. Given $U_n = 2 - \frac{1}{3}U_{n-1}$, find an expression for $U_{n+1} - U_n$.

A. $\left(-\frac{1}{3}\right)[U_n - U_{n-1}]$ C. $\left(-\frac{1}{3}\right)[U_{n-1} - U_{n-2}]$

B. $2[U_{n-1} - U_{n-2}]$ D. $\left(-\frac{1}{3}\right)^n [U_n - U_{n-1}]$

Use the sequence $x_1, x_2, x_3, \ldots$ to answer Question 2 – 5.

2. Find an expression for x_2 if the sequence is an A.P

A. $x_2 = \frac{x_1 + x_3}{2}$ C. $x_2 = x_3 - x_1$

B. $x_2 = \sqrt{x_1 + x_2}$ D. $x_2 = \frac{x_1 + x_3}{3}$

3. Find an expression for X_2 if the sequence is a G.P

A. $X_2 = \frac{x_3}{x_1}$ C. $x_2 = \sqrt{\frac{x_3}{x_1}}$

B. $x_2 = \sqrt{x_1 x_3}$ D. $x_2 = \sqrt{x_1 + x_3}$

4. Find the value of X_2 when $U_n = U_{n-1} + \left(\frac{1}{3}\right)^{n-1} \forall n > 1$ *and* $x_1 = 1$.

A. $\frac{1}{3}$ C. 1

B. $4\frac{1}{3}$ D. $1\frac{1}{3}$

5. What type of sequence is described by U_n?

A. G.P C. mixed

B. A.P D. recurrence

If $2^x - 5^y = 3$ and $2^{x-3} + 5^{y-2} = 21$

Use this information to answer question 6 and 7

6. Find the value of x.

A. 1970 C. 7.645

B. 5.0 D. 2.301

7. Evaluate x+y

A. 8.283 C. 200.28

B. 10.928 D. 5.584

Given that (x-1) and (2x – 4) are factors of $2x^3 + \lambda x^2 + 4x + \mu$.

Use this information to answer Questions 8 to 10

8. Find the value of λ

A. -6 C. -4

B. 3 D. 4

9. Find the value of μ

A. -1 C. 1

B. O D. 2

10. Find the other factor

A. x C. 2x

B. x-1 D. x+1

11. The nth term of the first n odd numbers is given by $U_n = 2n-1$. Find an expression for S_n.

A. n^2 C. $(n+1)^2$

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B. ½n(n+1) D. n(n+1)

Given the geometric sequence
$3-\sqrt{2}, 11-6\sqrt{2}, \ldots$
Use this information to answer Questions 12 and 13

12. Find the Common ratio r.
A. $3+\sqrt{2}$ C. $3-\sqrt{2}$
B. $3-2\sqrt{2}$ D. $3+2\sqrt{2}$

13. Find the third term,
A. $45-29\sqrt{2}$ C. $19-\sqrt{2}$
B. $45-\sqrt{2}$ D. $19-12\sqrt{2}$

Given $g(x)=-2\sqrt{3-2x-x^2}$
Use this information to answer Questions 14 and 15.

14. Describe the curve of g(x).
A. Upper semi-circle with centre (-1, -2) and radius 1 unit.
B. Lower semi-circle with centre (1,2) and radius 1 unit.
C. Upper semi-circle with centre (1,2) and radius 2 units.
D. Lower semi-circle with centre (-1, -2) and radius 2 units.

15. What is the range of values of g(x).
A. $-3 \le y \le 1$ C. $-4 \le y \le -2$
B. $-2 \le y \le 1$ D. $-1 \le y \le 2$

16. The roots of the equation $x^2+6x+q=o$ are x and x -1. Find the value of q.
A. $\frac{-35}{4}$ C. $\frac{-2}{3}$
B. $\frac{2}{3}$ D. $\frac{35}{4}$

17. Find $x, if\ \ln(\times+1)+\ln(\times-1)=0$
A. 4 C. $2^{\frac{1}{2}}$
B. 8 D. $2^{\frac{1}{4}}$

18. Given that $Sin\theta=\frac{3}{5}, find\ sec\theta$
A. $\frac{5}{4}$ C. $\frac{4}{5}$
B. $\frac{-4}{5}$ D. $\frac{-5}{4}$

19. Write $Sin35^\circ + Sin27^\circ$ as a product.
A. $-2Sin31^\circ$ C. $2sin31^\circ cost4^\circ$
B. $2Sin4^\circ Cost3^\circ$ D. $-2Cos4^\circ Sin31^\circ$

20. Find the total number of terms in the expansion of $[1-2x)^2]^{12}$.
A. 20 C. 25
B. 22 D. 24

21. If $Cost\theta=\frac{3}{5}$ *and* θ is an Quandrant IV, find Cost 2θ.
A. $\frac{-6}{5}$ C. $\frac{3}{5}$
B. $\frac{-7}{25}$ D. $\frac{9}{25}$

A matrix is given as $B=\begin{vmatrix} 1 & -2 & -5 \\ 2 & -1 & 0 \\ -4 & 5 & 6 \end{vmatrix}$
Use it is answer questions 22 to 25.

22. Find the trace of the matrix.
A. -6. C. 7
B. 6 D. 8

23. Find the sum of the minors of the diagonal elements of B.
A. -6 C. -14
B. −17 D. 3

24. Find the determinant of B.
A. −12 C. −2
B. −6 D. 6

25. Find the characteristics polynomial of B
A. $x^3+6x^2-14x-12$ C. x^3-8x^2-6x+6
B. x^3+7x^2+3x-2 D. $x^3-6x^2-17x+12$

26. State which of the following matrices is non-singular.
A. $\begin{pmatrix} 3 & -2 \\ 9 & 6 \end{pmatrix}$ C. $\begin{pmatrix} 34 & 119 \\ 26 & 9 \end{pmatrix}$
B. $\begin{pmatrix} 3 & 2 \\ 9 & 6 \end{pmatrix}$ D. $\begin{pmatrix} x & x^2 \\ x^2 & x^3 \end{pmatrix}$

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27. Find the values of x for which the matrix $\begin{pmatrix} x-2 & 1 \\ 2 & x-3 \end{pmatrix}$ has no inverse.
A. $-1, 2$ C. 1, 4
B. 3, 4 D. $-1, -2$

28. Find the limit of the function $f(x) = \frac{2x}{1+x}$ as $x \to \infty$.
A. -1 C. 2
B. $\frac{1}{x}$ D. $2x$

29. Find the values of x for which the expressions $\frac{2x+5}{x^2-x-6}$ does not exit.
A. 3, - 2 C. 3, 2
B. $-3, 1$ D. 3, -1

30. Factorize the expression $2x^3 + 3x^2 - 32x + 15$
A. (x-2)(x-3)(2x – 3) C. (x-3)(x-2)(2x+1)
B. (x+3) (x+5) (2x+1) D. (x – 3) (x+ 5) (2x – 1)

SECTION B

1. a. If α and β are the roots of $2x^2 - x - 2 = o$,
without solving the equation,
i. Find the value of $\alpha^2 - \beta^2$
ii. Show that $\alpha^3 - \beta^3 = \frac{5}{2}(\alpha^2 - \beta^2)$
iii. Find the equation whose roots are α^3 and β^3

b. Express 0.14 as an exponential series and hence write it in its fractional form.

2. a. Use matrix methods to solve the system of equations.
$3x + 2y - 2z + 8 = 0$
$x - 3y + z + 3 = 0$
$2x + y - 3z + 9 = 0$

b. Find the truth set of
$|x + 2| \geq |x + 1| + |x - 3|$

3. a. Sketch and describe completely the graph of $f(x) = \frac{x^3}{x^2-16}$

b. Decompose $f(ex) = \frac{x^2-5x-3}{2x^2+5x-12}$ into partial fractions.

4. Express $\frac{2x+1}{(x-1)(x+2)^2}$ as the sum of three partial fractions. When x is small so that x^3 and high powers of x may be neglected, obtain the expansion of the given functions in the form $a+bx+cx^2$. State the range of values of x for which the expansion is valid.

b. Given that $\log_b a + \log_c a = 2\log_b a \log_c a$, express a in terms of b and c.

2013/14

SECTION A

1. Simplify $(P \cap Q') \cup (P \cap Q)$, where P and Q are subsets of the universal set U.
   A. $\emptyset$ C. P
   B. Q D. μ

2. Solve $log_2\sqrt{2} = x$
   A. $x = \frac{-1}{2}$ C. x =1
   B. x=½ D. x=2

3. Solve the absolute value inequality $|4 - x| < -3$
   A. $\emptyset$ C. {x:x<1 or x > 7}
   B. $\{x : 0 < x < 1\}$ D. {x : x > 7}

4. Solve the absolute value inequality $|x + 3| > 4$
   A. (x : x < -7 or x > 1} C. { x: x >1}
   B. (x : x< – 1 or x > 1) D. { x: x < -7)

Use this information to answer questions 5 to 7
The polynomial defined as
$p(x) = x^3 + a_2x^2 + a_1x + a_o$ *has zeros* r_1, r_2 *and* r_3

5. Find a relation between the sum of the zeros and the coefficients of p(x),
   A. $r_1 + 1_2 + r_3 = a_1$
   B. $r_1 + r_2 + r_3 = -a_2$
   C. $r_1 + r_2 + r_3 = -a_0$
   D. $r_1 + 1_2 + r_3 = a_2$

6. Which of the following relationship is correct?
   A. $r_1 + r_1r_2 + r_1r_3 = a_1$
   B. $r_2 + r_1r_2 + r_2r_3 = a_1$
   C. $r_1r_2 + r_1r_3 + r_2r_3 = a_1$
   D. $r_1r_2r_3 = a_o$

7. If $p(x) = x^3 + 3x^2 - x - 3$ and – 1 is a zero of p(x), find the other zeros
   A. (1, 3) C. (- 3, - 1)
   B. (3, - 1) D. (- 3, 1)

8. Solve the inequality $-3 \leq 2x + 1 \leq 5$
   A. $-1 \leq x \leq 1$ C. $-1 \leq x \leq 3$
   B. $-2 \leq x \leq 2$ D. $-2 \leq x \leq 3$

A function f(x) is given by $f(x) = \frac{x^2-1}{x^2-4}$.
Use this information to answer questions 9 to 13.

9. Find the domain of f(x).
   A. (-2,2) C. (-2, 2) ∪ (-2, ∞)
   B. (-∞,-2) ∪ (-2, 2) D. (-∞, -2) ∪ (-2,2) ∪ (2, ∞)

10. Find the range of f(x).
    A. ($^1/_4$,1) C. $(-\infty, \frac{1}{4}] \cup (1, \infty)$
    B. $(-\infty, \frac{1}{2}]$ D. $\left(-\infty, \frac{1}{2}\right) \cup [1, \infty)$

11. What observation can you draw from the function?
    A. It is symmetrical about the origin
    B. It has y = 0 as the horizontal asymptote
    C. It is symmetrical about the y – axis
    D. It has x = 4 as the vertical asymptote.

12. At what point does the curve cuts the x – axis?
    A. (1 , 0) and (-1, 0) C. (-1,0)
    B. (1,0) D. (0, ¼)

13. At what point does the curve cuts the y – axis?
    A. (1,0) and (-1,0) C. (-1,0)
    B. (1,0) D. (0, ¼)

14. All the following are characteristics of tan x except that
    A. Its minimum and minimum values occurs at $\pm\infty$
    B. It is symmetrical about the origin
    C. It is periodic with period 2π
    D. It is asymptotic at $x = n\pi + \frac{\pi}{2}$

Use the equation $2|x| - |3y| = 9$ *to answer questions 15 and 16.*

15. Find the range of values of y.
    A. $\left(\frac{9}{2}, \infty\right)$ C. (∞, ∞)
    B. $\left(-\infty, \frac{-9}{2}\right)$ D. $[\frac{9}{2}, \infty)$

16. Find the range of values of x.

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A. $(0, \infty)$ C. $(-\infty, -3) \cup (3, \infty)$
B. $(-\infty, \infty)$ D. $(-\infty, 4\frac{1}{2}) \cup (4\frac{1}{2}, \infty)$

17. Solve $5log_x 2 - \frac{1}{2}log_x 8 = 2 - \frac{1}{2}log_x 2$
A. $x = \pm 4$ C. $x = 16$
B. $x = 2$ D. $x = 4$

18. If the fifth term of an arithmetic sequence is 14 and the tenth is 34, find the common difference.
A. 2 B. 3 C. 4 D. 5

19. Find the sum of the series $100 + 50 + 25 + \ldots$
A. 100 C. 305
B. 200 D. 400

20. Find the total number of terms in the expansion of $[(1 - 2x)^2]^{11}$
A. 20 B. 22
B. 23 D. 24

21. Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$ be a polynomial, where $a \in Z^+$. If p/q is a zero of p(x), which of the following is not true?
A. P is a factor of a_0
B. p is a factor of a_1
C. q is a factor of a_n
D. (p,q) = 1

22. Let M and N be matrices. If M and N nonsingular, then, for the product MN.
A. $|MN| \neq 0$
B. Its inverse does not exist
C. Its transpose does not exist
D. $|MN| = 0$

A matrix is given as $B = \begin{matrix} -1 & 2 & 1 \\ -7 & 4 & -1 \\ -4 & 2 & -1 \end{matrix}$

Use is to answer questions 23 to 26.

23. Find the trace of the matrix.
A. 2 C. 13
B. -2 D. -13

24. Find the sum of the diagonal minors.
A. 2 C. -2
B. 13 D. -13

25. Find the characteristics polynomial of B.
A. $p(x) = x^3 + 2x^2 + 13x - 2$
B. $p(x) = x^3 + -4x^2 + 13x + 2$
C. $p(x) = x^3 - 2x^2 + 15x + 2$
D. $p(x) = x^3 - 2x^2 + 13x + 2$

26. Find an expansion for B^{-1}
A. $-\frac{1}{2}(B^2 - 2B + 13I)$ C. $-2(B^2 + 2B + 13I)$
B. $B^2 - 2B + 13I$ D. $\frac{1}{2}(B^2 - 2B + 15I)$

A quadratic function f(x) has zero $m + {}^1/_n$ *and* $n + {}^1/_m$ *where m and n are roots of the equation* $x^2 + px + q = o$

Use it to answer questions 27 to 29

27. Find the sum of the roots
A. $\frac{-p(q+1)}{q}$ C. $\frac{-q(p+1)}{p}$
B. $\frac{-(q+1)}{p}$ D. $\frac{-p(q-1)}{q}$

28. Find the product of the roots
A. $\frac{-q(p+1)}{p}$ B. $\frac{(q^2-2q+1)}{p}$
C. $\frac{(q^2-2q-1)}{q}$ D. $\frac{(q^2+2q+1)}{q}$

29. Find the equation whose roots are $m + {}^1/_n$ and $n + {}^1/_m$
A. $x^2 + \frac{(q+1)}{p}x + \frac{(q^2-2q-1)}{q} = 0$
B. $x^2 + \frac{p(q-1)}{q}x + \frac{(q^2-2q+1)}{p} = 0$
C. $x^2 + \frac{p(q+1)}{q}x + \frac{(q^2+2q+1)}{pq} = 0$
D. $x^2 + \frac{p(q-1)}{q}x + \frac{(q^2+2q+1)}{q} = 0$

30. If θ is obtuse and $\sin\theta = {}^4/_5$, find $\cos\theta$
A. $-\frac{1}{4}$ C. $\frac{3}{5}$
B. $-\frac{3}{5}$ D. $\frac{3}{4}$

31. Factorize the expression $2x^3 + 3x^2 - 32x + 15$
A. (x-2)(x-3)(2x-3) C. (x-3)(x-2)(2x+1)

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B. (x+3)(x+5)(2x+1) D. (x-3)(x+5)(2x-1)

32. Find the values of x for which the expressions $\frac{2x+5}{x^2-x-6}$ does not exist.
A. 3, -2
B. -3, 1
C. 3,2
D. 3, -1

33. Find the limit of the function
$f(x) = \frac{2x}{1+x}$ as $x \to \infty$
A. -1
B. $^1/_x$
C. 2
D. 2x

The roots of the equation $3x^2+4x-5=0$ are α, β
Use if to answer questions 34 to 36

34. Find the value of $\frac{1}{\alpha}+\frac{1}{\beta}$
A. $^4/_5$
B. 2
C. $^{-4}/_3$
D. $^{-4}/_5$

35. Find the value of $\alpha^2+\beta^2$
A. $^{16}/_9$
B. $^{10}/_3$
C. $^{46}/_9$
D. $^{-4}/_3$

36. Find the value of $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$
A. $^{10}/_3$
B. $^{-46}/_{15}$
C. $^{16}/_9$
D. $^{-4}/_3$

Use the polynomial
$P(x) = x^5-4x^3+2x+3$ to answer questions 37 and 38

37. Find the remainder when p(x) is divided by (x-1)
A. 1
B. 2
C. 3
D. 4

38. Find the remainder when p(x) is divided by (x+2)
A. 0
B. -1
C. -2
D. 3

When the expression x^5+4x^2+ax+b is divided by x^2-1, the remainder is $2x+3$.
Use it to answer questions 39 to 40

39. Find the value of a
A. -1
B. 0
C. 1
D. 2

40. Find the value of b.
A. -1
B. 0
C. 1
D. 2

41. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$, find A.
A. (23 53 83)
B. $(23\ 83\ 53)^t$
C. (53 83 23)
D. $(23\ 53\ 83)^t$

42. Find the values of x for when the matric $\begin{vmatrix} x-2 & 1 \\ 2 & x-3 \end{vmatrix}$ has no inverse.
A. -1, 2
B. 3, 4
C. 1, 4
D. – 1, -2

43. State which of the following matrix is non-singular.
A. $\begin{pmatrix} 3 & -2 \\ 9 & 6 \end{pmatrix}$
B. $\begin{pmatrix} 3 & 2 \\ 9 & 6 \end{pmatrix}$
C. $\begin{pmatrix} 34 & 119 \\ 26 & 91 \end{pmatrix}$
D. $\begin{pmatrix} x & x^2 \\ x^2 & x^3 \end{pmatrix}$

44. Find the inverse of the matrix $\begin{pmatrix} x & -1 \\ 1 & x \end{pmatrix}$
A. $\frac{1}{2x-1}\begin{pmatrix} x & 1 \\ -1 & x \end{pmatrix}$
B. $\frac{1}{x^2+1}\begin{pmatrix} x & 1 \\ -1 & x \end{pmatrix}$
C. $\frac{1}{x^2-1}\begin{pmatrix} x & -1 \\ 1 & x \end{pmatrix}$
D. $\frac{1}{x^2-1}\begin{pmatrix} x & 1 \\ -1 & x \end{pmatrix}$

45. Evaluate the expression ${}^nC_{n-1}$
A. n
B. $n-1$
C. n(n-1)
D. n!

46. Evaluate the sum of the first twenty terms of the A.P -4 -1 + 2 + ...
A. 200
B. 249
C. 490
D. 530

Given that the 1st term of an A.P is 6 and the 7th term is -18.
Use it to answer questions 47 and 48.

47. Find the common difference, d
A. -6
B. -4
C. 2
D. 3

48. Find the sum of the first ten terms S_{10}.
    A. -230 C. 120
    B. -120 D. 230

49. Solve the equation $Sin\frac{1}{2}\theta = \frac{1}{2}$ for values of θ from O° to 360 inclusive.
    A. $60^{\circ}, 300^{\circ}$ C. $45^{\circ}, 270^{\circ}$
    B. $30^{0}, 330^{0}$ D. $60^{0}, 330^{0}$

50. Given that $Sin(x+45^{\circ}) = 2Cos(x+45^{\circ})$, find the value of tan x.
    A. $^{-2}/_{3}$ C. $^{2}/_{3}$
    B. $^{-1}/_{3}$ D. $^{1}/_{3}$

SECTION B

1. If $\propto and\ \beta$ are roots of $2x^2$-x-2 = 0, without solving the equation.
   i. Find the value of $\propto^2 - \beta^2$
   ii. Show that $\propto^3 - \beta^3 = {}^5/_2 (\propto^2 - \beta^2)$
   iii. Find the equations whose roots are $\propto^3$ and β^3

   b. Sketch and describe completely the graph of f(x) $= \frac{x^3}{x^2-16}$

   c. Prove that secx – tanx = tan $\left(\frac{\pi}{4} - \frac{1}{2}x\right)$

2. a. Use matrix methods to solve the system of equations
   $3z-y + 2\times+17 = 0$
   $2z + 6y - 10 = 0$
   $-z - 2y + \times + 3 = 0$

   b. Solve the equation $sec\theta tan\theta$ =2 given solution for $0^o \leq \theta \leq 360^o$

   c. A sequence is defined by the statement.
   $U_1 = \frac{5}{2}, U_2 = 2 - \frac{1}{3}U_n - 1. n > 1.$

   Find:
   i. an expression for U_n in terms n only.
   ii. Lim $\frac{S_n}{n}$ *where* S_n is the sum of the first n terms of the sequence

3. a. Find the truth set of $|\times +2| \geq |\times +1| + |\times -3|$
   b. A rational function f(x) is given by $f: x \rightarrow \frac{2}{1-x^2}$

   i. sketch the graph of f(x).
   ii. Decompose f(x) into partial fractions.
   Hence, find the expansion of f(x) as far as the term in x^4
   iii. Deduce that $\frac{1}{1-x^2} = \sum_{k=0}^{\infty} x^{2k}$
   c. Prove the following sets identities and illustrate them using Venn diagrams.
   i. $A \cap (B \cup C) = \cup (A \cap B) \cup (A \cap B)$
   ii. $(A \cup B \cup C)^I = A^I \cap B^I \cap C^I$

2012/13

SECTION A

1. Simplify $(P \cap Q^l) \cup (P \cap Q)$, *where* P and Q are subsets of the universal set ∪
   A. ∅ C. P
   B. Q D. μ

2. Solve $log_2\sqrt{2} = x$
   A. $x = -\frac{1}{2}$ C. $x = 1$
   B. $x = \frac{1}{2}$ D. x=2

3. Solve the absolute value of inequality $|4 - x| < -3$
   A. ∅ *B. {X:0<x<1)*
   C. (x: x < 1 or x > 7) *D. (x:x>7)*

4. Solve the absolute value inequality $|x + 3| > 4$
   A. {x:x < - 7 or x > 1} B. {x:x<-1 or x> 1}
   C. {x:x> 1} D: { x: x< - 7}

The polynomial defined as $p(x) = x^3 + a_2 x^2 + a_1X + a_0$ has zero $r_1, r_2 and\ r_3$
Use the information to answer question 5 to 7

5. Find relation between the sum of the zero and the coefficients of p(x)
   A. $r_1 + r_2 + r_3 = a_1$ C. $r_1 + r_2 + r_3 = -a_0$
   B. $r_1 + r_2 + r_3 = -a_2$ D. $r_1 + r_2 + r_3 = a_2$

6. Which of the following relationship is correct?
   A. $r_1 + r_1r_2 + r_1r_3 = a_1$
   B. $r_2 + r_1r_2 + r_2r_3 = a_1$
   C. $r_1r_2 + r_1r_3 + r_2r_3 = a_1$
   D. $r_1r_2r_3 = a_0$

7. If $p(x)=x^3+3x^2-x-3$ and -1 is a zero of p(x), find the other zeroes
   A. (1,3) C. (-3, -1)
   B. (3, -1) D. (-3, 1)

8. Solve the inequality $-3 \leq 2x + 1 \leq 5$
   A. $-1 \leq x \leq 1$ C. $-2 \leq x \leq 2$
   B. $-1 \leq x \leq 3$ D. $-2 \leq x \leq 3$

A function f(x) is given by $f(x) = {}^{x}/_{x^2 - 1}$
Use it to answer questions 9 to 10.

9. Which of the following is a vertical asymptote?
   A. x+1=0 C. 2x+1=0
   B. x=0 D. x-2 = 0

10. Find the horizontal asymptote of f(x)
    A. x – axis C. y = 1
    B. y = -1 D. y = 2

11. Given $P(x) = (x-a)^3$, the following are all true except that
    A. x = a has multiplicity 1
    B. (x – a)2 is a factor of P(x)
    C. p(x) is of degree 3
    D. (x-a) is a factor of p(x)

12. Under what condition is a function of (x) considered to be odd?
    A. f(-x)=f(x) C. f(-x) = -f(x)
    B. kf(x) = f(kx) D. f(x) =k

13. Find the coefficient of the term in x^3 in $(x - {}^{2}/_{x^2})^{11}$
    A. -220 C. -1760
    B. 495 D.1980

14. All the following are characteristics of tanx except that it is
    A. an add function
    B. symmetrical about the origin
    C. periodic with period 2π
    D. asymptotic at $x=n\pi + \pi/2$

15. One of the property of sex c is that
    A. it is symmetrical about the y-axis
    B. it is an odd function
    C. its axes of symmetry are $x = \pi$
    D. it is periodic with period π

16. What is the total number of terms in the expansion of $[(a+b)^3]^{17}$
A. 52 C. 50
B. 51 D. 20

17. Let A be n x m matrix and B a n x k. What is the dimension of A(A^TB)
A. n x m C. n x k
B. m x m D. k x m

18. Find x *if* $\cos(x-30^o)=0.5$
A. 60° C. 300°
B. 240° D. 330°

19. Find the determinant of A =
$$\begin{vmatrix} 8 & 4 & -4 \\ 5 & -5 & -5 \\ 7 & 1 & -11 \end{vmatrix}$$
A. 240 C. 480
B. 400 D. 560

20. Let $p(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_2\times^2 a_1+a_0$ be a polynomial, where a $\in Z^+$. If $\frac{p}{q}$ is a zero of P(x), which of the following is not true?
A. p is a factor of a_o. C. q is a factor of a_n
B. p is a factor of a_1 D. $\frac{p}{q}$ is rational

21. Find the exact value of $(1.02)^5$
A. 0.02 C. 1.041
B. 0.014 D. 0.0014

22. If m and N are non-singular matrices, then for MN,
A. $|MN|\neq 0$
B. its inverse does not exist
C. its transpose does not exist
D. $|MN|=0$

A binomial expression is given as $(x+2)^{10}$.
Use it to answer questions 23 and 24

23. Find the 5th term of the expansion of expression in descending power of x.
A. $5040x^5$ C. $3360x^6$
B. $360x^6$ D. $13440x^4$

24. Find the 5th term of the expansion of the expression in ascending powers of x.
A. $3360x^6$ C. $360x^4$
B. $1344x^4$ D. $130x^6$

25. Find the truth set of $\log_{10}(2x^2+5x+97)=2$
A. {-1/3, -3} C. {- ½, -1}
B. {1/3, 3} D. {½, -3}

26. For what value of k does the equation $4x^2-8x+k=0$ have equal roots?
A. k=o C. k=4
B. k=2 D. k=8

A quadratic function f(x) have zeroes $m+{}^1/_n$ and $n+{}^1/_m$ where m and n are roots of the equation $x^2+px+q=0$
Use it to answer questions 27 to 29.

27. Find the sum of roots
A. $\frac{-p(q+1)}{q}$ C. $\frac{-q(q+1)}{p}$
B. $\frac{-(q+1)}{p}$ D. $\frac{-p(q-1)}{q}$

28. Find the product of the roots.
A. $\frac{-q(p+1)}{p}$ C. $\frac{q^2-2q-1)}{q}$
B. $\frac{q^2-2q+1)}{p}$ D. $\frac{(q^2+2q+1)}{q}$

29. Find the equation whose roots are $m+{}^1/_n$ and $n+{}^1/_m$
A. $x^2+\frac{(q+1)}{p}x+\frac{(q^2-2q-1)}{q}=0$
B. $x^2+\frac{p(q-1)}{q}x+\frac{(q^2-2q+1)}{p}=0$
C. $x^2+\frac{p(q+1)}{q}x+\frac{(q^2+2q+1)}{q}=0$
D. $x^2+\frac{p(q-1)}{q}x+\frac{(q^2+2q+1)}{q}=0$

30. If θ is obtuse, and $\sin\theta=\frac{4}{5}$, find $\cos\theta$
A. ${}^3/_4$ C. ${}^3/_5$
B. ${}^{-3}/_5$ D. ¾

SECTION B

1. a. If α *and* β are the roots of $2x^2 - x - 2 = 0$, without solving the equation,

   i. Find the value of $\alpha^2 - \beta^2$ without solving the equation.

   ii. Show that $\alpha^3 - \beta^3 = \frac{5}{2}(\alpha^2 - \beta^2)$

   iii. Find the equation whose roots are α^3 *and* β^3

   b. Factorize completely the polynomial $p(x) = 6x^3 + 13x^3 + 13x^2 - 19x - 12$

2. a. Use matrix methods to solve the system of equation.

$$3Z - y + 2 \times + 17 = 0$$
$$2Z + 6y - 10 = 0$$
$$-Z - 2y + \times + 3 = 0$$

   b. Given the matrices.

$$A = \begin{pmatrix} 3 & 2 & -2 \\ 1 & -3 & 1 \\ 2 & 1 & -3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 8 & 4 & -4 \\ 5 & -5 & -5 \\ -5 & 1 & -11 \end{pmatrix}$$

   Find AB and deduce the inverse a A.

   c. If $x = (x\ 1)\ x \in R$,
   $P = \begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix}$ *and* X^t is the transpose of X, find the truth set of $(XP)X^t = 0$

3. a. Solve the equation $Sec\theta tan\theta = 2$ giving solution for $0^o \leq \theta \leq 360^0$.

   b. Prove that $\tan 3\theta = \frac{3t - t^3}{1 - 3t^2}$ where $t = tan\theta$. Hence, or otherwise show that $\tan {}^{\pi}/_{12} = 2 - \sqrt{3}$

4. a. Sketch and describe completely the curve. $f: x \to -2 - \sqrt{3 - 2x - x^2}$

   b. The sequence $3 - \sqrt{2}, 11 - 6\sqrt{2}, \ldots$ is a geometric progression. Find the common ration and the sixth term.

2011 / 12
SECTION A

1. Rewrite $m^r = n$ as logarithmic equation.
   A. $log_n r = m$ C. $log_n m = r$
   B. $log_m r = n$ D. $log_m n = r$

2. Find the sum of the series
   $^4/_9 + {}^2/_9 + {}^1/_9 + {}^1/_{18} + \cdots + {}^1/_{144}$
   A. $^{127}/_{36}$
   B. $^{127}/_{81}$
   C. $^{127}/_{144}$
   D. there are too many numbers to add

3. What is the eight term of the geometric sequence whose third term is $^3/_4$ and whose sixth term is 10?
   A. 36 C. 45
   B. 40 D. 49

Use the information below to answer questions 4 and 5.
A function f(x) is given by f(x) = |x| − 4

4. State the range of f(x)
   A. {y/y>-4} C. y/y>4}
   B. $\{y/y \geq 4\}$ D. $\{y/y \geq -4\}$

5. Find the intercepts of f(x)?
   A. (0, -4), (0,4), (0,-2)
   B. (-4,0), (4, 0), (0, -4)
   C. (-4, 0), (0, 4), (0, -4)
   D. (-4, 0), (8, 0), (0, -4)

6. Find the vertical asymptote (if any exists) for the curve of the function f(x) = $\frac{x^3-4x^2+7x-4}{x-1}$
   A. $x = -1$ C. y = x -1
   B. $x = 1$ D. *no vertical asymptote*

7. .....logarithms are logarithm to the base e.
   A. Standard C. Natural
   B. Common D. Exponential

8. Decompose $^1/_{x^2-5x+5}$ into partial fractions.
   A. $^1/_{x-2} + {}^1/_{x-3}$ C. $^1/_{x-2} - {}^1/_{x-3}$
   $^1/_{x-3} - {}^1/_{x-2}$ D. $^{-1}/_{x-2} - {}^1/_{x-3}$

9. Solve the inequality $|\frac{3-x}{2}| \leq -3$
   A. $(-\infty, -3] or [9, \infty)$
   B. $(-\infty, -3] or (9, \infty)$
   C. $(-\infty, -3)$ *or* $(9, \infty)$
   D. No solution

10. Determine the sum x+y for the system
    5x – 8y = 29
    2x+2y = -4
    A. -2 C. -4
    B. -3 D. -5

11. What is the domain for f(x) = log (x+4)?
    A. $(-\infty, -4) \cup (-4, \infty)$ C. (-4,∞)
    B. $(-\infty, -4)$ D. [-4, ∞)

12. Rewrite as a single logarithm ln x-2ln y+ln z
    A. $In\frac{xz}{y^2}$ C. $In(\frac{xz}{2y})$
    B. $In(\frac{xz}{y})^2$ D. $\frac{lnxz}{2lny}$

13. Find the solution set of $\begin{vmatrix} 3-x & x+1 \\ 5-x & x+2 \end{vmatrix}$=-6
    A. $x = \frac{1}{5}$ C. $x = \frac{5}{3}$
    B. $x = \frac{-7}{5}$ D. $x = \frac{7}{3}$

14. What is the quotient for $(2x^3 - x^2 + 2) \div (x + 3)$?
    A. $2x^2 - 7x - 21$ C. $2x^2 - 7x - 23$
    B. $2x^2 + 5x + 15$ D. $2x^2 - 7x + 21$

15. Solve for x *if* $log_5(2x - 7) = 1$
    A. x=4 C. x=6
    B. x=5 D. x=-7

16. Determine the values of x for which the sequence x, x+1, 2x+1.... is geometric.
A. -1, -4 C. 1, -4
B. -1, 4 D. 1, 4

17. What is the determinant for
$A = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & -2 & 1 \end{vmatrix}$
A. 3 C. 5
B. 4 D. 6

18. If $log_x + log_x = \log 36$, then
A. x=6 C. x= -6,6
B. x =18 D. x=-1, 18

19. Simplify $\log_5 5^{2t}$
A. 2 C. 2t
B. t D. 10t

20. What is the fifth term of the sequence where $a_n = n^2$?
A. 10 C. 20
B. 15 D. 25

21. Determine the vertical asymptote of the function
$h(x) \frac{x^3+1}{x^2+4}$
A. -4 C. 2
B. -2 D. h(x)

22. What is the vertex for $y=4x^2-6x+5$?
A. $(^3/_2, 5)$ C. (3, -4)
B. $(^3/_4, ^{11}/_4)$ D. (3, 14)

23. What is the slant asymptote for the graph of the graph of $r(x) = \frac{2x^2+x-1}{x+2}$
A. y = 5 C. y=2x+5
B. y=2x-3 D. there is no slant asymptote

24. Determine the 3rd term of the binomial expansion $(2x-3y^2)^5$.
A. $10x^3y^4$ C. $360x^3y^4$
B. $60x^3y^4$ D. $720x^3y^4$

25. All the following are rational numbers except rational numbers except
A. 5.13333 C. $\sqrt{25}$
B. π D. $\frac{25}{7}$

Use the following information to answer questions 26 to 27 given that
A =The set of all natural numbers less than 10
B= {4,6}
C = The solution of the equation $x^2 - 9x+18 = 0$

26. Evaluate $(A \times B) \cap (B \times C)$
A. {(4,6), (3,6)}
B. {(4,6), (3,6), (6,6)}
C. {(4,6),),(3,6),(6,3),(6,6)}
D. {(4,6),(6,3)}

27. Evaluate $(B \times A) \cap (B \times C)$
A. { (3,4),(6,6)} C. { (4,3)(6,6)]
B. {(3,4), (4,6), (6,6) }D. { (6,4), (4,3),(6,6) }

28. Solve $2\sin^2\theta = 1+2\sin\theta$ for $0 \leq \theta \leq \pi$
A-0. 366 C. 1.366
C. 0.366 D. -1.366

29. Solve for t in the equation $0.1^{-(t^2+t)} = 0.01$
A. -1. -2 C. -1, 2
B. 1, -2 D. 1,2

30. If the matrices A, B and C are all of dimension $m \times m$,
$[(AB)c]^t$ can be expressed as
A. $A^tB^tC^t$ C. $C^tB^tA^t$
B. $B^tC^tA^t$ D. $B^tA^tC^t$

SECTION B

1. a. Use matrix inversion to solve the system
   3x+11z=4+7y
   x-2y+3z=1
   -2+8y=4+16z

   b. write 0.67585858...as exponential series and deduce its fraction form.

   c. state and prove the zero multiplication theorem.

2. a. prove that $\frac{tan^2\theta+2}{1+tan^2\theta} = 1 + Cos^2\theta$
   b. Find all the asymptotes and sketch the rational
   function $f(x) = \frac{x^3}{x^2-1}$
   c. determine if the sequence 9,99,999,9999... is
   geometric.

3. a. Use the binomial theorem to expand $(\frac{a}{3} - 2b^3)^5$.
   b. Show that $\frac{x^n+x^{n+2}}{x+x^3} = x^{n-1}$
   c. Find all the solutions of the equation
   $3\cos2\theta + \text{Sin}\theta = 1$
   for values of $O \leq \theta \leq 360$

4. a. With the start of school approaching, Akofo and co. Stationeries is planning of having a sale of school materials. They have 600 notebook, 500 folders and 400 pens in stocks, and they plan on packing it in two different forms. In the first package, there will be 2 notebooks, 1 folders and 1 pen. The price of each package will be GH¢6.50 and GH¢7.00 respectively. How many packages should they put together of each type to obtain the maximum benefit?

   b. Evaluate $3|x + 1| \geq 1 - x$ and illustrate your answer on a number line.

   c. If A, B and C are sets, show that.
   $A - (B \cap C) = (A - B) \cup (A - C)$

2010/2011
SECTION A

1. If logarithm of any number is to base e, what is it called?
   A. standard logarithm
   B. common logarithm
   C. natural logarithm
   D. exponential logarithm

2. Which of the following is an odd function?
   A. $x^3 - 5x+3$ C. $4x^3 - 5x$
   B. $x^3 - x^2 - 6x$ D. $6x^4 - 3x^2+1$

3. Which of the following forms would be the partial fraction decomposition of $\frac{x^3+2x-1}{(x^2-4)^2(x^2+1)}$ represent?
   A. $\frac{Ax+B}{x^2} + \frac{(x+D)}{(x^2-4)^2} + \frac{E\times F}{x^x}$
   B. $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+2} + \frac{D}{(x+2)^2} + \frac{Ex+F}{x^2+1}$
   C. $\frac{A}{(x-2)^2} + \frac{B}{(x+2)^2} + \frac{(X+D)}{X^2+1}$
   D. $\frac{A}{x-2} + \frac{B}{x+2} + \frac{C}{x-1} + \frac{D}{x+1}$

4. Find the vertical asymptote (if any exist) for the curve of the function $f(x) = \frac{x^3-16}{x-4}$
   A. x= - 4 C. y = x+4
   B. x = 4 D. no vertical asymptote

5. Find the solution set of $\begin{vmatrix} 3-x & x+1 \\ 5-x & x+2 \end{vmatrix} = -6$
   A. $x = \frac{7}{5}$ C. $x = \frac{5}{3}$
   B. $x = 1$ D. $x = \frac{7}{3}$

6. A function is given by $f(x) = |x| - 4$. What is the range of f(x).
   A. {y:y>-4} C. $\{y: y > 4\}$
   B. $\{y: y \geq 14\}$ D. $\{y: y \geq -4\}$

7. Solve: $\frac{x+6}{4} - \frac{x-3}{7} < 9$
   A. x> 62 C. x< 106
   B. x < 62 D. x > 10

8. Find the determinant of the matrix
$$\begin{vmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{vmatrix}$$
   A. -11 C. 6
   B. -6 d. 11

9. Given that $A = \begin{vmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{vmatrix}$, determine the (3, 2) – cofactor of A
   A. -11 C. 6
   B. -6 D. 11

10. Which of the following is not a subset of Q?
    A. Z C. R
    B. W D. N

11. If log2= 0.3010 and log 3= 0.4770, find $\log \frac{1}{75}$
    A. -1.875 C. 0.875
    B. -.0875 D. 1.875

12. Find the oblique asymptote for the curve $g(x) \frac{x^2+5}{x+1}$
    A. y = -6 C. y = 6
    B. y = x-1 D. y = x+1

Use the information to answer questions 12 and 13.

Given that $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$ *and* $B = \begin{pmatrix} -5 & 3 \\ 0 & -2 \end{pmatrix}$

13. Determine the adjoint of (A+B)

21

A. $\begin{pmatrix} 8 & -5 \\ -3 & -4 \end{pmatrix}$ C. $\begin{pmatrix} -4 & 5 \\ 3 & 8 \end{pmatrix}$

B. $\begin{pmatrix} -4 & -5 \\ -3 & 8 \end{pmatrix}$ D. $\begin{pmatrix} 8 & 5 \\ 3 & -4 \end{pmatrix}$

14. Evaluate the invers of matrix A.

A. 0 C. $\begin{pmatrix} -4 & 5 \\ 3 & 8 \end{pmatrix}$

B. $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ D. the inverse does not exist

15. Solve $|2x + 1| > 7$

A. $-3<x<4$ C. $x< -3 \cup x>4$

B. $-4<x<3$ D. $x<-4 \cup x>3$

16. Decompose $\frac{1}{(x^2+5x+6)}$ into partial fractions.

A. $\frac{1}{x+2}+\frac{1}{x+3}$ C. $\frac{1}{x+2}+\frac{1}{x+3}$

B. $\frac{\frac{1}{x+3}}{\frac{1}{x+3}}-\frac{1}{x+2}$ D. $-\frac{1}{x+2}-$

17. Find the domain of the function $g(x) = \sqrt{4-x^2}$

A. $-2<x<2$ C. $0\leq x\leq 2$

B. $0<x<2$ D. $-2\leq x\leq 2$

18. Let A be an $n \times m$ matrix and B, an n x k matrix. What is the dimension of $A(A^TB)$?

A. $n \times m$ C. $n \times k$

B. $m \times m$ D. $k \times m$

Use the information to answer question 19 and 20 Given that the first term of an A.P is 3 and the 20th term is 117.

19. Find the common difference of the sequence.

A. 3 C. 5

B. 4 D. 6

20. Determine the sum of the first 19 terms.

A. 111 C. 113

B. 112 D. 114

21. Which of the following is a rational number; $\sqrt{0.16}, \pi, \sqrt{2}$ and $\sqrt{50}$

A. $\sqrt{0.16}$ C. π

B. $\sqrt{12}$ D. $\sqrt{50}$

22. Find the coefficient of the term a^7b^3 in the expansion of $(a-3b)^{10}$.

A. -3240 C. 120

B. -120 D. 3240

23. If A = {a,b}, B={b,c} and C={d,e,f}, how many elements would be contained in the set A×B×C

A. 6 C. 12

B. 7 D. 16

24. Find the truth values of x if $4^{(x+1/2)}+4^{(3/2-x)} = 17$

A. X = - ½, - $^3/_2$ C. x = ½ , - $^3/_2$

B. - ½ , $^3/_2$ D. x = - ½ , - $^3/_2$

25. Find the truth of set of $\cos\theta = 0$ for $0\leq \theta \leq 2\pi$

A. $\{\theta:\theta = 0, \frac{\pi}{2}\}$ C. $\{\theta:\theta = \frac{\pi}{2}, \frac{3\pi}{2}\}$

B. $\{\theta:\theta = 0, \frac{3\pi}{2}\}$ D. $\{\theta:\theta = \frac{\pi}{2}, \frac{\pi}{4}\}$

26. Evaluate $\log(\log 1000^{1000})$

A. 0 C. 1.4313

B. 3.4771 D. 0.9542

27. Find the intercepts of the functions f(x) =|x|-4 on both axes

A. (0, -4), (0.4), (0.-2) C.

B. (-4,0), (4,0), (0, -4)
C. (-4,0), (0,4), (0.-4)
D. (-4, 0), (8, 0), (0, -4)

28. Solve the inequality $|\frac{3-x}{2}| \geq 3$
A. $(-\infty, -3]$ *or* $[9, \infty)$
B. $(-\infty, -3]$ *or* $(9, \infty)$
C. $(-\infty, -3)$ *or* (9∞)
D. $(-\infty, -3)$*or* $[9\infty)$

Use the information to answer questions 29 and 30
Let U = {Natural numbers less than 17.
A=$\{x \in U: x^2 - 11x + 28 = 0\}$
B = $\{x \in u: x \; is \; a \; factor \; of \; 12\}$
C = {3,6}

29. Find $(AxB) \cap (CxB)$
A. $\emptyset$ C. {(3,4),(6,4)}
B. {(4,3), (4,6)} D.{(4,3), (7,3), (4,6), (7,6)}

30. How many elements would the set $(A \times B) \cap (A \times C)$ contain?
A. 2 C. 4
B. 3 D. 5

SECTION B

1. a. The notes of a piano, as measured in cycles per seconds, form a geometric sequence.
i. If A is 400 cycles per seconds, and A^1 which has 12 notes higher than A is 800 cycles per second, find the constant ratio.

ii. Find the cycle per second for C, three notes higher than A.

b. Solve for *x* if *log (x+3)+log x = 1*

c. Simplify the expression $\frac{30Cos^2\theta Sin\theta}{65in^2\sigma Cos\theta}$ assuming that $\sin\theta \neq 0 \; and \; Cos\theta \neq 0$

2. a. Decompose $f(x) = \frac{x^3-7+17x-17}{x^2-6x+9}$

b. If A, B and C are sets, show that:
$A - (B \cap C) = (A - B) \cup (A - C)$

c. Find all values of x in the interval $[0, 2\pi]$ such that *sinx=Sin2x*.

3. a. Use the cramer's rule to solve the system
3x+11z=4+7y
x- 2y+3z = 1
-2x +8y =4 +16z

b. Solve the equation $4tan^2\theta + sec\theta - 1 = 0$,
giving all solution in the interval $180^O < \theta <$ 180° to the nearest degree.

c. Show that $\frac{x^n+x^{n+2}}{x+x^3} = x^{n-1}$

4. Find all the asymptotes and sketch the rational function $f(x) = \frac{x}{x^2-x-2}$

b. If $\log_9(a+3) - \log_9 b = c + {}^1/_2$ and $\log_3(a-3) - \log_3 b = c-1$, write an expression for a^2.

c. Write $0.\dot{2}\dot{3}$ as an exponential series and deduce the fractional form.

2009/10
SECTION A

1. Which of the following statements about matrix operations is not true?
   A. matrix addition is commutative
   B. matrix multiplication is commutative
   C. matrix multiplication is associative
   D. matrix addition is associative

2. The polynomial $9zy^4y^9 + 9z^9y^7+4y^9$ is a trinomial in two variables with degree...
   A. 9 C. 13
   B. 27 D. 16

3. Let an = $(-1)^n n^2$, find S_4
   A. 16 C. 11
   B. 10 D. 13

4. Find the truth set of x if $\log_2(3x^2 + 8x - 1) = 1$
   A. $\{x:x=-3,-1/3\}$ C. $\{x:x=-3,1/3\}$
   B. $\{x:x=3,1/3\}$ D. $\{x:x=3,-1/3\}$

5. Evaluate the determination of the matrix
   $A = \begin{vmatrix} 3 & -2 & 0 \\ 2 & 5 & 8 \\ 5 & 3 & 5 \end{vmatrix}$
   A. 97 C. 3
   B. 63 D. -57

6. Solve $|4 - x| < -3$
   A. $\{x>1\}$ C. $\emptyset$
   B. $\{x>7\}$ D. $\{1<x<7\}$

7. Evaluate $\tan\left(\frac{\pi}{2} - \theta\right)$
   A. Cot θ C. $\tan\frac{\pi}{2}$
   B. -tanθ D. $\cot\frac{\pi}{2}$

8. Find the vertical asymptote (if it exist) for the curve whose equation is given as
   . $y = \frac{x^2+x-6}{x+3}$
   A. The vertical asymptote is the line x=-3
   B. The vertical asymptote is the line x=2
   C. The vertical asymptote is the line
   D. The curve does not have a vertical asymptote.

9. Find the turning point of the parabola $x^2 = x - 5$
   A. $(\frac{1}{2}, \frac{19}{4})$ C. $(-1/2, 21/4)$
   B. $(-1/2, 19/4)$ D.
   $(1/2, -21/4)$

10. Decompose $\frac{1}{(x+3)(x+2)}$ into partail fraction
    A. $\frac{1}{(x+2)} + \frac{1}{x+3}$ C. $\frac{1}{x+2} - \frac{1}{x+3}$
    B. $\frac{1}{x+3} + \frac{1}{x+2}$ D. $\frac{1}{x+2} - \frac{1}{x+3}$

11. Find the common difference for the arithmetic sequence *x-5b, x-3b, x-b....*
    A. -2b C. 2b
    B. 8b D. -8b

12. Solve $|x + 5| = |3x - 4|$
    A. $x=\frac{9}{4} or x = -\frac{1}{4}$ C. $x = \frac{9}{2} or\ x = -\frac{1}{4}$
    B. $x = -\frac{9}{2} or\ x = \frac{9}{2}$ D. $x = \frac{9}{4} or\ x = \frac{1}{4}$

13. Evaluate the determinant of the matrix
    $B = \begin{vmatrix} 1/2 & 1/2 \\ 1/8 & -3/4 \end{vmatrix}$
    A. $-5/16$ C. $-1/16$
    B. $-7/16$ D. $-4/16$

14. Which of the following fractions is an irrational number.
A. $\frac{1}{16}$ C. $\frac{2}{52}$
B. $\frac{15}{101}$ D. $\frac{4}{32}$

15. Solve for $\sin\theta$ in $25\sin^2\theta = 1+25\sin\theta$ for $0 \leq \theta \leq 2\pi$
A. -0.366 C. 0.366
B. 1.366 D. -1.366

16. Find the coefficient of x^3 in the expansion of the binomial expression $(1+3x)^{-2}$
A. -648 C. 648
B. 108 D. -108

17. Find the vertical asymptote of the curve whose equation is given as $y=\frac{3x^3+5x^2-26x+8}{x^2+2x-8}$
A. the vertical asymptotes are the lines = 2 and x = 4
B. the vertical asymptotes are the lines x= -2 and x = -4
C. the vertical asymptotes are the line x=2 and x = -4
D. the curve does not have a vertical asymptote.

18. Find the value of x if $x = \log_{10} 0.001$
A. x= –10 C. x = –0.001
B. x = –3 D. x= –2

19. Solve the inequality $\left|\frac{3-x}{2}\right| \geq 3$
A. $(-\infty, -3]$ or $[9, \infty)$ C. $(-\infty, -3)$ or $(9, \infty)$
B. $(-\infty, -3]$ or $(9, \infty)$ D. $(-\infty, -3)$ or $[9, \infty)$

20. Solve for x in the equation 2x + A =B.
where $A = \begin{vmatrix} 1 & -2 \\ 0 & 3 \end{vmatrix}$ and $B \begin{vmatrix} -3 & 4 \\ 2 & 1 \end{vmatrix}$
A. $\begin{vmatrix} -2 & 3 \\ 1 & -1 \end{vmatrix}$ C. $\begin{vmatrix} -8 & 12 \\ 4 & -4 \end{vmatrix}$
B. $\begin{vmatrix} -4 & 6 \\ 2 & -2 \end{vmatrix}$ D. $\begin{vmatrix} -2 & 2 \\ 2 & 4 \end{vmatrix}$

21. Find the common difference for the arithmetic sequence $x + \sqrt{3}, x\sqrt{12}, x + \sqrt{27}, \ldots$
A. $\sqrt{9}$ C. $\sqrt{3}$
B. $\sqrt{15}$ D. $\sqrt{12}$

22. What is the value of x in $2^x + 2^{-x} = 2$?
A. 0 C. 2
B. 1 D. 3

23. How many terms would the expansion of $(30x+20b)^{20}$ have?
A. 20 C. 30^{20}
B. 20^{20} D. 21

24. Find the value of y if $\begin{vmatrix} 2-y & y-4 \\ 5-y & y-1 \end{vmatrix} = 15$
A. $y=\frac{1}{2}$ C. $y=\frac{1}{4}$
B. y = 2 D. y = 4

25. Given $\tan\theta = {}^5/_4$ and $\cos\theta > o$. *find* $\sec\theta$.
A. $\frac{\sqrt{41}}{5}$ C. $\frac{-5}{\sqrt{41}}$
B. $\frac{\sqrt{41}}{4}$ D. $\frac{-\sqrt{41}}{5}$

26. If $\ln z = 24$, *find* z.
A. $Z = 2.65\text{x}10^{10}$ C. Z = – 3.18
B. Z = 1.38 D. $Z = 1\text{x}10^{24}$

27. Solve the inequality $|2x-3| \leq 4$
A. $[{}^{-1}/_2, {}^7/_2)$ C. $({}^{-1}/_2, {}^7/_2)$
B. $({}^{-1}/_2, {}^7/_2]$ D. $[{}^{-1}/_2, {}^7/_2]$

28. Find the horizontal asymptotes of the curve whose equation is given as $y=\frac{2x^2-5x-3}{3x^2-7x-6}$
A. y= 0 C. $y = {}^1/_2$
B. $y= {}^2/_3$ D. y = 1

29. Find the sum of the first three terms of the sequence with the nth term equal to 2.

A. 0 C. 6
B. 4 D. 2

30. Which quadrant should θ lie, to make the expression.
. $cos\ \theta = -\sqrt{1 - sin^2\theta}$ *true if* $\sin\theta > O$
A. 1st quadrant C. 3rd quadrant
B. 2nd quadrant D.4th quadrant

SECTION B

1. a. If A is a 3×3 matrix and I is a 3×3 identify matrix, prove that IA = A and AI = A

b. Decompose the rational function $\frac{2x^2-5x+4}{x^3}$ into partial fractions

c. Prove that $\sin\theta = \frac{2tan\frac{1}{2}\theta}{1+tan^2\frac{1}{2}\theta}$

2. Find the sum of the first n terms of the sequence whose n^{th} term is $2^n - 2n + 1$.
b. Given that $\log_\theta(p + 2) + \log_\theta q = r - 1/3$ and $\log_2(p - 2) - \log_2 q = 2r + 1$.
Show that $p^2 = 4 + 32^r$. If r = 1, find the possible values of p and q.

c. Your broker tells you of two investments she thinks are worthwhile. She advises a new issue of P. Coles stocks which should yield 20% over the next year, and then to balance your accounts. She advices Cape Coast Metropolitan bonds with 10% yearly yield. The stock to bond ratio should be no less than 3 to 1. If you have no more than 100,000 Ghana Cedis to invest and donot want to invest more than 70,000 Ghana cedis in P. Cores or less than 20,000 Ghana cedis in bonds, how much should be invested in each to maximize your return?

3. Use binomial series expansion to approximate $(2.99)^{1/3}$ correct to five decimal places.

b. Solve the inequality below, expressing the solution set in interval notation. $\left|\frac{x+4}{x-5}\right| \leq 3$

c. Solve that $\frac{Sex20+Cot20=}{Sec20}$

4. a. Solve the system of equations below
$x - 2y - 5z = -12$
. $2x - y = 7$ *using*
$5y + 6z = 4x + 1$

Using matrix approach

b. Sketch the graph of the function $y + 1\frac{-3}{x-2}$

c. Prove that $\frac{\sec^2 2\theta - \tan^2 2\theta}{\tan 2\theta - \sec 2\theta} = \frac{\cos 2\theta}{1+\sin 2\theta}$

SOLUTIONS TO SECTION A

2017/2018

1. Given $\frac{5m+10}{5} = n$

$$m + 2 = n \quad \text{--- (1)}$$

Then $\frac{9m+54}{9} = y$

$$m + 6 = y$$

$$but\ m + 2 = n$$

$$and\ y = m + 2 + 4$$

$$y = n + 4$$

$$ans: D$$

2. Given $y = x^2 + bx + 256$

Since h and k are real, $b^2 - 4ac \geq 0$

$$b^2 - 4(1)(256) \geq 0$$

$$b^2 \geq 1024$$

$$b \geq \pm\sqrt{1024} = \pm 32$$

Hence the least is -32. Ans: A

3. $S_n = \frac{a(1-r^n)}{1-r}$

$$S_6 = \frac{a(1-r^6)}{1-r} \quad and \quad S_3 = \frac{a(1-r^3)}{1-r}$$

$$\frac{S_6}{S_3} = \frac{a(1-r^6)}{1-r} \times \frac{1-r}{a(1-r^3)} = \frac{1-r^6}{1-r^3}$$

$$\frac{1-r^6}{1-r^3} = 9$$

$$r^6 - 9r^3 + 8 = 0$$

Let $x = r^3$

$$x^2 - 9x + 8 = 0$$

$$x = 8,1$$

$$but\ x = r^3\ so\ r^3 = 8 \implies r = 2$$

$$also, r^3 = 1 \implies r = 1$$

Ans. C

4. Given $2|-2x - 2| - 3 = 13$

$$|-2x - 2| = 8$$

$$(-2x - 2)^2 = 8^2$$

$$4x^2 + 8x + 4 = 64$$

$$4x^2 + 8x + 60 = 0 \implies x = 3, -5$$

Ans. C

5. $S_n = 2n = \frac{n}{2}[2a + (n-1)d]$

$$S_{2n} = 3n = \frac{2n}{2}[2a + (2n-1)d]$$

$$S_{3n} = \frac{3n}{2}[2a + (3n-1)d]$$

$$\implies 4 = (2a + nd - d) \ldots\ldots (1)$$

$$3 = (2a + 2nd - 2d) \ldots\ldots (2)$$

Now (1)-(2)

$$1 = -nd \implies d = \frac{-1}{n}$$

Put $d = \frac{-1}{n}$ *into* (1)

$$4 = 2a + n\left(\frac{-1}{n}\right) - \left(\frac{-1}{n}\right)$$

$$4 = 2a - 1 + \frac{1}{n}$$

$$a = \frac{2}{5} - \frac{1}{2n}$$

$$S_{3n} = \frac{3n}{2}\left|2\left(\frac{2}{5} - \frac{1}{2n}\right) + (3n-1)\left(\frac{-1}{n}\right)\right| = 3n$$

Ans. D

6. Area of first square $(u_1) = l^2 = 4^2 = 16$

Area of second square $(u_2) = \frac{1}{2}l^2 = 8$

Area of third square $(u_3) = \frac{1}{4}l^2 = 4$

$u_1, u_2, u_3, \ldots$

Since it's a G.P to infinity $a = 16,\ r = \frac{1}{2}$

$$S_\infty = \frac{a}{1-r} = \frac{16}{1-\frac{1}{2}} = 32$$

Ans. B

7. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

Minimum elements in $n(B \cap C) = 8$

Minimum elements in $n(A \cap B) = 7$

Minimum elements in $n(A \cap C) = 7$

$min.\ n(A \cap B \cap C) = 7$

$min.\ n(A \cup B \cup C) = 7 + 8 + 8 - 7 - 7 - 8 + 7 = 8$

Ans. A

8. Given $(x-2)^n = (-2+x)^n$

$u_6 = -8064x^5 \ldots . (1)$

$u_{r+1} = \binom{n}{r} a^{n-r}b^r$

$u_{5+1} = x^5 \ldots . (2)$

Comparing (1) and (2)

$-8064 = \binom{n}{5}(-2)^{n-5}$ *is true when* $n = 10$

Ans. C

9. Given $3kx^2 - 18tx + (2k+3) = 0$

Product of roots $= \alpha\left(\frac{1}{\alpha}\right) = \frac{2k+3}{3k}$

$1 = \frac{2k+3}{3k} \implies k = 3$

Ans. C

10. Given $(a+b)^{15}$

For a^2b^{13}

$u_{r+1} = \binom{n}{r}a^{n-r}b^r$ comparing $a^2b^{13},\ r = 13$

$u_{13+1} = \binom{15}{13}a^{15-13}b^{13} = 105a^2b^{13}$

Ans. A

11. For zeros, $2 + \frac{1}{\sqrt{2}}$ *and*, $2 - \frac{1}{\sqrt{2}}$

$\left(x - 2 - \frac{1}{\sqrt{2}}\right)\left(x - 2 + \frac{1}{\sqrt{2}}\right) = 0$

$2x^2 - 8x + 7 = 0$

Ans. C

12. Given $x^3 + x^2 - 7x - 3$

If $x = 3$, *then* $x^3 + x^2 - 7x - 3 = 0$

So $x = -3$ *is a factor. hence* $(x+3)$ *is a factor*

Dividing $x^3 + x^2 - 7x - 3$ by $x + 3$ gives $x^2 - 2x - 1$

Hence $x^2 - 2x - 1 = 0$ *gives* $x = 1 \pm \sqrt{2}$

Ans. C

13. $P(x) = a_3x^3 + a_2x^2 + a_1x + a_0$

$P(x) = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma$

$\alpha + \beta + \gamma = -\frac{a_2}{a_3}$ by comparison

Ans. C

14. Comparing p(x) in Q13, $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{a_1}{a_3}$

Ans. B

15. $P(x) = x^3 + 3x^2 - x - 3$, $x = 1$

Let a and b represent the other roots

$$a + b - 1 = -3 \ldots\ldots (1)$$

$$ab = -3 \ldots\ldots (2)$$

Solving (1) and (2), $b = 1, -3$

Ans. C

16. $S_\infty = \frac{a}{1-r} = 4 \ldots. (1)$

$u_2 = ar = \frac{3}{4} \ldots\ldots (2)$

From (1), $a = 4(1-r) \ldots\ldots (3)$

Putting (3) into (2)

$$4(1-r)r = \frac{3}{4} \quad \Rightarrow r = \frac{3}{4}, \frac{1}{4}$$

If $r = \frac{1}{4}$ $\quad \Rightarrow a = 4\left(1 - \frac{1}{4}\right) = 3$

Ans. D

17. $\sum_{k=0}^{\infty} 5\left(\frac{2}{3}\right)^k = \sum_{k=0}^{\infty} 5\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)^{k-1}$

$$\sum_{k=0}^{\infty} \frac{10}{3}\left(\frac{2}{3}\right)^{k-1} = \frac{a}{1-r} \quad where\ a = \frac{10}{3}, \quad r = \frac{2}{3}$$

$$\frac{a}{1-r} = \frac{\frac{10}{3}}{1-\frac{2}{3}} = 10$$

Ans. C

18. $u_{r+1} = \binom{n}{r} a^{n-r} b^r$

$$= \binom{12}{r} (2x^3)^{12-r} (4)^r$$

Thus $x^{30} = x^{3(12-r)} \quad \Rightarrow r = 2$

Ans. A

19. Given $x^2 + mx + 24 = 0$

When $x = 1.5$ *then* $m = -17.5$

Ans. B

20. $u_4 + u_{12} = 20$

$$a + 3d + a + 11d = 20$$

$$2a + 14d = 20$$

But $S_{15} = \frac{15}{2}(2a + 14d) = \frac{15}{2}(20) = 150$

Ans. C

21. $\log_a x = \log_{a^k} x^k$, $a = k = 2$

$$\log_2 x = \log_{2^2} x^2$$

$$\log_2 x = \frac{\log_2 x^2}{\log_2 4}$$

Simplifying this gives $\log_2 x = \log_2 x$ which means x is equal to all integers greater than 0. The question is actually a property which is true for all positive real values of x. Hence, all options are correct.

22. Given $M^2 - 3M + 5 = 0$

$\Rightarrow M^2 - tr(M) + |M| = 0$

Comparing $tr(M) = 3$, $|M| = 5$

From the option $M = \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}$

Ans. C

23. If $M = \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}$ $|M| = 5$

$$M^{-1} = \frac{1}{|M|}\begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0.4 & 0.6 \\ -0.2 & 0.2 \end{pmatrix}$$

Ans. B

24. No answer

25. No answer

26. $f(x) = \frac{x-2}{x^3-5x}$

For domain, $x^3 - 5x = 0, \quad x = \pm 2, 0$

$D_f = (-\infty, -2.2) \cup (-2.2, 0) \cup (0, 2.2) \cup (2.2, \infty)$

Ans. B

27. for vertical asymptotes

$$x^3 - 5x = 0, \quad x = \pm 2, 0$$

Ans. C

28. for horizontal asymptote, y=0. Ans: A

29. Given $Y \subseteq X$. This implies Y is contained in X. Thus, all elements of Y are elements of X but not vice versa. Thus, $X' \subseteq Y'$

From the options, $Y' \cap Z' \subseteq X' \cap Z'$

Ans. C

30. Given $(Z \times N) \cap (N \times Z)$ where N=natural numbers and Z=Integers

Since $N \subseteq Z$ then $N \times N \subseteq Z \times N$ and $N \times N \subseteq N \times Z$

Therefore, $(N \times N) \cap (N \times N) \subseteq (Z \times N) \cap (N \times Z)$

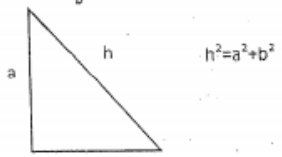
$(N \times N) \subseteq (Z \times N) \cap (N \times Z)$

Ans. C

2016/17

SECTION A

1. Refer to 2017/18 Question 1
2. Refer to 2017/18 Question 2
3. Refer to 2017/18 Question 3
4. Refer to 2017/18 Question 4
5. Given $\tan x = \frac{a}{b}$, by Pythagoras theorem



$h^2=a^2+b^2$

$\Rightarrow h = \sqrt{a^2 + b^2}$

$\sin 2x = 2 \sin x \cos x$

$=2\left(\frac{a}{\sqrt{a^2+b^2}}\right)\left(\frac{b}{\sqrt{a^2+b^2}}\right)$

$=\frac{2ab}{\sqrt{a^2+b^2}}$

Ans : A

6. $\tan 2x = \frac{2\tan x}{1-\tan^2 x} = \frac{2\left(\frac{a}{b}\right)}{1-\left(\frac{a}{b}\right)^2} = \frac{2ab}{b^2-a^2}$

Ans: B

7. Given 2,5,8,...

S_{2n}= ½(2n)[2(2)+(2n-1)3]

And 57, 59, 61, ...

S_n= ½n[2(57)+(n-1)2]

$\therefore S_{2n} = S_n$

$\Rightarrow$ ½(2n)[2(2)+(2n-1)3] = ½n[2(57)+(n-1)2]

Solving gives n=11

Ans: B

8. Refer to 2017/18 Question 8
9. Given $2\left|\frac{2}{3}x + 1\right| \geq 4$

$$\left|\frac{2}{3}x + 1\right| \geq 2$$

$$\Rightarrow \frac{2}{3}x + 1 \geq 2 \; and \; \frac{2}{3}x + 1 \leq -2$$

$x \geq \frac{3}{2}$ and $x \leq -\frac{9}{2}$

Ans: A

10. Refer to 2017/18 Question 10.
11. Given 1,-4, 9, -16,...

If n=1

$$U_1 = 1$$
$$U_2 = -4 = (2)^2(-1)^{2+1}$$
$$U_3 = 9 = (3)^2(-1)^{3+1}$$
$$\Rightarrow U_n = (n)^2(-1)^{n+1}$$

no answer in option

12. Refer to 2017/18 Question 11
13. Given

$$\sin\theta + \csc\theta = 2$$
$$\sin\theta + \frac{1}{sin\theta} = 2$$
$$sin^2\theta + sin\theta + 1 = 0$$

$$\sin\theta = 1$$

$\sin^2\theta + \csc^2\theta = 1^2 + (1/1)^2 = 2$

Ans: B

14. Given tanx=-24/7

But tan$x = \frac{2\tan(\frac{1}{2}x)}{1-\tan^2(\frac{1}{2}x)}$

$$-\frac{24}{7} = \frac{2\tan\left(\frac{1}{2}x\right)}{1-\tan^2\left(\frac{1}{2}x\right)}$$

$24\tan^2\left(\frac{1}{2}x\right) - 14\tan\left(\frac{1}{2}x\right) - 24 = 0$

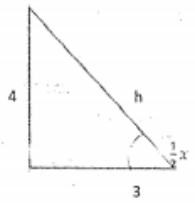
$$\tan\left(\frac{1}{2}x\right) = \frac{4}{3}, -\frac{3}{4}$$

Since x is in quadrant II, ½ x is in quadrant I, thus tan(½ x) is positive

$\therefore \tan^2\left(\frac{1}{2}x\right) = \frac{4}{3}$, $h^2 = 4^2 + 3^2 = 25$

h=5

$$\sin\left(\frac{1}{2}x\right) = \frac{4}{5}$$



15. Given $S_{2n}=3S_n$

$1/2(2n)[2a + (2n-1)d] = 3[1/2n(2a + (n-1)d]$

$n[2a+2nd-d] = 3/2n[2a+nd-d]$

$2a + 2nd - d = 3a\ 3/2nd - 3/2d$

$(2n-1)d - (3/2n-3/2)d = a$

Simplifying these;

$d = \frac{2a}{1+n}$

Ans: B

16. Refer to 2017/18 Q16

17. Refer to 2017/18 Q17

18. Refer to 2017/18 Q18

19. Refer to 2017/18 Q19

20. Refer to 2017/18 Q20

21. Refer to 2017/18 Q21

22. Refer to 2017/18 Q22

23. Refer to 2017/18 Q23

24. Refer to 2017/18 Q24

25. Refer to 2017/18 Q25

26. Refer to 2017/18 Q26

27. Refer to 2017/18 Q27

28. Refer to 2017/18 Q28

29. Refer to 2017/18 Q29

30. Refer to 2017/18 Q30

2014 / 15
SECTION A

1. $U_n = 2 - \frac{1}{3}U_{n-1}$,

$\Rightarrow U_{n+1} = 2 - 1/3\ [U_{(n+1)-1}]$

$= 2 - {}^1/_3 U_n$

$\therefore U_{n+1} - U_n = 2 - \frac{1}{3}Un - (2 - \frac{1}{3}U_{n-1})$

$= 2 - 1/3\ U_n - 2 + {}^1/_3\ U_{n-1}$

$= -1/3\ U_n + {}^1/_3\ Un\text{-}1$

$= (-{}^1/_3)\ [U_n - U_{n-1}]$

$\therefore$ *Answer is A*

2. Given $x_1, x_2, x_3, \ldots$ *as A.P* common difference, $d = x_2 - x_1 = x_3 - x_2$

$= x_2 - x_1 = -x_2$

$2x_2 = x_3 + x_1$

$X_2 = \frac{x_3 + x_1}{2}$

$\therefore$ *Answer is A*

3. $x_1, x_2, x_3, \ldots$ as $G.P$ common ration r = $\frac{x_3+x_1}{x_1} = \frac{x_3}{x_2}$
   $\Rightarrow x^2_2 = x_1x_3$
   $X_2 = \sqrt{x_1x_3}$
   ∴ *Answer is D ~ B*

4. Given $U_n = U_n + \left(\frac{1}{3}\right)^{n-1}$ $\forall n > 1$ *and* $x_1 =$ 1
   $X_2 = U_2 = U_{2-1} + (^1/_3)^{2-1}$
   $= U1 + (^1/_3)$
   $= 1 + ^1/_3$
   $X_2 = 1^1/_3$
   ∴ *Answer is D*

5. U_n is a recurrence sequence.
   ∴ *Answer is D*

6. Given $2^x - 5^y = 3$
   $2^{x-3} + 5^{y-2} = 21$
   Let $2^x = a$, $5^y = b$
   $= a - b = 3$ -------- (1)
   Also, $\frac{a}{2^3} + \frac{b}{5^2} = 21$

   $\Rightarrow 25a + 5b = 4200$ ---------- (2)
   Solving (1) and (2) simultaneously, a = 128, b = 125.
   But $a = 2^x$
   $= 128 = 2^x$
   $2^7 = 2^x$
   ∴ x = 7
   Also $b = 5^y$
   $125 = 5^y$
   $5^3 = 5^y$
   ∴ y = 3
   ∴ *Answer is C*

7. From Q.6, x = 7 and y = 3
   ∴ x + y = 7+3 = 10
   ∴ *Answer is B*

8. Given $2x^3 + \lambda x^2 + 4x + \mu$
   Since (x-1) and (2x - 4) are factors,
   $\Rightarrow$ x = 1 and x = 2 are zeros
   ∴ $2(1)^3 + \lambda(1)^2 + 4(1) + \mu = 0$
   $\lambda + \mu = -2 - 4$
   $x + u = -6$ – (1)
   $2(2)^3 + \lambda(2)^2 + 4(2) + \mu = 0$
   $16 + 4\lambda + 8 + \mu = 0$
   $4\lambda + \mu = -24$ – (2)
   From (1) and (2),
   $\lambda = -6$, $\mu = 0$
   ∴ *Answer is A*

9. From solution to Q8.
   ∴ *Answer is B*

10. Now the polynomial is $2x^3 - 6x^2 + 4x$,
    $\Rightarrow$ x = 0 is a zero, thus is the other factor
    ∴ *Answer is A*

11. Given $U_n = 2n - 1$
    $Sn = \sum_{n=1}^{x} U_k$
    But $U_k = 2k - 1$
    ∴ $Sn = \sum_{n=1}^{\infty} 2k - 1$
    $= \sum_{n=1}^{\infty} 2k - \sum_{n=1}^{x} 1$
    $= 2 \sum_{n=1}^{} k - \sum_{n=1}^{} 1$
    $= 2(^1/_2 n((n+1) - n$
    $= (n(n+1) - n$
    $= n^2 + n - n$
    $= n^2$
    ∴ *Answer is A*

12. Given $3 - \sqrt{2}, 11 - 6\sqrt{2}, \ldots$
    Common ratio r = $\frac{11-6\sqrt{2}}{3-\sqrt{2}}$

$r = \frac{11-6\sqrt{2}}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}}$

$r = \frac{33+11\sqrt{2}-18\sqrt{2}-6(2)}{9-2}$

$r = \frac{21-7\sqrt{2}}{7}$

$r = 3-\sqrt{2}$

∴ *Answer is C*

13. Since the sequence is a GP, $Un = ar^{n-1}$, a =

$3-\sqrt{2}$

$r = 3-\sqrt{2}$

$\therefore U_3(3-\sqrt{2})(3-\sqrt{2})^{3-1}$

$=(3-\sqrt{2})(3-\sqrt{2})^2$

$=(3-\sqrt{2})^3$

$=U_2 = 45-29\sqrt{2}$

∴ *Answer is A*

14. Given $g(x) = -2-\sqrt{3-2x-x^2}$

Let $y = -2-\sqrt{3-2x-x^2}$

$(y+2)^2 = \left(-\sqrt{3-2x-x^2}\right)^2$

$x^2+y^2+2x+4y+1=0$

Comparing to equation of circle $x^2+y^2+2gx+2fy+c=0$

$g=1, f=2, c=1$

$\therefore centre\ (-g,-f) = (-1,-2)$

$r = \sqrt{g^2+f^2-c} = \sqrt{(1)+(2)^2-1}$

$r = \sqrt{1+4-1}$

$r = \sqrt{4}$

$r = 2$

Now since the radical function in g(x) has a negative beside it, the curve is the lower part of a circle.

∴

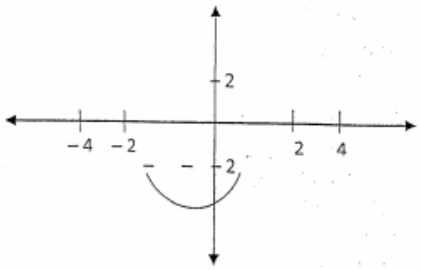
the curve is a lower circle with centre $(-1,-2)$ and radius 2.

∴ *Answer is D*

15. Given $g(x) = -2-\sqrt{3-2x-x^2}$

Let $y = -2-\sqrt{3-2x-x^2}$

Where r = 2 and c(-1, 2)



The range of g(x) are the values of y that satisfies the semi-circle above.

$\therefore -2 \le y \le -4$ *is the range*

∴ *Answer is C*

16. Given $x^2+6x+q=0$

Since roots are x and x – 1

$= x + -1 = -\frac{b}{a} = -6$

$\therefore xx-1 = -6$

$2x = -5$

$X = -\frac{5}{2}$

$x(x-1) = \frac{c}{a} = q$

$q = x^2 - x, put\ x = -\frac{5}{2}$

$= (\frac{-5}{2})^2 - (\frac{-5}{2})$

$q = \frac{35}{4}$

∴ *Answer is D*

17. $\ln(x+1)+\ln(x-1)=0$

$\ln(x+1)(x-1)=0$

$\ln(x^2-1)=0$

$= x^2-1 = e^0$

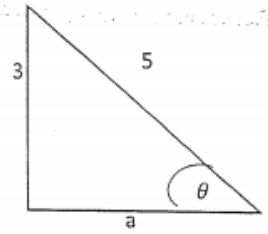
$x^2-1=1$

$x^2=2$

$x = 2^{½}$

∴ *Answer is C*

18. Given $\text{Sin}\theta = \frac{3}{5}$ representing on right-angle triangle gives.



From Pythagoras theorem,
$5^2 = 3^2 + a^2$
$a^2 = 2^5 - 9$
$a^2 = 16$
$a = 4$
$\Rightarrow Cos\theta \frac{4}{5}$
$\therefore Sec\,\theta = {}^{1}/_{Cos\theta} = {}^{5}/_{4}$
$\therefore$ *Answer is A*

19. Given Sin 35° + Sin 27°
we know from the factor formulae that,
$\text{Sin}\,x \times \text{Sin}\,y = \text{Sin}\left(\frac{x+y}{2}\right) Cos\left(\frac{x+y}{2}\right)$
$\text{Sin}\,35° + \text{Sin}\,27° = 2\text{Sin}\left(\frac{35+27}{2}\right) Cos\left(\frac{35-27}{2}\right)$
$= 2\,\text{Sin}\,31°\,\text{Cos}\,4°$
$\therefore$ *Answer is C*

20. Given $[(1-2x)^2]^{12}$
$= (1-2x)^{24}$
Since the total number of terms is the power of the binomial plus 1, the answer is 25.
$\therefore$ *Answer is C*

21. Given $Cos\theta = {}^{3}/_{5}$, *and* θ is in quadrant IV.
$\therefore Cos2\theta = Cos^2\theta - Sin^2\theta$
$= 2\text{Cos}^2\theta - 1$
$= 2({}^{3}/_{5})^2 - 1$
$= {}^{-7}/_{25}$
$\therefore$ *Answer is B*

22. Given $P = \begin{matrix} 1 & -2 & -5 \\ 2 & -1 & 0 \\ -4 & 5 & 6 \end{matrix}$
Trace is the summation of the leading diagonals
TR(B) = 1 – 1 +6 = 6
$\therefore$ *Answer is B*

23. Sum = $M_{11} + M_{22} + M_{33}$
$= \begin{vmatrix} -1 & 0 \\ 5 & 6 \end{vmatrix} + \begin{vmatrix} 1 & -5 \\ -4 & 6 \end{vmatrix} + \begin{vmatrix} 1 & -2 \\ 2 & -1 \end{vmatrix}$
$= (-6-0) + (6-20) + (-1+4)$
Sum = -6 – 14 + 3
= -17
$\therefore$ *Answer is C*

24. $|B| = \begin{vmatrix} -1 & 0 \\ 5 & 6 \end{vmatrix} + 2\begin{vmatrix} 2 & 0 \\ -4 & 6 \end{vmatrix} - 5\begin{vmatrix} 2 & -1 \\ -4 & 5 \end{vmatrix}$
$= (-6-0) + 2(12-0) - 5(10-4)$
= - 6 + 24 – 30 = -12
$\therefore$ *Answer is A*

25. Characteristics polynomial is $B(x) = x^3 - tr(B)x^2 + (M_{11} + M_{22} + M_{33})x - |B|$
$B(x) = x^3 - 6x^2 - 17x + 12$
$\therefore$ *Answer is A*

26. A non singular matrix is a matrix with a non zero determinant. Since $\begin{vmatrix} 3 & -2 \\ 9 & 6 \end{vmatrix} = 36$
$\therefore$ *Answer is A*

27. If a matrix has no inverse, its determinant is zero.
$\begin{vmatrix} x-2 & 1 \\ 2 & x-3 \end{vmatrix} = 0$
$(x-2)(x-3) - 2 = 0$
$x^2 - 3x - 2x + 6 - 2 = 0$
$x^2 - 5x + 4 = 0$
$x = 1, 4$
$\therefore$ *Answer is C*

28. Given $f(x) = \frac{2x}{1+x}$
$\lim_{x \to \infty} \frac{2x}{1+x} = \lim_{x \to \infty} \frac{{}^{2x}/_{x}}{{}^{1}/_{x} + {}^{x}/_{x}}$

$= \lim_{x \to \infty} \frac{2}{1/x+1}$

as $\to \infty$

$= 2$

$\therefore$ *Answer is C*

29. Given $\frac{2x+5}{x^2-x-6}$

This function does not exist if , $x^2 - x - 6 = 0$

x = 3, -2

$\therefore$ *Answer is C*

30. Given $2x^3 + 3x^2 - 32x + 15$

using the calculator, the zeros are $= -5, 3, \frac{1}{2}$

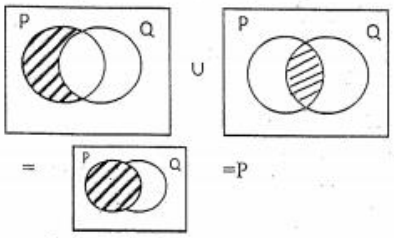
$\Rightarrow$ the factors are $(x+5)(x-3)(x-\frac{1}{2})$

$\therefore$ *Answer is D*

2013 / 14

SECTION A

1. Given $(P \cap Q^1) \cup (P \cap Q)$



∴ *Answer is C*

2. Given $Log_2\sqrt{2} = x$

$\frac{1}{2}log_2 2 = x$

x = ½

∴ *Answer is B*

3. Given $|4 - x| < -3$

-(-3) < 4 – x < - 3

3 < 4 – x < -3

∴ $3 < 4 - x$ *or* $4 - x > -3$

$x < 4 - 3$ $-x < -3 - 4$

$x < 1$ $-x < -7$

$x > 7$

∴ $x: x < 1$ *or* $x > 7$

Answer is C

4. Given $|x + 3| > 4$

x+3>4 or x +3< - 4

x > 4 – 3 x < -4 – 3

x > 1 x < - 7

∴ $x: x < -7$ *or* $x > 1$

Answer is A

5. Given $p(x) = x^3 + a_2x^2 + a_1x + a_0$

Zeros are r1, r2, r3

∴ *the* polynomial could be of the form.

$x^3 - (r_1+r_2+r_3)x^2 + (r_1r_2+r_1r_3+r_2r_3)x - r_1+r_2+r_3 = 0$

by comparison

$r_1 + r_2 + r_3 = -a_2$

$r_1r_2+r_1r_3+r_2r_3 = a_1$

$r_1r_2r_3 = -a_0$

∴ *Answer is B*

6. From the solution to Q5.

Answer is C

7. Given $p(x) = x^3 + 3x^2 - x - 3$ since -1 is a zero, let $r_1 = -1$

Comparing to $p(x) = x^3+a_2x^2+a_1x+a_0$

$a_2=3$ $a_1 = -1$, $a_0 = -3$.

From solution to Q5, $-1 + r_2 + r_3 = -3$

$r_2 + r_3 = -2$ ------- (1)

$(-1)r_2r_3 = -(-3)$

$r_2r_3 = -3$

$r_2 = -3/r_3$ ------ (2)

Put (2) into one

$-3/r_3 + r_3 = -2$

$-3 + r^2_3 = -2r_3$

$r^2_3 + 2r_3 - 3 = 0$

$r_3 = 1, -3$

If $r_3 = 1$, $r_2 = -3$

If $r_3 = -3$, $r_2 = 1$

∴ *other zeros are* 1 *and* -3

Answer is D

8. Given $-3 \leq 2x + 1 \leq 5$

$-3 - 1 \leq 2x \leq 5 - 1$

$\frac{-4}{2} \leq \frac{2x}{2} \leq \frac{4}{2}$

$-2 \leq x \leq 2$

∴ *Answer is C* ~~B~~

9. Given $f(x) = \frac{x^2-1}{x^2-4}$

Let $x^2 - 4 = 0$

$x = \sqrt{4} = \pm 2$

∴ $x: (-\infty, -2) \cup (-2,2) \cup 2, \infty)$

Answer is D

10. Let $y = \frac{x^2-1}{x^2-4}$

$yx^2 - 4y = x^2 - 1$

$yx^2 - x^2 = 4y - 1$

$x^2(y-1) = 4y - 1$

$x^2 = \frac{4y-1}{y-1}$

$x = \sqrt{\frac{4y-1}{y-1}}$

for real values, $\frac{4y-1}{y-1} \geq 0$

$(y-1)^2 \left|\frac{4y-1}{y-1}\right| \geq 0$

$(y-1)(4y-1) \geq 0$----------*

$let\ (y-1)(4y-1) = 0$

y = 1, y = ¼

0 ½ 2

½ 1

if y = o, put into eqn(*)

(0-1) (4(0) -1) = (-1)(-1) = 1 ≥ 0

if y = ½

(1/2 -1)(4(½) -1) = (½)(1) = - ½ ≱ 0

If y = 2

(2-1) (4(2) – 1) = (1)(7) = 7≥ 2

∴ *since* $y = 0$ *and* $y = 2$

Satisfies the inequality,

$= y \leq {}^1/_4 \ or\ y \geq 1$

But y = 1 makes eqn (*) undefined.

$\therefore y \leq \frac{1}{4} or > 1$

$= (-\infty, -{}^1/_4] \cup (1, \infty)$

∴ *Answer is B*

11. Since $f(x) = \frac{(-x)^2-1}{(-x)^8-4}$

$= \frac{x^2-1}{x^8-4}$

F(-x) = f(x).

f(x) is even, thus symmetrical about the y – axis.

Answer C

12. At points where curve cuts x –axis, y = or

$\therefore 0 = \frac{x^2-1}{x^2-4}$

0 = x2 – 1

(x-1)(x+1) = 0

X = 1, -1

= p(1,0) and p(-1,0)

Answer is A

13. At point where curve cuts y = axis, x = O

$\therefore y = \frac{(0)^2-1}{(0)^2-1} = {}^{-1}/_{-1} = {}^1/_4$

$\therefore p(0, \frac{1}{4})$

Answer D

14. For characters of tan x, it is not periodic with period x2.

Answer is C

15. Since the equation is linear , y will be defined for all real numbers.

∴ Range of values of y is $(-\infty, \infty)$

Answer is B~~.~~ C

16. Since the equation is linear, x will be defined for all real numbers.

∴ Range of values of x is $(-\infty, \infty)$

Answer is B

17. Given

$5log_x 2 - \frac{1}{2}log_x 8 = 2 - \frac{1}{2}log_x 2$

$5log_x 2 - \frac{1}{2}log_x 2^3 = 2 - \frac{1}{2}log_x 2$

$5log_x 2 - \frac{3}{2}log_x 2 + \frac{1}{2}log_x 2 = 2$

$4log_x 2 = 2$

$4log_x 2 = \frac{1}{2}$

$x^{\frac{1}{2}} = 2$

x = 4

∴ Answer is D

NB: Log of a negative base is undefined as x cant be negative.

18. Given $U_5 = 14$, $U_{10} = 34$
but $U_n = a+(n-1)d$
$U_5 = a+4d = 14$
$U_{10} = a + 9d = 34$
Solving simultaneously given a = -2, d =4
∴ Answer is C

19. Given 100 + 50+25…
Since this is a G.P, we sum to infinity.
$\therefore S_\infty = \frac{a}{1-r} = \frac{100}{1-\frac{1}{2}}$
= 200
∴ Answer is B

20. Given $[(1-x)^2]^{11}]$
$= (1-x)^{22}$
but total number = 22+1 = 23
∴ Answer is C

21. Given $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots a_2 x^2 + a_1 x + a_0$, if $^p/_q$ is a zero, then
p is a factor of a_o and q is a factor of a_n
∴ Answer is B.

22. Given M and N are nonsingular then the product MN is also non-singular, this its determinant is not equal to zero.
∴ Answer is A

23. Given $B = \begin{vmatrix} -1 & 2 & 1 \\ -7 & 4 & -1 \\ -4 & 2 & -1 \end{vmatrix}$
Since trace of B is the summation of the leading diagonals.
TR(B) = -1 + 4 – 1 = 2
Answer is A

24. Sum = M11 + M22 + M33
$= \begin{vmatrix} 4 & -1 \\ 2 & -1 \end{vmatrix} + \begin{vmatrix} -1 & 1 \\ -4 & -1 \end{vmatrix} + \begin{vmatrix} -1 & 2 \\ -7 & 4 \end{vmatrix}$
= (-4 + 2) + (1 + 4) +(- 4 + 14)
= - 2 +5+10
= 13
∴ Answer is B

25. The characteristic polynomial is
$B(x) = x^3 - tr(B)x^2 + (M_{11}+M_{22}+M_{33})x - |B|$
but $|B| = \begin{vmatrix} -1 & 2 & 1 \\ -7 & 4 & -1 \\ -4 & 2 & -1 \end{vmatrix} = -2$
$B(x) = x^3 - 2x^2 + 13x + 2$
∴ Answer is D

26. For an expression of B^{-1}, we have from Q25, the polynomial $B^3 - 2b^2 + 13B + 2I = 0$
$= B^3 - 2B^2 + 13B = -2I$
$- ½ (B^3 - 2B^2 + 13B) = I$
$[- ½(B^2 - 2B + 13I] = I$
Comparing to $BB^{-1} = I$
$= B - 1 = - ½ (B^2 - 2B + 13I)$
∴ Answer is A

27. Given m and n are roots of x2 + px + q = o
$\therefore m + n = -p, mn = q$
$\Rightarrow m + \frac{1}{n} + n + \frac{1}{m} = m + n + \frac{1}{n} + \frac{1}{m}$
$\Rightarrow m + n = \frac{m+n}{nm}$
$\Rightarrow - p \frac{-p}{q}$
$\Rightarrow \frac{-qp-p}{q}$
$\Rightarrow \frac{-p(q+1)}{q}$
∴ Answer is A

28. Product of root $= (m + \frac{1}{n})(n + \frac{1}{m})$
$= (\frac{mn+1}{n})(\frac{nm+1}{m})$
$= \frac{(nm+1)(nm+1)}{nm}$
$= \frac{(q+1)(q+1)}{q}$
$= \frac{(q+1)^2}{q} = \frac{q^2+2q+1}{q}$
∴ Answer is D

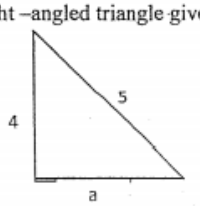
29. Equation whose roots are m+ ½ and n + $^1/_m$ is
$x^2 - (sum) x + product = 0$

$x^2 - \left(\left(\frac{-q(q+1)}{q}\right) \times \frac{q^2+2q+1}{q} = 0\right.$

$x^2 + \left(\frac{p(q+1)}{q}\right) \times + \frac{q^2+2q+1}{q} = 0$

∴ Answer is C

30. Given θ is abuse, $\sin\theta = {}^4/_5$, representing in right –angled triangle gives



Using Pythagoras theorem,

$5^2 = 4^2 + a^2$

$a^2 = 25 - 16$

$a^2 = 9$

$a = \sqrt{9} = 3$

$\text{Cos}\theta \frac{3}{5}$,

since θ is obtsuse, thus falls in the 2nd quadrant, $\text{Cos}\theta = -{}^3/_5$

∴ Answer is B

31. Given $2x^3+3x^2 - 32x + 15$

if x = 3, we have

$2(3)^3 + 3(3)^2 - 32(3) + 15$

$= 2(27) + 27 - 96 + 15$

$= 96 - 96 = 0$

∴ (x – 3) is a factor

$$\begin{array}{r} 2x^2 + 9x - 5 \\ (x-3)\overline{)2x^3 + 3x^2 - 32x + 15} \\ -(2x^3 \quad 6x^2) \\ \hline 9x^2 - 32x + 15 \\ -(9x^3 - 27x \\ \hline -5x + 15 \\ -\frac{(-5x+15)}{--} \end{array}$$

Now,

∴ $2x^3 + 3x^2 - 32x + 15 = (x - 3)(2x^2+9x-5$

but $2x^2 + 9x - 5 = 2x^2 + 10x - x - 5$

$= 2x(x+5) - 1(x + 5)$

$= (2x - 1)(x+5)$

∴ $2x^3 + 3x^2 - 32x + 15 = (x - 3)(2x + -1)x+5)$

∴ Answer is D

32. Given $\frac{2x+5}{x^2-x-6}$, for values of x for which the function does not exist, we let

$x^2 - x - 6 = 0$

$x = 3, -2$

∴ Answer is A

33. Given $f(x) = \frac{2x}{1+x}$

$\underset{x \to \infty}{Lim} f(x) = \underset{x \to \infty}{Lim} \frac{2x}{1+x}$

$= \underset{x \to \infty}{Lim} \frac{2x/x}{1/x + x/x}$

$= \underset{x \to \infty}{Lim} \frac{2}{\frac{1}{x}+1}$

$= {}^2/_1 = 2$

∴ Answer is C

34. Given α and β as roots of $3x^2 + 4x - 5 = 0$

Sum $(x + \beta) = {}^n/_b = -{}^4/_3$

Product $(x\beta) = {}^c/_a = -{}^5/_3$

$\therefore \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta+\alpha}{\alpha\beta} = \frac{-4/3}{-5/3}$

$= {}^4/_5$

∴ Answer is A

35. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (-{}^4/_3)^2 - 2(-{}^5/_3)$

$= \frac{16}{9} + \frac{10}{3}$

$= \frac{46}{9}$

∴ Answer is C

36. ${}^\alpha/_\beta + {}^\beta/_\alpha = \frac{\alpha^2+\beta^2}{\alpha\beta}$

$= \frac{46}{7} = \frac{46}{9} x \frac{-3}{5}$

$= \frac{-46}{15}$

∴ Answer is B

37. Given $p(x) = x^5 - 4x^3 + 2x + 3$
and $(x - 1)$ is divisor
then remainder = p(1)
$= (1)^5 - 4(1)^3 + 2(1) + 3$
$= 1 - 4 + 2 + 3$
$= 1 - 4 + 2 + 3$
$= 2$
∴ Answer is B

38. Given $p(x) = x^5 - 4x^3 + 2x + 3$ and divisor $= x + 2$
then remainder = p(-2)
$= (-2)^5 + 4(-2)^3 + 2(-2) + 3$
$= -32 + 32 - 4 + 3$
$= -1$
∴ Answer is B

39. Given $x^5 + 4x^2 + ax + b$
divisor $= x^2 - 1$
∴ $x^2 - 1 = 0$
$x = 1, -1$
remainder $= 2x + 3$
∴ p(1) = remainder at $x = 1$
P(-1) = remainder at $x = -1$
$= (1)^5 + 4(1)^2 + a(1) + b\ 2(1) + 3$
$1 + 4 + a + b = 2 + 3$
$a + b = 5 - 5$
$a + b = 0 \quad \ldots \quad (1)$
$(-1)^5 + 4(-1)^2 + a(4) + b = 2(-1) + 3$
$-1 + 4 - a + b = -2 + 3$
$-a + b = 1 - 3$
$-1 + b = -2 \quad \ldots \quad (2)$
from (1) and (2),
$a = 1, b = -1$
∴ Answer is C

40. From solution to Q39, answer is A

41. Given $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} B = \begin{pmatrix} 7 \\ 8 \end{pmatrix}$

$$AB = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \end{pmatrix}$$

$$= \begin{matrix} 1(7) + 2(8) \\ 3(7) + 4(8) \\ 5(7) + 6(8) \end{matrix} = \begin{pmatrix} 23 \\ 53 \\ 83 \end{pmatrix}$$

$AB = (23 \quad 53 \quad 83)^t$
∴ Answer is D

42. Given $\begin{vmatrix} x-2 & 1 \\ 2 & x-3 \end{vmatrix}$
Since matrix has no inverse, its determinant is zero.
∴ $\begin{vmatrix} x-2 & 1 \\ 2 & x-3 \end{vmatrix} = 0$
$(x - 2)(x - 3) - 2 = 0$
$x^2 - 3x - 2x + 6 - 2 = 0$
$x^2 - 5x + 4 = 0$
∴ $x = 4, \quad x = 1$
Answer is C

43. Non-singular matrix is a matrix whose determinant is not equal to zero. Since
$\begin{vmatrix} 3 & -2 \\ 9 & 6 \end{vmatrix} \neq 0$
Answer is A

44. Let $M = \begin{pmatrix} x & -1 \\ 1 & x \end{pmatrix}$
$M^{-1} = \frac{1}{|M|} \begin{pmatrix} x & 1 \\ -1 & x \end{pmatrix}$
$|M| = x^2 - (-1) = x^2 + 1$
∴ $M^{-1} \frac{1}{x^2+1} \begin{pmatrix} x & 1 \\ -1 & x \end{pmatrix}$
∴ Answer is B

45. Given ${}^nC_{n-1} = \frac{n!}{[n-(n-1)]!(n-1)!}$
${}^nC_{n-1} = \frac{n!}{1!(n-1)!} = \frac{n(n-1)!}{(n-1)!} = n$
∴ Answer is A

46. Given the A.P $-4 \ -1 + 2 \ldots$
$S_n = \frac{1}{2}n[(2a + (n - 1)d]$
but $d = -1 - (-4) = -1 + 4 = 3$

$\therefore S_{20} = ½ (20)[2(-4)+(20-1)3]$
$= 10 (-8 +57)$
$=10(49) = 490$
$\therefore$ Answer is C

47. Given $U_1 = 6$, $U_7 = -18$
$U_n = a + (n-1)d$
$U_1 = a = 6$
$U_7 = a + (7-1)d = -18$
$a + 6d = -18$
$6 + 6d = -18$
$\frac{6d}{6} = \frac{-24}{6}$
$D = -4$
$\therefore$ Answer is B

48. $S_n = ½n (2a + (n-1)d)$
$S_{10} = ½ (10)[2(6) + (10-1)(-4)]$
$= 5 [12 - 36]$
$= 5 (-24)$
$S_{10} = -120$
Answer is B

49. Given $\text{Sin } ½\theta = ½$
$½\theta = \text{Sin}^{-1}(½)$
$½\theta = 30$
$\theta = 30(2) = 60^0$

but $\text{Sin}½\theta$ is positive, thus in the 2nd quadrant,
$\therefore ½\theta_2 = 180 - ½\theta$
$= 180 - ½(60)$
$= 180 - 30 = 150$
$\theta_2 = 2(150) = 300^0$
$\therefore$ Answer is A

50. $\frac{Sin(x+45^0)}{Cos(x+45)} = \frac{2(Cos(x+45)}{Cos(x+45)}$

$\tan(x + 45) = 2$

$\frac{tanx+\tan 45}{1-tanxtan45} = 2$

$\frac{tanx+1}{1-tanxtan45} = 2$

$\frac{tanx+1}{1-tan} = 2$

$\tan x + 1 = 2(1-\tan x)$

$\tan x + 1 = 2 - 2\tan x$

$\tan x + 2\tan x = 2-1$

$3\tan x = 1$

$\tan x = {}^1/_3$

$\therefore$ Answer is D

2012 / 13
SECTIONA

1. Refer to 2013/14 Question 1
2. Refer to 2013/14 Question 2
3. Refer to 2013/14 Question 3
4. Refer to 2013/14 Question 4
5. Refer to 2013/14 Question 5
6. Refer to 2013/14 Question 6
7. Refer to 2013/14 Question 7
8. Refer to 2013/14 Question 8
9. Given $f(x) = \frac{x}{x^2-1}$
   for vertical asymptote $x^2 - 1 = 0$
   $(x-1)(x+1) = 0$
   $\therefore x = 1, \ x = -1$
   Answer is A

10. For horizontal asymptote since degree of denominator is greater than that of numerator, the horizontal asymptote is y = 0. That is, x – axes.
    ∴ Answer is A

11. Given $p(x) = (x-a)^3$, $x = a$
    does not have a multiplicity of 1
    Answer is A

12. A function is odd if $f(-x) = -f(x)$
    Answer is C

13. No term in x^3 for $(x - {}^{2}/_{x})^4$
    Bonus question

14. Refer to 2013/14 Question 14.
15. Since sec x is an even function it is symmetric about the y-axis.
    Answer is A.

16. Given $[(a+b)^3]^{17} = (a+b)^{3\times17}$
    $=(a+b)^{51}$
    = total number of terms is
    $51 + 1 = 52$
    Answer is A

17. Given A is n ×m
    B is n × k
    $= A^T = m \times n$
    $\therefore ATB = (n\times m)\times(m\times K)$
    $=n\times k$
    Answer is C

NB: for multiplication of two matrix , the dimension of the resulting matrix is the number of rows of first matrix by number of columns of second matrix.

18. Given Cos (x -30) = 0.5
    Let y = x – 30
    Cos y = 0.5
    Y, - Cos^{-1} (0.5) = 60
    But y_1 = x – 30
    60 = x – 30
    x = 60+30 = 90
    since cosine is positive in quadrant iv,
    $y_2 = 360 - \cos^{-1}(0.5)$
    = 360 – 60
    $y^2 = 300$
    but y_2 = x -300
    x = 30 = 300
    x = 300 +30
    $x = 330^0$
    ∴ $x = 90^0$, $x = 330^0$
    Answer is D

19. Given $\begin{vmatrix} 8 & 4 & -4 \\ 5 & -5 & -5 \\ 7 & 1 & -11 \end{vmatrix} = A$
    $\therefore |A| = 8\begin{vmatrix} -5 & -5 \\ 1 & -11 \end{vmatrix} - 4\begin{vmatrix} 5 & -5 \\ 7 & -11 \end{vmatrix} - 4\begin{vmatrix} 5 & -5 \\ 7 & 1 \end{vmatrix}$

$= 8(60) - 4(-20) - 4(40)$
$= 480 + 80 - 160$
$= 480 + 80 - 160$
$|A| = 400$

Answer is B

20. Refer to 2013/14 Question 21.

21. Given $(1.02)^5$, using calculator
$(1.02)^5 = 1.1040808$
$= 1.1041$
Answer is C

22. Refer to 2013/14 Question 22

23. Given $(x + 2)^{10}$
$U_{r+1} = {}^nC_r a^{n-r} b^r$
For 5th term, $r = 4$
$\therefore U_{4+1} = U_5 = {}^{10}C_4 (x)^{10-4} (2)^4$
$U5 = 210(16)\, x^6$
$U_5 = 3360x^6$
Answer is C

24. For ascending powers of x , we use $(2+x)^{10}$
$\therefore U_{r+1} = {}^nC_r a^{n-r} b^r$
$U_{4+1} = {}^{10}C_4 (2)^{10-4} (x)^4$
$= 210\,(2)x^4$
$U_5 = 13440\, x^4$
Answer is B

25. Given $\text{Log}_{10}(2x^2+5x+97)=2$
$= 2x^2+5x+97=10^2$
$2x^2 + 5x + 97 - 100 = 0$
$2x^2 + 5x - 3 = 0$
$x = ½, -3$
Answer is D

26. Given $4x^2 - 8x + k=0$
for equal roots, $b^2 - 4ac = 0$
$= (-8)^2 - 4(4)k(0)$
$64 - 16k = 0$
$\frac{16k}{16} = \frac{64}{16}$
Answer is C

27. Refer to 2013/14 Question 27
28. Refer to 2013/14 Question 28
29. Refer to 2013/14 Question 29
30. Refer to 2013/14 Question 30

2011 / 12
SECTION A

1. Given $m^r = 2$

   $= \log m^r = \log n$

   $R\log m = \log n$

   $r = \frac{\log n}{\log m}$

   $r = \log_m n$

   answer is D

2. Given $\frac{4}{9}+\frac{2}{9}+\frac{1}{9}+\frac{1}{18}+\cdots\frac{1}{144}$

   since this a a G.P with $a = \frac{4}{9}, r = \frac{1}{2}$

   $Sn = a\left(\frac{1-r^n}{1-r}\right)$

   We find n from last term

   $Un = ar^{n-1} = \frac{1}{144}$

   $\frac{4}{9}\left(\frac{1}{2}\right)^{n-1} = \frac{1}{144}$

   $\frac{\left(\frac{1}{2}\right)^n}{\frac{1}{2}} \quad \frac{1}{64}$

   $(½)n = \frac{1}{128}$

   $(½)^n = (½)^7$

   $\therefore n = 7$

   $= Sn = \frac{4}{9}\left[\frac{1-\frac{1}{2}^7}{1-\frac{1}{2}}\right]$

   $= \frac{4}{9}\left[\frac{1-\frac{1}{218}}{1-\frac{1}{2}}\right]$

   $Sn = 8/9(1-{}^1/_{128})$

   $= {}^8/_9\,({}^{127}/_{128})$

   $= \frac{127}{144}$

   Answer is C

3. Given $U_3 = {}^5/_4$, U6 =10

   $= U_3 = ar^2 = {}^5/_4$ (1)

   U6 = ar5 = 10 - (2)

   (2) – (1)

   $= \frac{ar^5}{ar^2} = \frac{10}{\frac{5}{4}}$

   $r^3 = 8$

   $r3 = 2^3$

   r = 2

   put r = 2 into (1)

   a (2)2 = ${}^5/_4$

   $a = \frac{{}^5/_4}{4}$

   $a = {}^5/_{16}$

   $\therefore U_8 = ar7 = = \frac{5}{16}(2)^7 = 40$

   Answer is B

4. Given f(X) = |x|-4

   Let y = |x| – 4

   |x|=y+4

   = y+4 ≥ 0

   $y \geq -4$

   answer is D

5. For intercept of f(x) = |x|-4

   Let x = 0 (y – intercept)

   y = f(x) = |0|-4 = -4

   ∴ p(0 , -4)

   Let y = 0 (x – intercept)

   y = f(x) = |x|- 4 = 0

   |x| = 4

   $x = 4, x = -4$

   ∴ (4, 0), (-4, 0)

   ∴ intercepts are (0, -4), (4, 0), (-4, 0)

   Answer is B

6. Given $f(x) = \frac{x^3-4x^2+7x-4}{x-1}$

$f(x)\frac{(x^3-4x^2+7x-4}{x-1}$

$f(x)\frac{(x-4)(x^2+3x+4)}{x-1}$

$f(x) = x^2 - 3x + 4$

f(x) has no vertical asymptote
Answer is **D**

7. Natural logarithms are to the base e-
Answer is C

8. Given $\frac{1}{x^2-5x+6} = \frac{1}{(x-2)(x-3)}$

$\frac{1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$

$1 = A(x-3) + B(x-2)$

When $x = 3$

$1 = A(2,-3) + B(2-2)$

$1 = -A$

$A = -1$

$\frac{1}{x^2-5x+6} = \frac{-1}{x-2} + \frac{1}{x-3}$

Answer is B

9. $\left|\frac{3-x}{2}\right| \leq -3$

$-(-3) \leq \frac{3-x}{2} \leq -3$

$= 2(3) \leq 3 - x \leq 2(-3)$

$6 - 3 \leq -x \leq -6 - 3$

$\frac{3}{-1} \leq \frac{-x}{-1} \leq \frac{-9}{-1}$

$-3 \geq x \geq 9$

$x \leq -3, x \geq 9$

$= (-\infty, -3] \cup [9, \infty)$

Answer is A

10. Given $5x - 8y = 29$

$2x + 2y = -4$

Solving simultaneously gives

$x = 1,\ y = -3$

$\therefore x + y = 1 - 3 = -2$

Answer is A

11. $f(x) = \log(x+4)$

for domain, $x + 4 > 0$

$= (-4, \infty)$

Answer is C

12. Given $\ln x - 2\ln y + \ln z$

$\ln x - \ln y^2 + \ln z$

$\ln\left(\frac{x}{y^2}\right) + \ln z$

$= \ln\left(\frac{xz}{y^2}\right)$

Answer is A

13. Given $\begin{vmatrix} 3-x & x+1 \\ 5-x & x+2 \end{vmatrix} = -6$

$(3-x)(x+2) - (5-x)(x+1) = -6$

$3x + 6 - x^2 - 2x - (5x + 5 - x^2) = -6$

$6 + x - x^2 - 5 - 4x + x^2 = -6$

$1 - 3x = -6$

$3x = 1 + 6$

$\frac{3x}{3} = \frac{7}{3}$

$x = \frac{7}{3}$

Answer is D

14. Given $\frac{2x^3-x^2+2}{x+3}$

$x+3\overline{)\ 2x^3-x^2+2}\ \big)\ 2x^2-7x+21$

$\frac{(2x^3+6x^2}{-7x^2+2}$

$\frac{-(7x^2-2/x}{\frac{2}{x}+2}$

$\frac{-(21x+63}{-61}$

∴ the quotient is $2x^2 - 7x + 21$

Answer is D

15. Given $\log_5(2x-7) - 1$

$= 2x - 7 = 5$

$2x = 5 + 7$

$\frac{2x}{2} = \frac{12}{2}$

$x = 6$

answer is C

16. Given $x, x+1, 2x+1, \ldots$

for geometric sequence

$\frac{x+1}{x} = \frac{2x+1}{x+1}$

$(x+1)^2 = x(2x+1)$

$x^2 + 2x + 1 = 2x^2 + x$

$x^2 - x - 1 = 0$

$x = 1 - 618, -0.618$

No answer is options.

NB: answer becomes B, if question is $x, x+2, 2x+1, \ldots$

17. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & -2 & 1 \end{bmatrix}$

$\therefore |A| = 1\begin{vmatrix} 0 & 1 \\ -2 & 1 \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} + 0\begin{vmatrix} 1 & 0 \\ 3 & -2 \end{vmatrix}$

$= (0+2) - (1-3) + 0$

$|A| = 2 - (-2)$

$= 4$

Answer is B

18. If $\log x + \log x = 36$

$= \log x^2 = \log 36$

$\sqrt{x^2} = \sqrt{36}$

$x = 6$

Answer is A

19. Given $\log_5 5^{2t} = 2t\log_5 5$

$= 2t$

Answer is C

20. Given $a^n = n^2$

For 5th term, $a_5 = 5^2 = 25$

Answer is D

21. Given $h(x) - \frac{x^3+1}{x^2+1}$

For vertical asymptote $x^2 + 4 = 0$

Since $x \in R$, there is no vertical asymptote.

Answer is D

22. Given $y = 4x2 - 6x + 5$ by completing of square

$= y = 4\,4\left(x^2 - \frac{3x}{2} + \frac{5}{4}\right)$

$= 4\left|x^2 - \frac{3}{2}x + \left(\frac{-3}{4}\right)^2 + \frac{5}{4} - \left(\frac{-3}{4}\right)^2\right|$

$= 4\left|(x - \frac{3}{4})^2 + \frac{11}{16}\right|$

$= 4\left|(x - \frac{3}{4})^2 + \frac{11}{16}\right|$

$= y = 4(x - \frac{3}{4})^2 + 11/4$

$= (p, q) = (3/4, 11/4)$

Answer is B

23. $R(x) = \frac{2x^2+x-1}{x+2}$

For slant asymptote,

$(x+2)\frac{2x-3}{2x^2+x-1}$

$\frac{-(2x^2+4)}{-3x-1}$

$\frac{-(-3x-6)}{5}$

$= y = 2x - 3$ is the slant asymptote

Answer is B

24. Given $(2x - 3y^2)\,5$

$U_{r+1} = {}^nC_r a^{n-r} b^r$

For 3rd term, $r = 2$

$= U_{2+1} = U_3 = {}^5C_2(2x)^{5-2}(-3y^2)^2$

$U_3 = 10(2x)^3(9y^4)$

$= 10(8x^3)(9y^4)$

$U_3 = 720x^3y^4$

Answer is D

25. π is not a rational number.
Answer is B

26. Given A{1,2,3,4,5,6,7,8,9}
B = {4,5}
C: $x^2 - 9x + 18 = 0$
x = 6,3
∴ C{3, 6}
A x B = {(a,4), (1,6),(2,4), (2,6),
(3,4),(3,6),(4,4),(4,6),(5,4),
(5,6), (6,4),(6,6), (7,4),(7,6)
(8,4), (8,6),(9,4),(9,6)}

B×C = {(4,3), (4,6), (6,3) (6,6)}

$\therefore A \times B \cap B \times C = \{(4,6), (6,6)$

Answer is C

27. B x a = {(4,1), (4,2), (4,3),...1
(4,9, (6,1), (6,2), (6,3), ...(6,9)}

B x c = {4,3,), (4,6),(6,3),(6,6)}
∴ b x a ∩ b x c = (4, 6), (6,3), (6,6), (6,6)

No answer in options.

28. Given $25\sin^2\theta = 1 + 2\sin$
Let y = Sin θ
$2y^2 = 1 + 2y$
$2y^2 - 2y - 1 = 0$
Y = 1.366, -0.366
Since θ lies between -1 and 1
= Sinθ = - 0.366
Answer is A

29. Given $0.1^{-(t^2+t)} = 0.01$
= Log $0.1^{-(t^2+t)}$ = Log 0.01
$-(t^2+t)\log 0.1 = \log 0.01$
$-(t^2 + t(-1) = -2$
$T^2+t = -2$
$T^2+t+2=0$
No solution
NB: answer becomes B, if question is
$0.1^{-(t^2+t)} = 0$

30. Given $[(AB)C]^T$ = ct (AB)
= c'b'a'

Answer is C

2010 / 11
SECTION A

1. If logarithm is to the base e, it is called Natural logarithm.
   Answer is C

2. If a function f(x) is an odd function, then f(x) = -f(- x).
   This means $4x^3 - 5x$ is an odd function.
   Answer is C

3. Given
   $\frac{x^3+2x-1}{(x^2-4)^2(x^2+1)} = \frac{x^3+2x-1}{(x-2)^2(x+2)^2(x^2+1)}$

   $\frac{A}{(x-2)} + \frac{B}{(x-2)^2} + \frac{C}{(x+2)} + \frac{D}{(x+2)^2} + \frac{Ex+F}{x^2+1}$
   Answer is B

4. Given $f(x) = \frac{x^3}{x-4}$
   For vertical asymptote, x -4 = 0
   ∴ x = 3
   Answer is B

5. Given $\begin{vmatrix} 3-x & x+1 \\ 5-x & x+2 \end{vmatrix} = -6$
   = (x - x)(x +2) – (x + 1) (5- x) = -6
   $3x + 6 - 2^x - 2x - (5x - x^2 + 5 - x) = -6$
   $x + 6 - x^2 - 4x + x^2 - 5 + 6 = 0$
   $-3x + 7 = 0$
   $\frac{-3x}{-3} = \frac{-7}{-3}$
   $x = {}^7/_3$
   Answer is D

6. Refer to 2011 / 12 question 4.

7. $\frac{x+6}{4} - \frac{x-3}{7} < 9$
   $7(x+6) - 4(x-3) < 28(9)$
   $7x + 42 - 4x - 4x + 12 < 252$
   $3x < 252 - 42 - 12$
   $\frac{3x}{3} < \frac{198}{3}$
   $x = \frac{198}{3}$
   No answer in option

8. $A = \begin{vmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{vmatrix}$

   $|A| = \begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} - 3\begin{vmatrix} 4 & -1 \\ 5 & 2 \end{vmatrix} - 2\begin{vmatrix} 4 & 1 \\ 5 & -3 \end{vmatrix}$

   = 2 – 3 – 3 (8+5) – 2 (-12 – 5)

   = -1 -3 (13) – 2 (-17)

   $|A| = -6$

   Answer is B

9. For (3,2) – co factor of A, we find the mirror at (3,2) and attach a sign.
   From
   $\begin{vmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{vmatrix}$
   ∴ Minor at (3,2) = $M32 = \begin{vmatrix} 1 & -2 \\ 4 & -1 \end{vmatrix}$
   = - 1 +8
   =7
   ∴ co factor at (3,2) = $(-1)^{3+2}(7)$
   = - 7
   No answer in option

10. Rational numbers are denoted by Q, Integers by Z, whole number by W,

Real numbers by R and Natural number by N.
Hence, R is not a subset of Q.
Answer is C

11. Given $\log\frac{1}{75} = log1 - log75$
$= -\log(3\times25)$
$= -[\log3 + \log25]$
$= -[\log3 + \log(\frac{100}{4})]$

$= -[\log3 + \log100 - \log4]$
$= -[\log3 + 2 - 2\log2]$
$= -(0.4770 + 2 - 2(0.3010)]$
$\text{Log}\frac{1}{75} = -1.875$
Answer is A

12. iven $g(x) = \frac{x^2+5}{x+1}$
$\therefore x+1 \frac{x-1}{\sqrt{x^2+5}}$
$\frac{-(x^2+x)}{-x+5}$
$\frac{-(-x-1)}{6}$

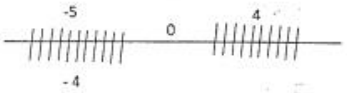
$\therefore y = x - 1$

Answer is B

13. Given A $A = \begin{vmatrix}1 & 2\\ 3 & 6\end{vmatrix}, B \begin{vmatrix}-5 & 3\\ 0 & 2\end{vmatrix}$
$A+B = \begin{vmatrix}1 & 2\\ 3 & 6\end{vmatrix} + \begin{vmatrix}-5 & 3\\ 0 & 2\end{vmatrix}$
Adj. $(A+B) = \begin{vmatrix}8 & -5\\ -3 & -4\end{vmatrix}$
Answer is A

14. $A-1 = \frac{1}{|A|}\begin{pmatrix}6 & -2\\ -3 & 1\end{pmatrix}$
$|A| = 6-6 = 0$
Since
$|A| = 0$, inverse of A do not e x it.
Answer is D.

15. Given $|2x+1| > 7$
$(2x+1)^2 > 7^2$
$4x^2 + 4x + 1 > 49$
$4x^2 + 4x + 1 - 49 > 0$
$4x^2 + 4x - 48 > 0$
Let $4x^2 + 4x - 48 > 0$
$x = 3, -4$



if $(0)2 + 4(0) - 48 = -48 > 0$

$\therefore x < -4$ or $x > 3$

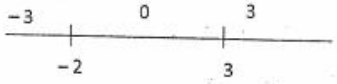
$= x < -4 \cup x > 3$

Answer is D

16. $\frac{1}{x^2+5x+6} = \frac{1}{(x+2)(x+3)} = \frac{A}{x+2} + \frac{B}{x+3}$
$\therefore 1 = A(x+3) + B(x+2)$
If $x = -2$
$A = 1$
If $x = -3$
$B = -1$

$\therefore \frac{1}{x^2+5x+6} = \frac{1}{x+2} - \frac{1}{x+3}$
Answer is C

17. Given $g(x) = \sqrt{4-x^2}$ for domain,
$4 - x^2 \geq 0$
$= (2-x)(2+x) \geq 0$
Let $(2-x)(2+x) = 0$
$X = 2, x = -2$



If $x = 0$
$\therefore -2 \leq x \leq 2$

Answer is D

18. Refer to 2012/13 question 17.

19. Given U1 = 3, u20 = 117

Un = a+(n – 1) d
= U1 = 3 = a
U20 = 117 = 3 + 19d
$= \frac{19d}{19} = \frac{14}{19}$
Answer is D

20. Sn= ½ n [2a + (n – 1)d]
S19 = ½ (19) [2(3)+(19 – 1)6]
= 9.5 (6+108)
S19 = 1083
No. 1083

No answer in option
No: Answer becomes A if question is find the 19th term.

21. Rational functions have definite decimals. Hence, $\sqrt{0.16}$ is a rational number.
Answer is A

22. Given (a – 3b) 10
$Ur + 1 = {}^{n}{}_{r}ca^{n-r}b^{r}$
Comparing to $a^7 b^3$, $r=^3$
$\therefore U_{3+1} = U_4 = {}^{10}C_3(a)^{10-3}(-3b)^3$
$= 120a^7 (-27b^3)$
$= -\ 3240\ a^7b^3$
Answer is A

23. Given A = {a, b}, B = {b, c}
C= {d,e,f}
For element in A × B× C = 2×2×3
Answer is C

24. Given $4(x + ½) + 4(^3/_2 - x) = 17$
$= 4\ ½\ (4^x) + 4\ ^3/_2\ (4 - x) = 17$
Let $y = 4^x$
$2y + \frac{8}{y} = 17$
$2y^2 + 8 = 17y$
$2y^2 - 17y + 8 = 0$
$\therefore y = 8, ½$

But $y = 4^x$

$\therefore 4^x = 8$ $\quad$ $4^x = ½$

$2^{2x} = 2^3$ $\quad$ $2^{2x} = 2^{-1}$

$x = {}^3/_2$ $\quad$ $x = -½$

Answer is B

25. Given Cos θ = on o≤ θ
$\theta = Cos^{1-}(0)$
$\therefore O_1 = 9_O = \frac{\pi}{2}$
but cosine is also positive in 4 quadrant,
$\theta = 360 - O1$
$= 360 - 90$
$= 270$
$\frac{3\pi}{2}$
Answer is C

26. Given $\log(\log 1000^{1000})$
= log (1000 log1000)
Log (1000(3)]
Log (3000) = 3.4771
Answer is B

27. Refer to 2017 / 12 Question

28. Given $\left|\frac{3-x}{2}\right| \geq 3$

$= \frac{3-x}{2} \geq 3$ *or* $\frac{3-x}{2} \leq -3$

$3 - x \geq 6$ 3- x ≤ -6

$-x \geq 3$ $-x \geq -9$

$x \leq -3$ $x \leq 9$

$= (-\infty, -3]$ or $(9, \infty]$

Answer is A

29. Given A: $x^2 = 11x + 28 = 0$

x = 7, 4

∴ A= {4, 7}

B = {1 ,2,3,4,6,12}

C = {3, 6}

∴ A x B = {4,1),
(4,2),(4,3),...,(4,12),
(7,1),(7,2),(7,3),...,(7,12)}

C ×B = {(3,1) (3,2),(3,3,),...,
(3,12)
(6,1), (6,2), (6,3), ...(6,12)}

∴ (A×B) ∩ (C×B) = ∅

30. A × C = {(4,3), (4,6), (7,3), (7,6)}

∴ (A×B) ∩ (A×)={4,3), (4,6)}

Hence, has 2 elements:

Answer is a

2009/10

SECTION A

1. Matric multiplication is not commutative.
   Answer is **B**

2. The term with the highest sum of e x ponents depicts the degree of the polynomial. The second term has 9+7 = 16 as the highest sum of e x ponents.
   Answer is **D**

3. Given $a_n = (-1)^n n^2$
   $\therefore S_4 = a_1 + a_2 + a_3 + a_4$
   $S4 = (-1)(1^2) + (-1)^2(2)^2 + (-1)^3(3)^2 + (-1)^4(4)^2$
   =-1 + 4-9+16
   S4 = 10
   ∴ Answer = B

4. Given $Log_2(3x^2 + 8x - 1) = 1$
   $\Rightarrow 3x^2 + 8^2 x - 1 = 2^1$
   $3x^2 + 8x - 1 - 2 = 0$
   $3x^2 + 8x - 3 = 0$
   $x = \frac{1}{3}, -3$
   $\{x : x = -3, {}^1/_3\}$
   Answer is C

5. Given $A = \begin{matrix} 3 & -2 & 0 \\ 2 & 5 & 8 \\ 5 & 3 & 5 \end{matrix}$
   $\therefore |A| = 3\begin{vmatrix} 5 & 8 \\ 3 & 5 \end{vmatrix} + 2\begin{vmatrix} 2 & 8 \\ 5 & 5 \end{vmatrix} + 0\begin{vmatrix} 2 & 5 \\ 5 & 3 \end{vmatrix}$
   = 3(25-24) +2(10-40)+0
   =3 –60+0= -57
   Answer is D

6. Given $|4 - x| < -3$
   $-(-3) < 4 - x < -3$
   $3 < 4 - x < -3$
   First, 3< 4 - x
   X < 1

   Also, 4- x <3
   - x <-3 – 4
   $\frac{-x}{-1} < \frac{-7}{-1}$
   $\Rightarrow$ x >7
   Now possible answers are x <1 or x >7
   ∴ *Answer is B*

7. Given $\tan(\frac{\pi}{2} - \theta)$,
   NB: from transformation about y-a x is.
   Tan $(90 - \theta) = \cot\theta$
   Tan $(90 + \theta) = -\cot\theta$
   ∴ *Answer is A*

8. Given $y = \frac{x^2+x-6}{x+3}$
   $\Rightarrow y = \frac{(x+3)(x-2)}{x+3}$
   $y = x - 2$
   This shows that curve has no vertical asymptote.
   ∴ *Answer is D*

9. Given $x^2 = x - 5$
   $\Rightarrow x^2 - x + 5 = 0$
   For turning points, we use completing squares.
   $\therefore x^2 - x + ({}^{-1}/_2)^2 - ({}^{-1}/_2)^2 + 5 = 0$
   $(x - {}^1/_2)^2 - {}^1/_4 + 5 = 0$
   $(x - 1/2)^2 + {}^{19}/_4 = 0$
   comparing to a $(a-p)^2+q=0$
   $\Rightarrow (p, q) = ({}^1/_2, {}^{19}/_4)$
   ∴ *Answer is* **A**

10. $\frac{1}{(x-3x-2)} = \frac{A}{(x+3)} + \frac{B}{(x+2)}$
    $1 = A(x+2) + B(x+3)$
    If $x = -2$
    $B = 1$
    If $x = -3$
    $A = -1$

$\frac{1}{(x-3(x+2)} = \frac{-1}{(x+3)} + \frac{1}{(x+2)}$

∴ *Answer is C*

11. Given x -5b, x -3b, x -b,...
For A.P, common difference, $d=a_2-a_1$

$\therefore d = x - 3b - (x - 5b)$

$= x - 3b - x + 5b$

$d = 2b$

∴ *Answer is C*

12. Given $|x + 5| = |3x - 4|$

$= (x+5)^2 = (3x-4)^2$

$x^2 + 10x + 25 = 9x^2 - 24x - 16$

$8x^2 - 34x - 9 = 0$

$X = {}^9/_2, -1/4$

∴ *Answer is B*

13. Given $B = \begin{matrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{8} & -\frac{3}{4} \end{matrix}$

$|B| = \frac{1}{2}\left(-\frac{3}{4}\right) - \frac{1}{2}\left(\frac{1}{8}\right)$

$= \frac{-3}{8} - \frac{1}{16}$

$= \frac{-6-1}{16}$

$= \frac{-7}{16}$

∴ *Answer is B*

14. Irrational numbers are numbers whose decimal equivalent don't terminate or recur. Thus, 2/52 is an irrational number
∴ *Answer is C*

15. $2Sin^2\theta = 1 + 2Sin\theta$

Let $x = Sin\theta$

$\Rightarrow 2x^2 = 1 + 2x$

$2x^2 - 2x - 1 = 0$

$X = 1.3660, -0.3660$

Since Sin θ lies between -1 and 1

$\therefore \Rightarrow Sin\theta = -0.3660$

∴ *Answer is A*

16. Given $(1+3x)^{-2}$
for coefficient of x^3, since the e x ponent is negative, we use the normal e x pansion instead of the general terms formula.

$\therefore (1 + x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \frac{n(n-1)(n-3)x^3}{31} + \ldots$

$= (1+3x)^{-2} = 1 - 2(3x) - \frac{2(-2-1)(3x)^2}{2!} + \frac{-2(-3)(-4)(3x)^3}{3!} + \ldots$

$(1+3x)^{-2} = 1 - 6x + 27x^2 - 108x^3 + \ldots$

∴ *Answer is D*

17. $y = \frac{3x^3+5x^2-26+8}{x^2+2x-8}$

$y = \frac{(x+4)(x-2)(3x-1)}{(x+4)(x-2)}$

$Y = 3x - 1$

This shows that he curve has no vertical asymptote

∴ *Answer is D*

18. $X = \log_{10} 0.001$

$10^x = 0.001$

$10^x = 10^{-3}, \Rightarrow X = 3$

∴ *Answer is B*

19. *Given* $\left|\frac{3-x}{2}\right| \geq 3$

$\Rightarrow \frac{3-x}{2} \geq 3$ *or* $\frac{3-x}{2} \leq -3$

$\therefore 3 - x \geq 6$ $\quad$ $3 - X \leq -6$

$-X \geq 3$ $\quad$ $-X \leq -9$

$, \Rightarrow X \leq -3$ $\quad$ $X \geq 9$

$\therefore (-\infty, -3] or [9, \infty)$

∴ *Answer is A*

20. $A = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$ $\quad B = \begin{pmatrix} -3 & 4 \\ 2 & 1 \end{pmatrix}$

Given $2X + A = B$

$= 2X = B - A$

Let $X = \begin{pmatrix} w & x \\ y & z \end{pmatrix}$

$\therefore \begin{pmatrix} w & x \\ y & z \end{pmatrix} = \begin{pmatrix} -3 & 4 \\ 2 & 1 \end{pmatrix} - \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$

$\therefore \begin{pmatrix} 2w & 2x \\ 2y & 2z \end{pmatrix} = \begin{pmatrix} -4 & 6 \\ 2 & -2 \end{pmatrix}$

By matriXcomparism

$2w = -4$ $\quad$ $2X = 6$

$\therefore w = -2$ $\quad X = 3$

$2y = 2$ $\quad 2z = -2$

$y - 1$ $\quad z = -1$

$\therefore X = \begin{pmatrix} w & x \\ y & z \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}$

$\therefore$ *Answer is A*

21. Given that $x + \sqrt{3}, x + \sqrt{12}, x + \sqrt{27}$

For A.P, common difference

$d = a_2 - a_1$,

$= x + \sqrt{12} - (x\sqrt{3})$

$= x + \sqrt{12} - x - \sqrt{3}$

$= \sqrt{12} - \sqrt{3}$

$d = \sqrt{3}$

$\therefore$ *Answer is C*

22. Given $2^x + 2^{-x} = 2$

let $u = 2^x$

$\Rightarrow u + u^{-1} = 2$

$u + \frac{1}{u} = 2$

$u^2 + 1 - 2u$

$u^2 - 2u + 1 = 0$

$\therefore u = 1$

But $u = 2^x$

$\Rightarrow 1 = 2^x$

$= 2O = 2^X$(by comparison)

$\therefore x = 0$

$\therefore$ *Answer is A*

23. Given $(30+206)^{20}$

The total number of term is the power of the binomial plus 1.

$\Rightarrow$ no. of terms is 21

$\therefore$ *Answer is* **21**

24. Given $\begin{vmatrix} 2-y & y-4 \\ 5-y & y-1 \end{vmatrix} = 15$

$=(2-y)(y-1) - (5-y)(y-4) = 15$

$2y - 2 - y^2 + y - (5y - 20 - y2 + 4y) = 15$

$3y - 2 - y^2 - 9y + 20 + y^2 = 15$

$-6y + 18 - 15 = 0$

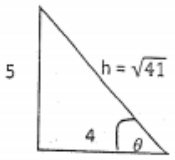
$6y = 3$

$Y = ½$

$\therefore$ *Answer is A*

25. Given $\tan\theta = -5/4$, $\cos\theta > 0$

Representing $\tan\theta = 5/4$ on a right-angled triangle gives.



$h^2 = 5^2 + 4^2 = 25 + 16$

$h^2 = 41$

$h = \sqrt{41}$

from the figure above,

$\cos\theta = \frac{4}{\sqrt{41}}$

but $\sec\theta = \frac{1}{\cos\theta} = \frac{\sqrt{41}}{4}$

$\therefore$ *Answer is C*

26. Given $\ln z = 24$

NB: $\ln z = \log_e z = 24$

$\Rightarrow e^{24} = z$

$\therefore Z = 2.6489 \times 10^{10}$

$=Z = 2.65 \times 10$

$\therefore$ *Answer is A*

27. Given $|2x - 3| \leq 4$

$\Rightarrow -4 \leq 2X - 3 \leq 4$

$-4 + 3 \leq 2x \leq 4 + 3$

$\frac{-1}{2} \leq \frac{2x}{2} \leq \frac{7}{2}$

$\frac{-1}{2} \leq X \leq \frac{7}{2}$

$\therefore x: [-\frac{1}{2}, \frac{7}{2}]$

$\therefore$ *Answer is D*

28. Given $y = \frac{2x^2 - 5x - 3}{3x^2 - 7x - 6}$

$\therefore y = \frac{2}{3}$

(since the degrees are the same)

$\therefore$ *Answer is C*

29. Given n^{th} term = 2

Since the nth term is a constant 2, it implies all terms are the same.

$\therefore S_3 = u_1 + u_2 + u_3 = 2 + 2 + 2 = 6$

$S_3 = 6$

$\therefore$ *Answer is C*

30. Given $\text{Cos}\theta = \sqrt{1 - sin^2\theta}$
Since $\sin\theta > 0$
$\Rightarrow \text{Cos}\theta = -\sqrt{+ve}$
$\text{Cos}\theta < 0$(-ve)
Using the quadrant signs,
$\text{Cos}\theta$ is negative in both 2nd and 3rd quadrant. but $\sin\theta > 0$ (positive), so we choose 2nd quadrant.
$\therefore$ *Answer is B*

SOLUTIONS TO SECTION B

2017/18

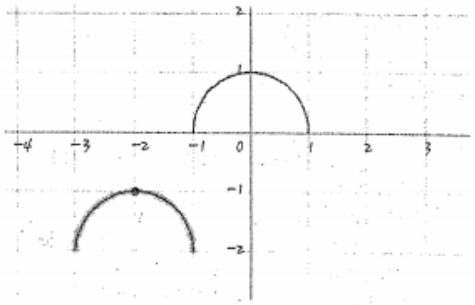
QUESTION 1

a. Given $f(x) = \sqrt{1 - x^2}$

let $y = f(x)$

$$y = \sqrt{1 - x^2}$$
$$y^2 = 1 - x^2$$
$$x^2 + y^2 = 1$$

This is a circle centered at (0,0) with radius 1. Thus the graph of $f(x)$ is the upper half of the circle $x^2 + y^2 = 1$.



Now, $h(x) = -2 + \sqrt{1 - (x + 2)^2}$

$$= -2 + f(x + 2)$$

Thus, the graph of h(x) is a translation of the graph of f(x) 2 units to the left followed by a shift down by 2 units.

(b) $\sqrt{\frac{1-x}{1+x}} = \frac{(1-x)^{\frac{1}{2}}}{(1+x)^{\frac{1}{2}}} = (1-x)^{\frac{1}{2}}(1+x)^{-\frac{1}{2}}$

But $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$

$$(1-x)^{\frac{1}{2}} = 1 + \frac{1}{2}(-x) + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}(-x)^2 + \cdots$$
$$= 1 - \frac{1}{2}x - \frac{1}{8}x^2 + \cdots$$
$$(1+x)^{-\frac{1}{2}} = 1 - \frac{1}{2}(x) + \frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)}{2!}(x)^2$$
$$= 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \cdots$$
$$\Rightarrow (1-x)^{\frac{1}{2}}(1+x)^{-\frac{1}{2}}$$
$$= \left(1 - \frac{1}{2}x - \frac{1}{8}x^2 + \cdots\right)\left(1 - \frac{1}{2}x + \frac{3}{8}x^2 + \cdots\right)$$
$$= 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{1}{2}x + \frac{1}{4}x^2 - \frac{1}{8}x^2$$
$$= 1 - x + \frac{1}{2}x^2$$
$$\Rightarrow \sqrt{\frac{1-x}{1+x}} = 1 - x + \frac{1}{2}x^2$$

(c) Refer to 2014/15 question 2b.

QUESTION 2

Given $P = \begin{vmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix}$

(a) The characteristics equation is given by

$$P(x) = x^3 - tr(P)x^2 + (P_{11} + P_{22} + P_{33})x - |P|$$

$$tr(P) = 0 - 1 + 1 = 0$$

$$P_{11} + P_{22} + P_{33} = \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 0 & -1 \\ 2 & -1 \end{vmatrix}$$

$$= -1 - 1 + 2 = 0$$

$$|P| = 0\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix}$$

$$= 0 + (2 - 2) + (0 + 1) = 1$$

$$P(x) = x^3 - 1$$

(b) The Matrix P is a zero of the p(x). So,

$$P^3 - I = 0$$

$$P^3 = I$$

$$P^3 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

(c) from b, we have

$$P^3 = I$$

$$P(P^2) = I$$

Comparing to

$$P(P^{-1}) = I$$

$$\Rightarrow P^{-1} = P^2$$

$$P^{-1} = \begin{vmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix}^2 = \begin{vmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix}\begin{vmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & 1 & -1 \\ 0 & -1 & 2 \\ 1 & 0 & 2 \end{vmatrix}$$

(d) Given

$$-y + z = 2$$

$$2x - y + 2z = 5$$

$$x + z = 4$$

$$\begin{vmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix}\begin{vmatrix} x \\ y \\ z \end{vmatrix} = \begin{vmatrix} 2 \\ 5 \\ 4 \end{vmatrix}$$

Comparing $PX = b$

$$\Rightarrow P = \begin{vmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix},\ X = \begin{vmatrix} x \\ y \\ z \end{vmatrix},\ b = \begin{vmatrix} 2 \\ 5 \\ 4 \end{vmatrix}$$

From $PX = b$, $\Rightarrow X = P^{-1}b$

$$\Rightarrow X = \begin{vmatrix} -1 & 1 & -1 \\ 0 & -1 & 2 \\ 1 & 0 & 2 \end{vmatrix}\begin{vmatrix} 2 \\ 5 \\ 4 \end{vmatrix}$$

$$= \begin{vmatrix} -2 + 5 - 4 \\ 0 - 5 + 8 \\ 2 + 0 + 8 \end{vmatrix}$$

$$= \begin{vmatrix} -1 \\ 3 \\ 10 \end{vmatrix}$$

QUESTION 3

(a) Given $y = x^2 - 4x + 5$

For y intercept, $x = 0$

$$y = 0^2 - 4(0) + 5 = 5$$

For x intercept, $y = 0$

$x^2 - 4x + 5 \neq 0$ so no x intercept.

For turning point (p,q)

$$p = \frac{-b}{2a} \text{ and } q = \frac{4ac - b^2}{4a} \text{ where } a = 1,\ b = -4,\ c = 5$$

$$p = \frac{-(-4)}{2(1)} = 2$$

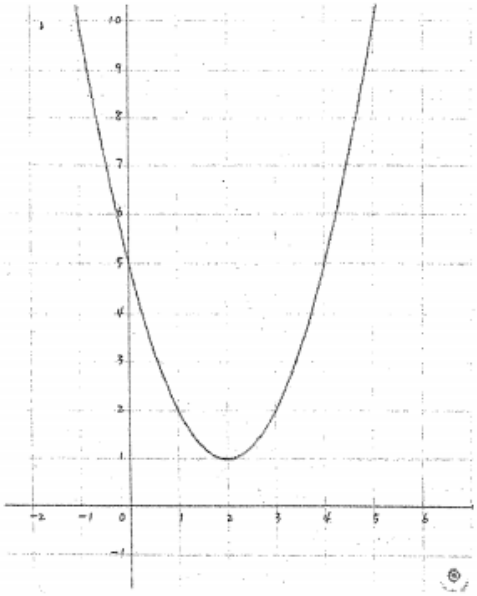
$$q = \frac{4(1)(5) - (-4)^2}{4(1)} = 1$$

$$(p, q) = (2,1)$$

Since $a > 0$, it implies that $(p, q) = (2,1)$ is a minimum turning point.

For domain:$\{x: x \in \mathbb{R}\}$

For range: $\{x: x \in \mathbb{R}\}$



(b) (i) From $y = x^2 - 4x + 5$

For translation 3 units left and 2 units upwards

$y = (x+3)^2 - 4(x+3) + 5 + 2$

$y = x^2 + 2x + 4$

Thus the translated curve is $x^2 + 2x + 4$

(ii) for the point of intersection, solve $y = x^2 - 4x + 5$ and $y = x^2 + 2x + 4$ simultaneously

$$x^2 - 4x + 5 = x^2 + 2x + 4$$

$$6x = -1$$

$$x = \frac{-1}{6}$$

Put $x = \frac{-1}{6}$ into $y = x^2 - 4x + 5$

$$y = \left(\frac{-1}{6}\right)^2 - 4\left(\frac{-1}{6}\right) + 5 = \frac{157}{36}$$

Therefore, the point of intersection is $\left(\frac{-1}{6}, \frac{157}{36}\right)$.

(iii) Translated curve is $y = x^2 + 2x + 4$

For the roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{2^2 - 4(1)(4)}}{2(1)} = \frac{-2 \pm \sqrt{-12}}{2}$$

$$= \frac{-2 \pm \sqrt{12}(\sqrt{-1})}{2} = \frac{-2 \pm 2\sqrt{3}i}{2} \quad (\sqrt{-1} = i)$$

$$= -1 \pm 2\sqrt{3}i$$

(c) Given $f(x) = \frac{x+1}{x^2-x-6}$

$$\frac{x+1}{x^2-x-6} = \frac{x+1}{(x+2)(x-3)} = \frac{A}{x+2} + \frac{B}{x-3}$$

$$\frac{x+1}{x^2-x-6} = \frac{A(x-3) + B(x+2)}{(x+2)(x-3)}$$

$$x + 1 = A(x-3) + B(x+2)$$

If $x = 3$

$$3 + 1 = A(3-3) + B(3+2)$$

$$B = \frac{4}{5}$$

If $x = -2$

$$-2 + 1 = A(-2-3) + B(-2+2)$$

$$-1 = -5A$$

$$A = \frac{1}{5}$$

$$\frac{x+1}{x^2-x-6} = \frac{1}{5(x+2)} + \frac{4}{5(x-3)}$$

QUESTION 4

(a) Given $u_1 = 1$, $u_{n+1} = u_n + \left(\frac{1}{4}\right)^n$, $\forall n \geq 1$

If n=1

$u_{1+1} = u_1 + \left(\frac{1}{4}\right)^1$

$u_2 = u_1 + \frac{1}{4} = 1 + \frac{1}{4} = \frac{5}{4}$

If n=2

$u_{2+1} = u_2 + \left(\frac{1}{4}\right)^2$

$u_3 = \frac{5}{4} + \left(\frac{1}{4}\right)^2 = \frac{21}{16}$

If n=3

$u_{3+1} = u_3 + \left(\frac{1}{4}\right)^3$

$= \frac{21}{16} + \left(\frac{1}{4}\right)^3 = \frac{85}{64}$

(b) For an expression u_n

$$u_n = \frac{ar^n}{r-b} + b^{n-1}\left|u_1 - \frac{ar}{r-b}\right|$$

But $u_{n+1} = u_n + \left(\frac{1}{4}\right)^n$

Comparing to $u_{n+1} = ar^n + bu_n$ we have $a = 1$, $b = 1$ and $r = \frac{1}{4}$

$$u_n = \frac{(1)\left(\frac{1}{4}\right)^n}{\frac{1}{4}-1} + (1)^{n-1}\left|1 - \frac{(1)\left(\frac{1}{4}\right)}{\frac{1}{4}-1}\right|$$

$$= \frac{-4}{3}\left(\frac{1}{4}\right)^n + 1 + \frac{4}{3}\left(\frac{1}{4}\right)$$

$$u_n = \frac{4}{3} - \frac{4}{3}\left(\frac{1}{4}\right)^n$$

(c) $\lim_{n\to\infty} u_n = \lim_{n\to\infty}\left|\frac{4}{3} - \frac{4}{3}\left(\frac{1}{4}\right)^n\right|$

$As\ n \to \infty$

$$= \frac{4}{3} - \frac{4}{3}\left(\frac{1}{4}\right)^{\infty}$$

$$= \frac{4}{3} - 0 = \frac{4}{3}$$

$\lim_{n\to\infty} u_n = \frac{4}{3}$

2016/17

Q1 a. Given f(x)=$\frac{2x}{x^2-4}$

For vertical asymptote, $x^2 - 4 = 0$

⇒X=2, x=-2

For horizontal asymptote, y=0

For domain, {x:x∈ $\mathcal{R}, x \neq \pm 2$}

For range, let y=$\frac{2x}{x^2-4}$ ⇒ $yx^2 - 4y = 2x$

$$yx^2 - 2x - 4y = 0$$

For real values, $b^2 - 4ac \geq 0$

$$\Rightarrow (-2)^2 - 4(y)(-4y) \geq 0$$

$4+16y^2 \geq 0$

Let $4+16y^2=0$

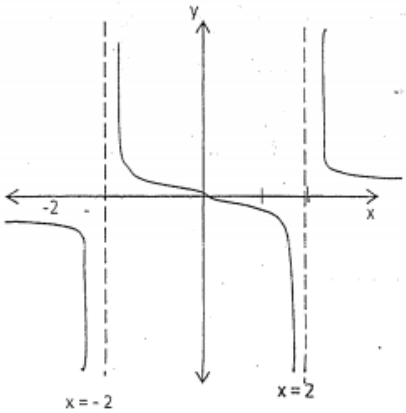
But $4+16y^2 \neq 0\ \forall\ y \in R$

∴Range is $\{y: y \in R\}$

For intercept,

x-intercept is (0,0)

y-intercept is (0,0)



For behavior of curve

As $x \to 2^+, y \to \infty$
$x \to 2^-, y \to -\infty$
$x \to -2^+, y \to \infty$
$x \to -2^-, y \to -\infty$

As $y \to 0^+, y \to \infty$
$y \to 0^-, y \to -\infty$

Q1 i. $f(x)=\frac{2x}{x^2-4}=\frac{A}{x-2}+\frac{B}{x+2}$

$$\Rightarrow \frac{2x}{(x-2)(x+2)}=\frac{A(x+2)+B(x-2)}{(x-2)(x+2)}$$
$$\therefore 2x = A(x+2)+B(x-2)$$

Solving gives A=1 and B=1

$F(x)=\frac{1}{x-2}+\frac{1}{x+2}=(x-2)^{-1}+(x+2)^{-1}$

$=[-2\left(1-\frac{x}{2}\right)]^{-1}+[2(1+\frac{x}{2})]^{-1}$

$f(x)=-\frac{1}{2}(1-\frac{x}{2})^{-1}+\frac{1}{2}(1+\frac{x}{2})^{-1}$

but $(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\cdots$

$\therefore (1-\frac{x}{2})^{-1} =$

$$1-\left(-\frac{x}{2}\right)+\frac{-(-2)\left(-\frac{x}{2}\right)^2}{2!}+\frac{-(-2)(-3)\left(-\frac{x}{2}\right)^3}{3!}+\frac{-1(-2)(-3)(-4)\left(-\frac{x}{2}\right)^4}{4!}+\frac{-1(-2)(-3)(-4)(-5)\left(-\frac{x}{2}\right)^5}{5!}+\cdots.$$
$$=1+\frac{x}{2}+\frac{x^2}{4}+\frac{x^3}{8}+\frac{x^4}{16}+\frac{x^5}{32}+\cdots$$

$$\therefore -\frac{1}{2}\left(1-\frac{x}{2}\right)^{-1}=-\frac{1}{2}-\frac{x}{4}-\frac{x^2}{8}-\frac{x^3}{16}-\frac{x^4}{32}-\frac{x^5}{64}-\cdots$$

Also,

$$(1+\frac{x}{2})^{-1}=1-\left(\frac{x}{2}\right)+\frac{-(-2)\left(\frac{x}{2}\right)^2}{2!}+\frac{-(-2)(-3)\left(\frac{x}{2}\right)^3}{3!}+\frac{-1(-2)(-3)(-4)\left(\frac{x}{2}\right)^4}{4!}+\frac{-1(-2)(-3)(-4)(-5)\left(\frac{x}{2}\right)^5}{5!}+\cdots.$$
$$=1-\frac{x}{2}+\frac{x^2}{4}-\frac{x^3}{8}+\frac{x^4}{16}-\frac{x^5}{32}+\cdots$$
$$\therefore -\frac{1}{2}\left(1-\frac{x}{2}\right)^{-1}=\frac{1}{2}-\frac{x}{4}+\frac{x^2}{8}-\frac{x^3}{16}+\frac{x^4}{32}-\frac{x^5}{64}+\cdots$$
$$\Rightarrow f(x)=[-\frac{1}{2}-\frac{x}{4}-\frac{x^2}{8}-\frac{x^3}{16}-\frac{x^4}{32}-\frac{x^5}{64}-\cdots]+[\frac{1}{2}-\frac{x}{4}+\frac{x^2}{8}-\frac{x^3}{16}+\frac{x^4}{32}-\frac{x^5}{64}+\cdots]$$
$$=\frac{1}{2}-\frac{x}{4}+\frac{x^2}{8}-\frac{x^3}{16}+\frac{x^4}{32}-\frac{x^5}{64}+\cdots$$

$$=-\frac{x}{2}-\frac{x^3}{8}-\frac{x^5}{32}-\cdots$$

$$for\ f(x)=-\frac{1}{2}\sum_{r=0}^{\infty}\frac{x^{2r+1}}{4^r}$$
$$\Rightarrow \sum_{r=0}^{\infty} U_r = U_0+U_1+U_2+U_3+\cdots$$

$$but\ U_r=-\frac{1}{2}(\frac{x^{2r+1}}{4^r})$$

If r=0

$$U_0=-\frac{1}{2}[\frac{x^{2(0)+1}}{4^{(0)}}]=-\frac{1}{2}x$$

If r=1

$$U_1=-\frac{1}{2}[\frac{x^{2(1)+1}}{4^{(1)}}]=-\frac{1}{8}x^3$$

If r=2

$$U_2=-\frac{1}{2}[\frac{x^{2(2)+1}}{4^{(2)}}]=-\frac{1}{32}x^5$$
$$\Rightarrow -\frac{1}{2}\sum_{r=0}^{\infty}\frac{x^{2r+1}}{4^r}=-\frac{x}{2}-\frac{x^3}{8}-\frac{x^5}{32}-\cdots$$

But $f(x)=-\frac{x}{2}-\frac{x^3}{8}-\frac{x^5}{32}-\cdots$

$$\Rightarrow f(x)=-\frac{1}{2}\sum_{r=0}^{\infty}\frac{x^{2r+1}}{4^r}$$

Q 2. Refer to 2017/18 Question 2
Q 3a. Refer to 2017/18 Question 3a
Q 3b. Refer to 2017/18 Question 3b

Q 3c. Given $f(x)=\frac{x^2-2x-6}{x+1}$

For vertical asymptote, x+1=0

$$\Rightarrow x=-1$$

For horizontal asymptote, no H.A

For oblique asymptote, y=x (equate y to the quotient of f(x))

For Domain; $\{x: x \in \mathbb{R},\ x \neq -1\}$

For range let $y=\frac{x^2-2x-6}{x+1}$

$$yx+y=x^2-x-6$$
$$x^2+x(-1-y)-6-y=0$$

For real values, $b^2 - 4ac \geq 0$

$$\Rightarrow (-1-y)^2 - 4(1)(-6-y) \geq 0$$

$$y^2 - 6y + 25 \geq 0$$

Let $y^2 - 6y + 25 = 0$

But

$$y^2 - 6y + 25 \neq 0 \quad \forall\ y \in R$$

$\therefore R_f = \{y : y \in R\}$

For intercept ,

x-intercept is (-2,0), (3,0)

y-intercept is (0,-6)

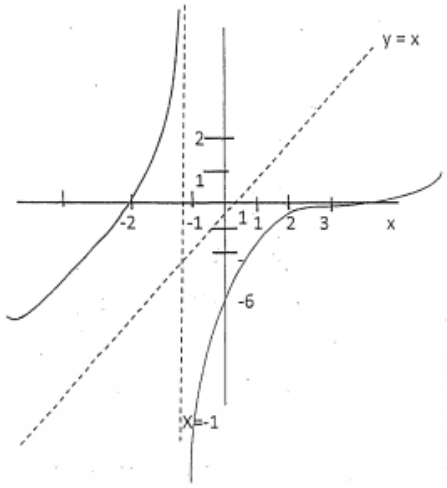
For behavior of curve ,

As $x \to -1^+, y \to -\infty$

$x \to -1^-, y \to \infty$

As $y \to x^+, y \to -\infty$

$y \to x^-, y \to \infty$



Q4. Given $U_1 = \frac{5}{2}, U_{n+1} = 2 - \frac{1}{3}U_n$

Comparing to , $U_{n+1} = ar^n + bU_n$

$$\Rightarrow a = 2, r = 1, b = -\frac{1}{3}$$

$$\therefore U_{n+1} - U_n = 2 - \frac{1}{3}U_n - U_n$$

$$= 2 - \frac{4}{3}U_n$$

But $U_n = \frac{ar^n}{r-b} + b^{n-1}\left|U_1 - \frac{ar}{r-b}\right|$

$$\therefore U_n = \frac{2(1)^n}{1-\frac{1}{3}} + \left(-\frac{1}{3}\right)^{n-1}\left|\frac{5}{2} - \frac{2(1)}{1-\frac{1}{3}}\right|$$

$$= \frac{3}{2} + \left(-\frac{1}{3}\right)^{n-1}\left|\frac{5}{2} - \frac{3}{2}\right|$$

$$= \frac{3}{2} + 3\left(-\frac{1}{3}\right)^n$$

$$\therefore U_{n+1} - U_n = 2 - \frac{4}{3}\left|\frac{3}{2} + 3\left(-\frac{1}{3}\right)^n\right|$$

$$= 4\left(-\frac{1}{3}\right)^n$$

$$\therefore U_{n+1} = U_n + 4\left(-\frac{1}{3}\right)^n$$

b. From $U_{n+1} = U_n + 4\left(-\frac{1}{3}\right)^n$

Comparing to $U_{n+1} = ar^n + bU_n$

$$b = 1, a = 4, r = -\frac{1}{3}$$

But $U_n = \frac{ar^n}{r-b} + b^{n-1}\left|U_1 - \frac{ar}{r-b}\right|$

$$\therefore U_n = \frac{4(-\frac{1}{3})^n}{-\frac{1}{3}-1} + (1)^{n-1}\left|\frac{5}{2} - \frac{4(-\frac{1}{3})}{-\frac{1}{3}-1}\right|$$

$$= \frac{-3(-\frac{1}{3})^n}{1} + \left|\frac{5}{2} - 1\right|$$

$$\therefore U_n = \frac{3}{2} - 3(-\frac{1}{3})^n$$

Q4c. For $\lim_{n\to\infty} \frac{S_n}{n}$

$$S_n = \sum_{k=1}^{n} U_k$$

But $U_k = \frac{3}{2} - 3(-\frac{1}{3})^k$

$=\frac{3}{2}-3(-\frac{1}{3})(-\frac{1}{3})^{k-1}$

$=\frac{3}{2}+(-\frac{1}{3})^{k-1}$

$$\therefore S_n = \sum_{k=1}^{n} |\frac{3}{2}+(-\frac{1}{3})^{k-1}|$$

$$\sum_{k=1}^{n} |\frac{3}{2}| + \sum_{k=1}^{n} (-\frac{1}{3})^{k-1}$$

$=\frac{3}{2}n+(\frac{1-(-\frac{1}{3})^n}{1-(-\frac{1}{3})})$

$=\frac{3}{2}n+\frac{3}{4}(1-(-\frac{1}{3})^n)$

$=\frac{3}{2}n+\frac{3}{4}-\frac{3}{4}(-\frac{1}{3})^n$

$$\lim_{n\to\infty}\frac{S_n}{n}=\lim_{n\to\infty}\frac{\frac{3}{2}n+\frac{3}{4}-\frac{3}{4}(-\frac{1}{3})^n}{n}$$

$=\lim_{n\to\infty}\frac{3}{2}+\frac{3}{4n}-\frac{3}{4n}(-\frac{1}{3})^n$

$as\ x\to\infty$

$$\lim_{n\to\infty}\frac{S_n}{n}=\frac{3}{2}$$

2015/2016

SECTION B

1. Given $f(x)=\frac{3x-9}{x^2-x-2}$

For vertical asymptotes, $x^2-x-2=0 \implies x=2,-1$

For horizontal asymptote, $y=0$ (since f(x) is a proper rational function)

For x-intercept, y=0

$\frac{3x-9}{x^2-x-2}=0 \implies x=3$ P(3,0)

For y intercept, x=0

$y=\frac{3(0)-9}{(0)^2-(0)-2}=4.5$ P(0,4.5)

For domain: $\{x: x\in\mathbb{R}, x\neq 2,-1\}$

For range, let $y=\frac{3x-9}{x^2-x-2}$. Then $yx^2+(-y-3)x-2y+9=0$

For real, $b^2-4ac\geq 0$

$$(-y-3)^2-4y(-2y+9)\geq 0$$

Simplifying gives $9y^2-30y+9\geq 0$

Let $9y^2-30y+9=0 \implies y=3,\frac{1}{3}$



By testing, $y\geq 3$ *and* $y\leq\frac{1}{3}$

Range $=\{y: y\in\mathbb{R},\ y\geq 3\ or\ y\leq\frac{1}{3}\}$

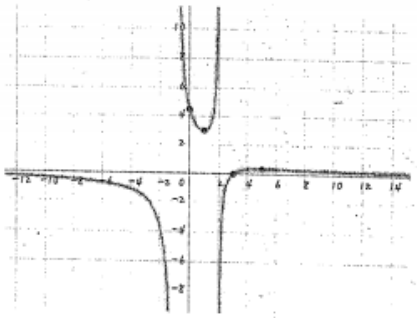
For behavior of the curve

As $x\to 2^+,\ y\to-\infty$

As $x\to 2^-,\ y\to\infty$

As $x\to -1^+,\ y\to\infty$

As $x\to -1^-,\ y\to-\infty$



From the range, f(x) is greater than or equal to 3 and also less than or equal to $\frac{1}{3}$. Thus ,it does not lie between $\frac{1}{3}$ and 3 which are the two certain values.

2014/15

SECTION B

1. a. Given $2x^2 - x - 2 = 0$

sum of ro/ot $(\alpha + \beta) = \frac{-(-1)}{2} = \frac{1}{2}$

product of root $(\alpha+\beta) = \frac{c}{a} = \frac{-2}{2} = -1$

i. $\alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$

but $(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$

$(\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$

but $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\therefore (\alpha - \beta)^2 = (\alpha + \beta)^2 - 2\alpha\beta - 2\alpha\beta$

$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$(\alpha - \beta) = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$

$= \alpha^2 - \beta^2 = (\alpha + \beta)\sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$

$= (\frac{1}{2})\sqrt{(\frac{1}{2})^2 - 4(-1)}$

$= (\frac{1}{2})\sqrt{\frac{1}{4} + 4}$

$= \frac{1}{2}\sqrt{\frac{17}{4}}$

$= \frac{1}{2}\frac{\sqrt{17}}{\sqrt{4}} = \frac{1}{2}\left(\frac{\sqrt{17}}{2}\right) = \frac{\sqrt{17}}{4}$

$\therefore \alpha^2 - \beta^2 = \frac{\sqrt{17}}{4}$

Given,

$\pi \alpha^2 - \beta^3 = \frac{5}{2}(\alpha^2 + \beta^2)$

But $(\alpha - \beta)^3 = \alpha^3 - 3\alpha^2\beta + 3\alpha\beta^2 - \beta^3$

$(\alpha - \beta)^3 = \alpha^3 - \beta^3 - 3\alpha\beta(\alpha - \beta)$

$a^3 - \beta^3 = (\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta)$

$\alpha^3 - \beta^3 = (\alpha - \beta)[(\alpha - \beta)^2 + 3\alpha\beta]$

but $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$= \alpha^3 - \beta^3 = (\alpha - \beta)[(\alpha + \beta)^2 - \alpha\beta]$

$= (\alpha - \beta)(\alpha + \beta)[(\alpha + \beta) - \frac{\alpha\beta}{\alpha+\beta}]$

$= (\alpha^2 - \beta^2)[\alpha + \beta) - \frac{\alpha\beta}{\alpha+\beta}$

But $\alpha + \beta = \frac{1}{2}$. $\alpha\beta = -1$

$\alpha^3 - \beta^3 = (\alpha^2 - \beta^2)[\frac{1}{2} - \frac{-1}{1/2}]$

$= (\alpha^2 - \beta^2)(\frac{1}{2} - (-2)]$

$= (\frac{1}{2} + 2)(\alpha^2 - \beta^2)$

$\alpha^3 - \beta^3 = \frac{5}{2}(\alpha^2 - \beta^2)$

As required

Product of roots $= (\alpha^2)(\beta^2)$

$= (x\beta)^3$

$= (-1)^3$

$= -1$

The equation is of the form $x^2 - (\alpha^2 + \beta^2)x + \alpha^2\beta^2 = 0$.

$\therefore x^2 - \frac{13}{8}x - 1 = 0$

$8x^2 - 13x - 8 = 0$

b. Given $0.1\dot{4} = 0.14444\ldots$

$= 0.14 = 0.1 + 0.04 + 0.04 + \ldots$

$= 0.1 + 4 \times 10^{-2} + 4 \times 10^{-3} + 4 \times 10^{-4} + \ldots$

$= 0.1 + 4 \times 10^{-2}(1 + 10^{-1} + 10^{-2} + 10^{-3} + \ldots$

$= 0.1 + 4 \times 10^{-2}$ (sum of infinity of a G.P)

$= 0.1 + 4 \times 10^{-2}\left[\frac{1}{1 - \frac{1}{10}}\right]$

$= 0.1 + 4/100\ (^1/_{9/10})$

$= 0.1 + 4/100\ (^{10}/_9)$

$= 0.1 + {}^4/_{90}$

$= {}^1/_{10} + {}^4/_{90}$

$= {}^{13}/_{90}$

$\therefore 0.14$ is e α pressed as $^{13}/_{90}$

2. given

a. $3x + 2y - 2z = -8$

$x - 3y + z = -3$

$2x + y - 3z = -9$

$$\begin{vmatrix} 3 & 2 & -2 \\ 1 & -2 & 1 \\ 2 & 1 & -3 \end{vmatrix} \begin{vmatrix} x \\ y \\ z \end{vmatrix} = \begin{vmatrix} -8 \\ -3 \\ -9 \end{vmatrix}$$

Using cramer's rule

$\alpha = \frac{|Ax|}{|A|}$ $\quad y = \frac{|Ay|}{|A|}$

$z = \frac{|Az|}{|A|}$

now $|A| = \begin{vmatrix} 3 & 2 & -2 \\ 1 & -2 & 1 \\ 2 & 1 & -3 \end{vmatrix}$

$|A| = 3\begin{vmatrix} -2 & 1 \\ 1 & -3 \end{vmatrix} - 2\begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} - 2\begin{vmatrix} 1 & -3 \\ 2 & 1 \end{vmatrix}$

$= 3(9 - 1) - 2(-3 - 1) - 2(+6)$

$= 3\,)8 = -2\,0\,-5) - 2)7)$

$= 24 + 10 - 14$

$|A| = 20$

Also $|Ax| = \begin{vmatrix} -8 & 2 & -2 \\ -3 & -2 & 1 \\ -9 & 1 & -3 \end{vmatrix}$

$|Ax| = -8\begin{vmatrix} -3 & 1 \\ 1 & -3 \end{vmatrix} - 2\begin{vmatrix} -3 & 1 \\ -9 & -3 \end{vmatrix} - 2\begin{vmatrix} 1 & -3 \\ 2 & 1 \end{vmatrix}$

$= -8\,(9 - 1) - 2(9 + ()) - 2(-3 - 27)$

$= -8\,(8)\ 2(18) - 2(-30)$

$= -64 - 36 + 60$

$|Ax| = -40$

$X = \frac{|Ax|}{|A|} = \frac{-40}{20} = -2$

also $|Ay| = \begin{vmatrix} 3 & -8 & -2 \\ 1 & 1 & 1 \\ -3 & 1 & -3 \end{vmatrix}$

$|Ay| = 3\begin{vmatrix} -3 & 1 \\ -9 & -3 \end{vmatrix} + 8\begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} - 2\begin{vmatrix} 1 & -3 \\ 2 & -9 \end{vmatrix}$

$= 3(9+9) + 8(-3 - 2) - 2(-9 + 6)$

$= 3(18) + 8(-5) - 2(-3)$

$= 54 - 40 + 6$

$|Ay| = 20$

$\therefore y = \frac{|Ay|}{|A|} = \frac{20}{20} = 1$

$$|Az| = \begin{vmatrix} 3 & 2 & -8 \\ 1 & -3 & -3 \\ 2 & 1 & -9 \end{vmatrix}$$

$$= 3\begin{vmatrix} -3 & 1 \\ 1 & -9 \end{vmatrix} - 2\begin{vmatrix} 1 & -1 \\ 2 & -9 \end{vmatrix} - 8\begin{vmatrix} 1 & -3 \\ 2 & 1 \end{vmatrix}$$

$= 3(27 + 3) - 2(-9 + 6) - 8(+6)$

$= 3(30) - 2(-3) - 8(7)$

$= 90 + 6 - 56$

$= 40$

$\therefore Z = \frac{|Az|}{|A|} = \frac{40}{20} = 2$

∴ the solution for the system of equations are x= -2, y = 1 and Z = 2

b. Given

$|x + 2| \geq |x + 2| + |x - 3|$

$|x + 2|^2 \geq (1x1 + 1x - 31)^2$

$[x + 2]^2 \geq |x + 1|^2 + 2|x + 1||x - 3| + |x - 3|^2$

$= [x + 2]^2 \geq |x + 1|^2 + 1|x \pm 3||x - 3|^2$ $x^2 4x + 4 \geq x^2 + 2x + 1 + x^2 - 6x + 9 + 21x + 1|x - 3|$

$x^2 + 4x + 4 \geq 2x^2 - 4x + 10 + 2|x + 1||x - 3|$

$2|x + 1||x - 3| \leq -x^2 + 8x - 6$

$1|x + 1||x - 3| \leq \frac{1}{2}(-x^2 + 8x - 6)$

$$|(x + 1)(x - 3) \leq -\frac{1}{2}x^2 + 4x - 3|$$

$= -(-½ x^2 + 4x - 3) \leq (x+1)(x-3) \leq \frac{-1}{2}x^2 + 4x - 3$

Taking the first ineguality given,

$½^{x2} - 4x + 3 \leq (x+1)(x-3)$

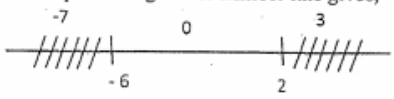
$½^{x2} - 4x + 3 \leq x2 - 3x + x - 3$

$-½ x2 - 2x + 6 \leq 0$

$= -x^2 - 4x + 12 = 0$

$= x = 2, -6$

Representing on the number line gives,



$= -(-7)\,2, 4(-7) + 12 = -9 \leq 0$

If $x = 0$

$-(0)2 - 4(0) + 12 = 12 \leq 0$

If $x = 3$

$-(3)2) - 4(3) + 12 = -9 \leq 0$

Since $x = -7$ and $x = 3$ satisfies the inequality.

$= x \leq -6$ and $x \geq 2$ satisfies the inequality.

Also, taking the second inequality.

gives $(x+1)9x-3) \leq -½ x^2 + 4x - 3$

$x^2 - 2x - 3 \leq -½ x2 + 4x - 3$

$2x^2 - 4x - 6 \leq -x^2 + 8x - 6$

$= 3x^2 - 12x \leq = 0$ - (**)

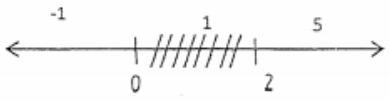
Let $3x^2 - 12x = 0$

$3x(x-4) = 0$

$3x = 0, \quad 4 - 4 = 0$

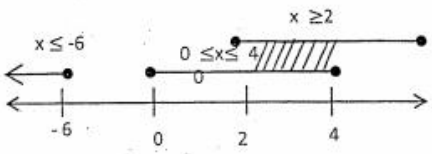
$x = 0 \qquad x = 4$

representing on the number line gives



If x = -1 (substitute into **)
$3(-1)^2 - 12(1) = 3 + 12 = 15 \leq 0$
If x = 1
$3(1)2 - 12(1) = 3 - 12 = -9 \leq 0$
If x =5
$3(5)2 - 12(5) = 75 - 60 \leq 0$
Since x = 1 satisfies the inequality **
$= 0 \leq x \leq 4$ satisfies the inequality

Now, the intersection of all possible solutions gives the overall truth set.



From the above number line, since x ≥ 2 and $0 \leq x \leq 4$ overlaps at $2 \leq x \leq 4$, it implies the truth set of the inequality un question is $2 \leq x \leq 4$

3.. a. Given $f(x) = \frac{x^3}{x^2-16}$

Let $y = \frac{x^3}{x^2-16}$

x – intercept is (0,0)

y- intercept is (0,0)

for vertical asymptote, $x2 - 16 = 0$

$(x-4)(x+4) = 0$

$X = 4, x = -4$

For oblique asymptote, we find the quotient.

$x^3 - 16\frac{x}{\sqrt{x^3}}$

$\frac{-(x^3-16x)}{16x}$

$\therefore = x$

for horizontal asymptote, f(x) has no horizontal asymptote

for Domain, $\{ x : x \in R, x \# 4\}$

for range, let $y = \frac{x^3}{x^2-16}$

$y(x^2 - 16) = x^3$

$yx^2 - 16y = x^3$

$yx^2(y - x) = 16y$

$\frac{x^2(y-x)}{y-x} = \frac{16y}{4-x}$

$x2 = \frac{16y}{y-x}$

$x = \sqrt{\frac{16y}{y-x}}$

for real values, $\frac{16y}{y-x} \geq 0$

$\therefore (y-x)^2 \frac{16y}{y-x} \geq 0$

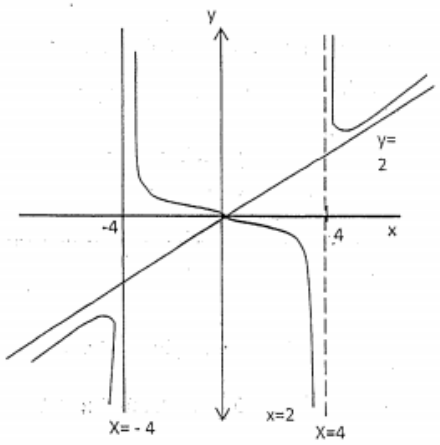
$y = 0, y \geq x$

since x =y is not defined, the range is $y \geq 0 \cup y > x$

for the behavior of curves,
As $x \to 4^+$, $y \to \infty$
$x \to 4^-$ $y \to -\infty$
$x \to -4^+$, $y \to \infty$
$x \to -4^-$, $y \to -\infty$

As $y \to x^+$ $x \to \infty$

$y \to x^-$, $x \to -\infty$



∴ the graph is symmetric about the origin with x ,y $\in R$ such that $x \neq \pm 4$, $y \geq 0$ and $y > x$

b. Given $f(x) = \frac{x^2-5x-3}{2x^2+5x-12}$

$\frac{x^2-5x-3}{2x^2+5x-12} = \frac{A}{x-4} + \frac{B}{2x-3}$

$\therefore x^2 - 5x - 3 = A(2x - 3) + B(x + x 4)$

Let $x = -4$

$1-4)^2 - 5(4) - 3 = A[2(-4)3] + B(-4,4)$

$16 + 20 - 3 = A(-8 - 3) + B(o)$

$$\frac{11A}{11} = \frac{33}{-11}$$

$A = -3$

Let $x = 0$

$(o)^2 - 5(0) - 3 = A[2(0) - 3] + B[0+4]$

$-3 = -3A+44$, but a $-3 - 3 = -3(-3) + 4B$

$-9-3 = 4b$

$\frac{-12}{4} = \frac{4B}{4}$

$B = -3$

$\therefore f(x) = \frac{-3}{x-4} + \frac{3}{2x-3}$

4. a.

$\frac{2x+1}{(x-1)(x+2)^2} = \frac{A}{x-1} + \frac{B}{x+2} \frac{C}{(x+2)^2}$

$\frac{2x+1}{(x+1)(x+2)^2} = \frac{A(x+2)^2 B(x-1)(x+2)(c(x-1)}{(x-1)(x+2)}$

$2 x 2 + 1 = A(x+2)^2 + B(x - 1)(x+2) + c(x - 1)$

Let $x = 1$

$2(1) + 1 = A(1+2)^2 + B(1+1)(1+2) + C(1-1)$

$2+1 = 9A$

$\frac{9A}{9} = \frac{3}{9}$

$A = {}^1/_3$

$2(-2) + 1A(-2+2)^2 + B(-2 - 1) + C(2 - 1)$

$14+1 = -3c$

$\frac{-3C}{3} = \frac{-3}{-3}$

$C = 1$

Let $x = 0$

$2(0) - 1 = A(0+2)^2\ B(0, -1)\ (0+2)\ C(0, -1)$

$-1 = 4A - 2B - C$, but $A = {}^1/_3$, $C=1$

$-1 = 4({}^1/_3) - 2B - 1$

$\frac{2B}{2} = \frac{\frac{4}{3}}{2}$,

$B = {}^4/_6 = {}^2/_3$

$\therefore \frac{2x+1}{(x-1)(x+2)^2} = \frac{1}{3(x-1)} + \frac{2}{3(x+2)} + \frac{1}{(x+2)^2}$

Now

$$\frac{2x+1}{(x-1)(x+2)^2} = 3^{-1}(x-1)^{-1} + 2\times$$

$$3^{-1}(x+2)^{-1} + (x+2)^2$$

$$= {}^1/_3(-1+x)^{-1} + {}^2/_3(2+x)^{-1} + (2+x)^{-2}$$

$$= {}^1/_3](-x)]^{-1} + {}^2/_3[]1+{}^x/_2)]^{-1}\,[2\{1+x/2)\}]^{-2}$$

$$= -{}^1/_3(1-x)^{-1} + {}^2/_{3\,x}\ {}^1/_2(1+{}^x/_2)^{-1} + {}^1/_4(1+{}^x/_2) - 2$$

$$= -{}^1/_3(1-x)^{-1} + {}^1/_3(11+{}^x/_2)^{-1} + {}^1/_4(1+{}^x/_2)^{-2}$$

but $(1+x)^n = 1+nx + \frac{n(n-1)}{21}x^2 + \cdots$

$$\therefore (1-x)^{-1} = 1-(-x) + \frac{(-1-1)(x)^2}{21} + \cdots$$

$$= 1+x+x^2+\ldots$$

$$(1+{}^x/_2)^{-2} = 1-2({}^x/_2) + \frac{-2(-2-1)(\frac{x}{2})^2}{2!} + \cdots$$

$$= 1 - x + {}^3/_4 x^2 - \ldots$$

Now, $-1/3(1-x) + {}^1/_3(1+{}^x/_2)^{-1} + {}^1/_4(1+{}^x/_2)^{-2}$

$$-{}^1/_3(1+x+x^2+\ldots) + {}^1/_3(1-{}^x/_2+{}^{x2}/_4\ldots)$$

$$+ {}^1/_4(1-x+{}^3/_4 x^2\ldots)$$

$$= -{}^1/_3 - {}^x/_3 - {}^{x2}/_3 + {}^1/_3 - {}^x/_6 + {}^{x2}/_{12} + {}^1/_4 - {}^x/_4$$

$${}^3/_6{}^{x2} + \ldots$$

$$= \frac{-1}{3} + \frac{1}{3} + \frac{1}{4} - \frac{x}{3} - \frac{x}{6} - \frac{-x}{4} - \frac{-x2}{3} + \frac{x2}{12} + \frac{3x2}{16} + \ldots$$

$$= \frac{1}{3} - \frac{3k}{4} - \frac{1}{16}x^2 + \cdots$$

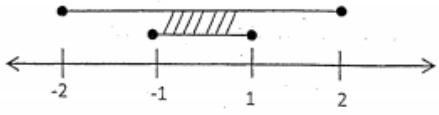
$$\therefore \frac{2x+1}{(x-1)(x+2)^2} = \frac{1}{4} - \frac{3x}{4}x - \frac{1}{16}x^2$$

Where $a = {}^1/_4$, $b = {}^{-3}/_4$, $c, = {}^{-1}/_{16}$

Also ,i. the e x pression is valid for $|x| < 1$

ii. $\left|\frac{x}{2}\right| < 1$, $= \left|\frac{x}{2}\right| < 1$, $\quad |x| < 2$,

representing both on number line gives



By taking the intersection gives $-1 < x < 1$

$= |x| < 1$.

Hence, the e x pansion is valid for $|x| < 1$.

b. Given $\log_b{}^a + \log_c{}^a = 2\log_a{}^b \log_c{}^a$

$$\frac{1}{\log_a b} + \frac{1}{\log_a c} = 2\ {}^1/_{\log_a^b}\ {}^1/_{\log_a^c}$$

$$\frac{\log_a c + \log_a b}{(\log_a^b)(\log_a^c)} = \frac{2}{(\log_a^b)(\log_a^c)}$$

$$\log_a c + \log_a b = 2$$

$$\log_a cb = 2$$

$$= a^2 = cb$$

$$a = \sqrt{cb}$$

2013/14

SECTION B

1. a. Refer to 2014/15 question 1a.
   b. Refer to 2014/15 Question 3a.

   c. Given

$$Secx - tanx = tan\left(\frac{\pi}{4} - \frac{-1}{2x}\right)$$

$$tan\left(\frac{\pi}{4} - \frac{-1}{2x}\right) = \frac{\tan\left(\frac{\pi}{4} - \frac{1}{2}x\right)}{1+tan\frac{\pi}{4}tan\frac{1}{2}x}$$

$$= \frac{1-1tan\frac{1}{2}x}{1+tan\frac{1}{2}x}$$

$$= \frac{1-\frac{sin\frac{1}{2}x}{cos\frac{1}{2}x}}{1+\frac{sin\frac{1}{2}x}{cos\frac{1}{2}x}}$$

$$= \frac{\frac{cos\frac{1}{2}x-sin\frac{1}{2}x}{cos\frac{1}{2}x}}{\frac{cos\frac{1}{2}x+sin\frac{1}{2}x}{cos\frac{1}{2}x}}$$

$$= \frac{cos\frac{1}{2}x-sin\frac{1}{2}x}{cos\frac{1}{2}x+sin\frac{1}{2}x} \times \frac{cos\frac{1}{2}x-sin\frac{1}{2}x}{cos\frac{1}{2}x+sin\frac{1}{2}x}$$

$$= \frac{(cos\frac{1}{2}x-sin\frac{1}{2}x)^2}{cos\frac{1}{2}x+sin\frac{1}{2}x}$$

but cos x $= \cos^{2}1/_{2}$ x $- \text{Sin}^{2}$ ½ x

=tan

$$\left(\frac{\pi}{4} - \frac{1}{2}x\right) = \frac{cos^{2}\frac{1}{2}x+sin^{2}\frac{1}{2}x-2sin\frac{1}{2}xcos\frac{1}{2}x}{cosx}$$

$$= \frac{1-2sin\frac{1}{2}xcos\frac{1}{2}x}{cosx}$$

but sin x $= 2\sin$ ½ x cos ½ x

$$= \frac{1-sinx}{cos\ x}$$

$$= \frac{1}{cos\ x} = \frac{sinx}{cos\ x}$$

$$\tan\left(\frac{\pi}{4} - \frac{1}{2}x\right) = secx - tanx$$

2. Given $3z - y + 2x = -17$

$-2z + 6y = 10$

$-Z - y + x = -3$

$$\begin{pmatrix} 3 & -1 & 2 \\ 2 & 6 & 0 \\ -1 & -2 & 1 \end{pmatrix} \begin{pmatrix} z \\ y \\ x \end{pmatrix} = \begin{pmatrix} -17 \\ 10 \\ -3 \end{pmatrix}$$

Using cramer's rule,

$$\text{Let A} = \begin{pmatrix} 3 & -1 & 2 \\ 2 & 6 & 0 \\ -1 & -2 & 1 \end{pmatrix}$$

$$\therefore z = \frac{|Az|}{|A|} \quad y = \frac{|Ay|}{|A|} \quad z = \frac{|Az|}{|A|}$$

$$\text{now } |A| = \begin{pmatrix} 3 & -1 & 2 \\ 2 & 6 & 0 \\ -1 & -2 & 1 \end{pmatrix}$$

$$|A| = -2\begin{vmatrix} -1 & 2 \\ -2 & 1 \end{vmatrix} + 6\begin{vmatrix} 3 & 2 \\ -1 & 1 \end{vmatrix} - 0\begin{vmatrix} 3 & -1 \\ -1 & -2 \end{vmatrix}$$

$$|A| = -2(-1+4) + (6(3+2)$$

$$+ -2\,(3) + 6(5)$$

$$= -6 + 30$$

$$|A| = 24$$

$$|A| = \begin{pmatrix} -17 & -1 & 2 \\ 10 & 6 & 0 \\ -3 & -2 & 1 \end{pmatrix}$$

$$= -10\begin{vmatrix} -1 & 2 \\ -2 & 1 \end{vmatrix} + 6\begin{vmatrix} -17 & 2 \\ -3 & 1 \end{vmatrix} - 0\begin{vmatrix} -17 & -1 \\ -3 & -2 \end{vmatrix}$$

$$= -10\,(-1+4) + 6\,(-17+6)$$

$$= -10\,(3) + 6(-11)$$

$$= -30 - 66$$

$$|A| = 96$$

$$\therefore Z = \frac{|Az|}{|A|} = \frac{-96}{24} = -4$$

$$|Ay| = \begin{pmatrix} 3 & -17 & 2 \\ 2 & 10 & 0 \\ -1 & -3 & 1 \end{pmatrix}$$

$$= -2\begin{vmatrix} -17 & 2 \\ -3 & 1 \end{vmatrix} + 10\begin{vmatrix} 3 & 2 \\ -1 & 1 \end{vmatrix} - 0\begin{vmatrix} -3 & -17 \\ -1 & -3 \end{vmatrix}$$

$$= -2(-7+6) + 10\,(3+2)$$

$$= -2\,(-11) + 10(5)$$

$$= 22 + 50$$

$$|Ay| = 72$$

$$\therefore y = \frac{|Ay|}{|A|} = \frac{72}{24} = 3$$

$$|Ax| = \begin{pmatrix} 3 & -1 & -17 \\ 2 & 6 & 10 \\ -2 & -2 & -3 \end{pmatrix}$$

$= -z\begin{vmatrix}-1 & -17\\ -2 & -3\end{vmatrix} + 6\begin{vmatrix}3 & -17\\ -1 & -3\end{vmatrix} - 10\begin{vmatrix}3 & -1\\ -1 & -2\end{vmatrix}$

$= -2(3.34) + 6(-9-17) - 10(-6-1)$

$= -2(-13) + 6(-26) - 10(-7)$

$= 62 - 156 + 170$

$|Ax| = -24$

$\therefore x = \frac{|Ax|}{|A|} = \frac{-24}{24} = -1$

∴ The solution for the system of equation is x = -1, y = 3 and z = -4

b. Given $\sec\theta\tan\theta = 2$

$\left(\frac{1}{\cos\theta}\right)\left(\frac{\sin\theta}{\cos\theta}\right) = 2$

$\frac{\sin\theta}{\cos^2\theta} = 2$

$\sin\theta = 2(1 - \sin^2\theta)$

$\sin\theta = 2 - 2\sin^2\theta$

$2\sin^2\theta + \sin\theta - 2 = 0$

Let $y = \sin\theta$

$\therefore 2y2 + y - 2 = 0$

$y = 0.7808, -1.2808$

but $y = \sin\theta$

if $y = 0.7808$

$= \sin\theta = 0.7808$

$\theta = \sin\text{-}1(0.7808)$

$\theta_1 = 51.3339^0$

Since sine is positive in 2nd quadrant,

$= \theta_2 = 180 - \theta_1$

$= 180 - 51.3339$

$\theta_2 = 128.6661^O$

$= \sin\theta = -2.808$

$= \sin\theta = -1.2808$

This is indeterminate since sine lies between -1 and 1.

$\therefore \theta = 51.339^O, 138\text{-}6661^0$

c. Given $U^1 = {}^5/_2$

$Un = 2 - {}^1/_3\, Un_{-1}$

Comparing to $Un = ar^{n-1} + bUn_{-1}$

A = 2 r = 1 b = 1/3

$\therefore Un = \frac{1}{1-b} + b^{n-1}\left|U_1 - \frac{1}{1-b}\right|$

$\frac{2}{1-\frac{-1}{3}} + \left(\frac{-1}{3}\right)^{n-1}\left|\frac{5}{2} - \frac{2}{1-\frac{-1}{3}}\right|$

$Un = {}^2/_4{}_2 + \left(-\frac{1}{3}\right)^{n-1}\left|\frac{5}{2} - \frac{2}{1-\frac{-1}{3}}\right|$

$= {}^3/_2 + \left(-\frac{1}{3}\right) n\text{-}1\left|\frac{5}{2} - \frac{3}{2}\right|$

$Un = {}^3/_2 + \left(\frac{-1}{3}\right)^{n-1}$

ii. $\lim_{n \to \infty} \frac{Sn}{n} = ?$

$Sn = \sum_{k-1} U_k$

But $Uk = 3/2 + \frac{3}{2} + \left(\frac{-1}{3}\right)k - 1$

$Sn = \sum_{k-1}^{n} \left|\frac{3}{2} + \left(-\frac{1}{3}\right)^{k-1}\right|$

$= \sum_{k=1}^{n} \frac{3}{2} + \sum_{k=1}^{n} \left(\frac{-1}{3}\right)^{k-1}$

$= 3/2n + \left(\frac{1-(-\frac{1}{3})^n}{1-\frac{1}{3}}\right)$

$= {}^3/_2 n + {}^3/_4\left[1 - ({}^{-1}/_3)^n\right]$

$Sn = {}^3/_2 n + {}^3/_4 - {}^3/_4\,({}^{-1}/_3)^n$

$\therefore \lim_{\infty} \frac{Sn}{n} = \lim_{n \to \infty} \frac{\left(\frac{3}{2}n + \frac{3}{4} - \frac{3}{4}\left(-\frac{4}{3}\right)^n\right)}{n}$

$= \text{Lim}\ {}^3/_2 + {}^3/_{4n} - {}^3/_{4n}\left(\frac{-1}{3}\right)$

$= \text{Lim}\ {}^3/_2 + {}^3/_{4n} - {}^3/_{4n}\,({}^{-1}/_3)^n \quad n \to \infty {}^3/_2$

3. a. Refer to 2014 /15 question 2b.

b. Given $f(x) = \frac{2}{1-x^2}$

Let $y = {}^2/_{1-x^2}$

y – intercept is (0,2)

for vertical asymptote, $1 - x^2 = 0$

$(1 - x)(1 + x) = 0$

$X = 1,\ x = -1$

For horizontal asymptote, y=o

For domain: { x : x ∈ R, x #±}

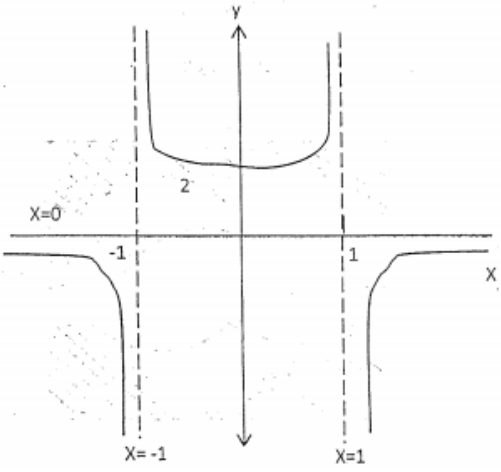
For behavior of the curve,

As $x \to 1^+, y \to -\infty$

$x \to 1^-, y \to \infty$

$x \to -1^+, y \to \infty$

$x \to -1^1, y \to -\infty$



ii. $f(x) = \frac{2}{1-x^2} = \frac{A}{1-x} + \frac{B}{1+x}$

$\frac{2}{1-x^2} = \frac{A(1+x)+B(1-x)}{1-x}$

$2 = A(1+x)+B(1x)$

if $x = 1$

$2 = A(1+1) + B(1-1)$

$\frac{2}{2} = \frac{2A}{2}$

$A = 1$

If $x = -1$

$2 = A(-1 + 1) + B(1—1)$

$\frac{2}{2} = \frac{2}{2}$

$B = 1$

$\therefore f(x) \frac{1}{1-x} = \frac{1}{1+x}$

Now $f(x) = (1-x)^{-1} + (1+x)^{-1}$

But $(1 + x)n = 1 + nx +$

$\frac{n(n-2)}{2!}x^{2+} \frac{n(n-n)(n-2)}{3!}x^2+\ldots$

$\therefore (1-x)^{-1} = 1 - (-x) - \frac{(-1-1)(-x)^2}{2!} +$

$\frac{-(-2(-3)(-x)^3}{3!} = \frac{-(-2)(-3(-4)(-x)^4+.}{4!} \ldots$

$(1+x)^{-1} = 1+x+x^2+x_3+x_4+\ldots$

$(1+x)^{-1} = 1 - x\frac{(-1-1)x^2}{2!} - \frac{-(-2)(-3)x^2}{3!}$

$\frac{-(-2)(-3)x^2}{4!}+\ldots$

$(1+x)^{-1} = 1 - x + x^2 - x^3 + x^3 + x^4 +$

…)

$\therefore f(x) = (1+x+x^2+x^3+x^4+\ldots)$

$(1 - x + x^3 - x^3 + x^4 + \ldots)$

$f(x) = 2 + 2x^2 + 2x^4 + \ldots$

Hence, $f(x)$ as far as the terms in x^4 is $2+2x^2+2x^4$

iii. For $\frac{1}{1-x^2} = \sum_{k=0}^{\infty} x^{2k}$

$f(x) = 2+2x^2+2x^4+2x^6+\ldots$

$= \frac{2}{1-x^2} = 2+2x^2+2x^4+2x^6..$

$\frac{2}{1-x^2} = 2(1 + \alpha^2 + \alpha^4 + \ldots$

$\frac{2}{1-x^2} = 1 + \alpha^2 + \alpha^4 + \ldots (1)$

For $\sum_{k=0}^{\infty} x^{2k}$, let $U_k = x^{2k}$ $k = 0$

If $k = 0$

$Un = x^{2(0)} = 1$

$U_2 = x^{2(2)} = x^4$

$U3 = x^{2(3)} = x^6$

$= \sum_{k=0}^{\infty} x^{2k} = U_0+U_1+U_2+U_3+\ldots$

$= 1 + x^2 + x^4 + x^6 + \ldots$

$$\therefore \sum_{k=0}^{\infty} x^{2k} = 1 + x^2 + x^4 + x^6 + \ldots \quad (2)$$

From (1) and (2), it shows that

$$\frac{1}{1-x^2} = \sum_{k=0}^{\infty} x^{2k}$$

c. Given

i. $An(A \cap B) \cup (A \cap C)$

Let $x \in A \cap (B \cup C$

Then $x \in A$ and $x \in (B \cup C)$

$= x \in A$ and $x \in$ or $x \in C$

$= x \in A$ and $x \in B$ or $x \in A$ and $x \in C$

$= x \in A \cap B$ or $x \in A \cap C$

$= x \in (A \cap B) \cup (A \cap C)$

$\therefore A A \cap (B \cup C) \leq (A \cap B) \cup (A \cap C)$

Conversely,

Let $x \in (A \cap B) \cup (A \cap C)$

Then $x \in A \cap B \cup A \cap C$

$= x \in A$ and $x \in B$ or $x \in A$ and $x \in C$

$= x \in A$ and $x \in B$ or $x \in C$

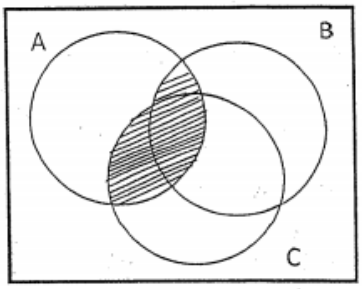
$= x \in A$ and $x \in B \cup C$

$= x \in A \cap (B \cup C)$

$\therefore (B \cap C) \cup (A \cap B) \leq A \cap (B \cup C)$

$= A \cap (B \cup C) = (A \cap B) \cup (A \cup C)$

Illustration on venn diagram



ii. Given $(A \cup B \cup C)^1 = A^1 \cup B1 \cup C^1$

let $x \in$ $(A \cup B \cup C)^1 =$

Then $x \subset A \cup B \cup C$

$= x \subset A^1$ and $x \cup B$ $x \subset C$

$= x \subset A^1$ and $x \in B^1$ and $x \in C^1$

$= x \in A^1 \cap B^1 \cap C^1$

$\therefore (A \cup B \cup C)^1 \leq A^1 \cap B^1 \cap C^1$

Conversely,

Let $\alpha \in A^1 \cap B^1 \cap C^1$

Then $\alpha \in A^1$ and $\alpha \in B^1$ and $\alpha \in C^1$

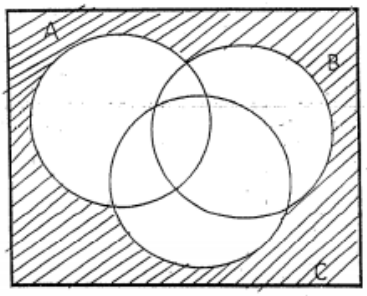
$= \alpha \subset A$ or $\alpha \subset B$ or $\alpha \subset C$

$= \alpha \subset (A \cup B \cup C)$

$= \alpha \leq (A \cup B \cup C)^1$

$= A^1 \cap B^1 \cap C^1 \leq (A \cup B \cup C)^1$

Illustrating on venn diagram



2012 /13

SECTION B

1. a. Refer to 2014/15 question 1a.

b. given $P(f) = 6x^3 + 13x^2 - 19x - 12$

let $x = -3$

$\therefore P(-3) = 6(-3)^3 + 13(-3)^2 - 19(-3) - 12$

$= 6(-27) + 13(9) + 57 - 12$

$= 162 + 117 + 57 - 12$

P9-3) $= 0$

Since P(-3) =, 0, it implies (x +3) is a factor

$$(x+3)\frac{6x^2-5x-4}{6x^3+13x^2-19x-12}$$

$$\frac{-(6x^3+18x^2)}{-5x^2-19x-12}$$

$$\frac{-(-5x^2-15x)}{-4x-12}$$

$$\frac{-(-4x12)}{---}$$

$= P(x) = (x + 3)(6x^2 - 5x - 4)$

Now $6x^2 - 5x - 4 = 6x^2 - 8x + 3x - 4$

$= 2x(3x - 4) + (3x - 4)$

$-(2x + 1(*3x - 4)$

$\therefore p(x) = (x + 3)(2+1)(3x - 4)$

2. a. Refer to 2013/14 question 2a

b. Given

$$\begin{vmatrix} 3 & 2 & -2 \\ 1 & -3 & 1 \\ 2 & 1 & -3 \end{vmatrix} \quad B = \begin{vmatrix} 8 & 4 & -4 \\ 5 & -5 & -5 \\ 7 & 1 & -11 \end{vmatrix}$$

$$AB = \begin{vmatrix} 3 & 2 & -2 \\ 1 & -3 & 1 \\ 2 & 1 & -3 \end{vmatrix} \begin{vmatrix} 8 & 4 & -4 \\ 5 & -5 & -5 \\ 7 & 1 & -11 \end{vmatrix}$$

$$= \begin{matrix} 3(8) + 2(5) - 2(7) & 3(4) + 2(-5) - 2 & 3(-4) + 2(-5) + 22 \\ 11(8) - 3(5) + 7 & 4 - 3(-5) + 1 & -4 - 3(-5) - 11 \\ 2(8) + 5 - 3(7) & 2(4) - 5 - 3 & 2(-4) - 5 + 33 \end{matrix}$$

$$= \begin{matrix} 24 + 10 - 14 & 12 - 10 - 2 & -12 - 10 + 22 \\ 8 - 18 + 7 & 4 + 15 + 1 & 4 + 15 - 11 \\ 16 + 5 - 21 & 8 - 5 - 3 & -8 - 5 + 33 \end{matrix}$$

$$AB = \begin{vmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{vmatrix}$$

$$\text{Now } AB = 20 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$= AB = 20I$

$\frac{AB}{20} = I$

$A(\frac{1}{20}B) = I$

by comparing to $A A^{-1} = I$

$= A^{-1} = {}^{1}/_{20}\, B$

$$\therefore A - 1 = \frac{1}{20}\begin{vmatrix} 8 & 4 & -4 \\ 5 & -5 & -5 \\ 7 & 1 & -11 \end{vmatrix}$$

c. Given

$x = (x\ 1) = x^1 = \begin{pmatrix} x \\ 1 \end{pmatrix}$

$p = \begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix}$

$= (x\,p)^{x1} = \left\{(x1)\begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}\right\}\begin{pmatrix} x \\ 1 \end{pmatrix} = 0$

but $(x1)\begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix} = [xu - 2 - 2x + 3]$

$\therefore (x\,p)\ x\ t = [x - 2 - 2x + 3]$

$\begin{vmatrix} x \\ 1 \end{vmatrix} = 0$

$= x(x -) + (x - 2 - 2x + 3)(1) = 0$

$x2 - 2x - 2x + 3 = 0$

$x2 - 4x + 3 = 0$

$x = 3, 1$

$\therefore$ the truth set is $\{x : x = 3,1\}$

3. a. Refer to 2013/14

b. Given $\tan 3\theta = \frac{3t - t^2}{1 - 3t^2}$

$\tan 3\theta = \tan(\theta + 30) = \frac{\tan\theta + \tan 2\theta}{1 - \tan\theta \tan 2\theta}$

but $\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$

$= \tan 3\theta = \frac{\tan\theta + \left(\frac{2\tan\theta}{1-\tan^2\theta}\right)}{1-\tan\theta\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)}$

$= \frac{\tan\theta(1-\tan^2\theta)+2\tan\theta}{1-\tan^2\theta}$

$\frac{1-\tan^2\theta-2\tan^2\theta}{1-\tan^2\theta}$

$= \frac{\tan\theta-\tan^2\theta+2\tan\theta}{1-\tan^2\theta} = \frac{1-\tan^2\theta}{(1-3\tan^2\theta}$

but $t = \tan\theta$

$\tan 3\theta = \frac{3t-t^3}{1-3t^2}$

for $\tan\theta \frac{\pi}{12} = 2 - \sqrt{3}$

$\tan 3\left(\frac{\pi}{12}\right) = \frac{3t-t^3}{1-3t^2}$

$\tan\frac{\pi}{4} = \frac{3t-t^3}{1-3t^2}$

$1 = \frac{3t-t^3}{1-3t^2}$

$1 - 3t^2 = 3t - t^3$

$t3 - 3t^2 = 3t - t^3$

$t3 - 3t^2 - 3t + 1 = 0$

if $t = -1$

$(-1)3 - 3(-1)^2 - 3(-2) + 2$

$-1 - 3 + 3 + 1$

$= 0$

$=$ (t +1) is a factor

$|+| \frac{t^2-4t+1}{t^3-3t^2-3t+1}$

$- \frac{t^3+t^2}{-4t^2-3t+1}$

$\frac{(-4t^2-4t)}{t+1}$

$\frac{(t+1)}{--}$

$= (t + 1)(t2 - 4t+1) = 0$

$= t = -2, t2 - 4t + 1 = 0$

$t = \frac{-(-4)\pm\sqrt{(-4)^2-(4)(1)(1)}}{2(1)}$

$= \frac{4\pm\sqrt{16-4}}{2}$

$t = \frac{4\pm\sqrt{12}}{2}$

$= \frac{4\pm2\sqrt{12}}{2} = 2 \pm \sqrt{3}$

$\therefore t = 2\sqrt{3}$ and $t = 2 - \sqrt{3}$

but $t = \tan\theta$

where$\theta = \frac{\pi}{12}$

$\tan\frac{\pi}{12} = 2 - \sqrt{3}$

4. a. Given

f(x) = - 2- $\sqrt{3-2x-x^2}$

let y = -2 - $\sqrt{3-2x-x^2}$

$(y+2)^2 = \left(-\sqrt{3-2x-x^2}\right)^2$

$= y^2 + 4y + 4 = 3 - 2x - x^2$

$= x^2 + y^2 + 2x + 4y + 4 - 3 = 0$

$x^2 + y_2 + 2x\,4y+1 = 0$

comparing $x^2 + y^2 + 2gx + 2fy + c = 0$

$= \frac{2g}{2} = \frac{2}{2}, \frac{2f}{2} = \frac{4}{2} < 1$

g = 2 + =2

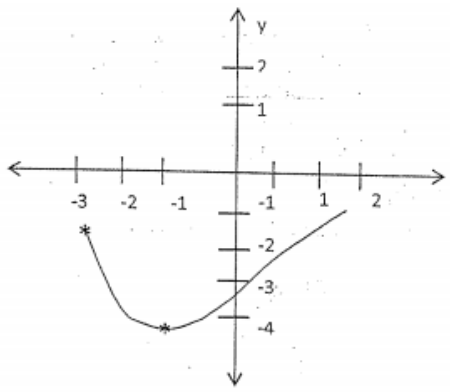
center (-g, -f) = (-1 , -2)

radius $\sqrt{g^2+f^2-c}$

$= \sqrt{(1)^2 + (2)^{-1}}$

$= \sqrt{4}$

Radius = 2



The curve depicts the lower half of a circle with center (1, -2) and radius 2.

b. $\sqrt{2}$, 11-$^{11-6}\sqrt{2}$

Since it a G.P common ration = $\frac{U_2}{U_1}$

$\therefore r \frac{11-6\sqrt{2}}{3-\sqrt{2}}$

$$= \frac{11-6\sqrt{2}}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}}$$

$$\frac{(11-6\sqrt{2}\,(3+\sqrt{2}}{(3)^2-(\sqrt{2})^2}$$

$$= \frac{33+11\sqrt{2}-18\sqrt{2}--(2)}{9-2}$$

$$= \frac{33-12-7\sqrt{2}}{7}$$

$$r = \frac{21-7\sqrt{2}}{7}$$

$$r = 3 - \sqrt{2}$$

from 6th term, $U6 = ar^5$ u6

$$= (3-\sqrt{2})^6$$

$$= (3)^6 + 6(3)^5(-\sqrt{2}) + 15(3)4(-\sqrt{2})^2 + 20(3)^3(-\sqrt{2})^3 + 15(3)2(-\sqrt{2})^4 + 6(3)(-\sqrt{2})^5 + (-\sqrt{2})^6$$

U6 = 129 – 145 1458$\sqrt{2}$+2430-1080 $\sqrt{2}$+ 540 – 72 $\sqrt{2}$ + 8

729 + 2430 + 540 + 8 – 1458 $\sqrt{2}$ – 1080 $\sqrt{2}$ – 72 $\sqrt{2}$

= 3707 – 2610$\sqrt{2}$

$\therefore U_6 = 3707 - 2610\sqrt{2}$

2011/12
SECTION B

1. a. Given

$3x - 7y + 1z = 4$

$x - 2y + 3z = 1$

$-2x + 8y - 16x = 4$

$$\begin{bmatrix} 3 & -7 & 11 \\ 1 & -2 & 3 \\ -2 & 8 & -16 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 4 \end{bmatrix}$$

Using Crammer's rule

Let $A = \begin{vmatrix} 3 & -7 & 11 \\ 1 & -2 & 3 \\ -2 & 8 & -16 \end{vmatrix}$

$\alpha = \frac{|Ax|}{A} \quad y = \frac{|Ay|}{|A|}$

$z = \frac{|Az|}{|A|}$

$$|A| = \begin{vmatrix} 3 & -7 & 1 \\ 1 & -2 & 3 \\ -2 & 8 & -16 \end{vmatrix}$$

$$|A| = 3\begin{vmatrix} -2 & 3 \\ 8 & -16 \end{vmatrix} + 7\begin{vmatrix} 1 & 3 \\ -2 & -16 \end{vmatrix} + 11\begin{vmatrix} 1 & -2 \\ -2 & 8 \end{vmatrix}$$

$= 3(32 - 24) + 7(16+6) 11(8-4)$

$3(8) + 7(-10) + 11(4)$

$24 - 70 + 44$

$x - 2$

$$|Ax| = \begin{vmatrix} 4 & -7 & 11 \\ 1 & -2 & 3 \\ 4 & 8 & -16 \end{vmatrix}$$

$$= 4\begin{vmatrix} -2 & 3 \\ 8 & -16 \end{vmatrix} + 7\begin{vmatrix} 1 & 3 \\ 4 & -16 \end{vmatrix} + 11\begin{vmatrix} 1 & -2 \\ 4 & 8 \end{vmatrix}$$

$|Ax| = 4(32 - 24) + 7(-16 + 2) + 11(16)$

$= 32 - 196 + 176$

$|Ax| = 12$

$\therefore x \frac{|Ax|}{|A|} = \frac{12}{-2} = -6$

$$|Ay| = \begin{vmatrix} 3 & 4 & 11 \\ 1 & 1 & 3 \\ -2 & 4 & -16 \end{vmatrix}$$

$$= 3\begin{vmatrix} 1 & 3 \\ 4 & -16 \end{vmatrix} - 4\begin{vmatrix} 1 & 3 \\ -2 & -16 \end{vmatrix} + 11\begin{vmatrix} 1 & 1 \\ -2 & 4 \end{vmatrix}$$

$=3(-16+2) - 4(-16+6) + 11(4+2)$

$= 3(-28) - 4(-10) + 11(0)$

$-84 + 40 + 66$

$|Ax| = 22$

$\therefore = y \frac{|Ay|}{|A|} = \frac{22}{2} = 11$

$$|Az| = \begin{vmatrix} 3 & -7 & 4 \\ 1 & -2 & 1 \\ -2 & 8 & 4 \end{vmatrix}$$

$$= 3\begin{vmatrix} -2 & 1 \\ 8 & 4 \end{vmatrix} + 7\begin{vmatrix} 1 & 1 \\ -2 & 4 \end{vmatrix} + 4\begin{vmatrix} 1 & 1 \\ -2 & 8 \end{vmatrix}$$

$= 3(-8 - 8) + 7(4+2) + 4(8 - 4)$

$3(-16) + 7(6) + 4(4)$

$= -48 + 42 + 16$

$Az = 10$

$Z = \frac{|Az|}{A} = \frac{10}{2} = 5$

b. Given 0.67585858...

Let y = 0.67585858...

$\therefore y = 0.67 + 0.0058 + 0.000058 + 0.00000058 + \ldots$

$= 0.67 + 5.8 \times 10^{-3} \times 5.810^{-5} + 5.8 \times 10^{-7}$

$= 0.67 + 5.8 \times 10^{-3} [1 + 10^{2} + 10^{-4} + \ldots]$

$= 0.67 + 5.8 \times 10 - 3 \left[\frac{1}{1 - \frac{1}{100}}\right]$

$= 0.67 + 5.8 \times 10 - 3 \left[\frac{1}{\frac{99}{100}}\right]$

$= \frac{67}{100} + \frac{58}{1000}\left(\frac{100}{99}\right)$

$= \frac{67}{100} + \frac{58}{900}$

$= \frac{9(67)+58}{900}$

$y = \frac{661}{900}$

$\therefore 0.67585858... = \frac{661}{900}$

c. The zero multiplication theorm states that the product of any number and zero is zero.

To prove this,

0.a = b – (1)
We know that, 0.a = (0+0) a
So,
(0+0) a = b
0.a + 0.a = b (2)
Subtracting (2) – (1) give 90.a+0.9)
–(0.a) –b-b
= 0.a+0.a – 0.a = 0
ie. 0.a + 0.a = 0.a
0.a = 0 - (3)
Comparing to (1)
B = 0
As result, 0.a = b = 0

2. Given

$\frac{tan^2\theta+2}{1+tan^2\theta} = 1 + Cos^2\theta$

$\frac{tan^2\theta+2}{1+tan^2\theta}\frac{Sin^2\theta}{Cos^2\theta} + 2$

$1 + \frac{Sin^2\theta}{Cos^2\theta}$

$= \frac{Sin^2\theta+2Cos^2\theta}{Cos^2\theta}$

$\frac{Cos^2\theta+2Sin^2\theta}{Cos^2\theta}$

$= \frac{tan^2\theta+2Cos^2\theta}{Cos^2\theta sm^2\theta}$

but $Sin^2\theta + Cos^2\theta = 1$

$= Sin^2\theta + 2Cos^2\theta$

$= (Sin^2\theta + cos^2\theta) + Cos^2\theta = 1 + Cos^2\theta$

$= 1 + Cos2\theta$

$\frac{tan^2\theta+2}{1+tan^2\theta} = 1 + cos^2\theta$

b. Given $f(x) = \frac{x^3}{x^2}$

Let $y = \frac{x^3}{x^2-1}$

x – intercept is (0,0)

for vertical asymptotes

$x^2 - 1 = 0$

$(x-1)(X+1) = 0$

For oblique asymptotes, we find the quotient.

$x^2 - 1 \overline{)\frac{x}{\sqrt{x^3}}}$

$\frac{-(x^2-x)}{x}$

$\therefore y = x$

For horizontal asymptotes, f(x) has no horizontal asymptotes for domain:

$\{x : x \in R, x \# \pm 1\}$

for behavior of curve.

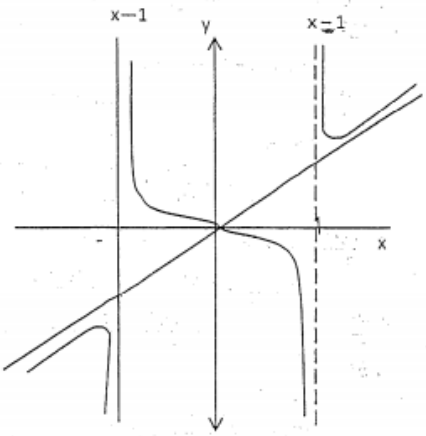
$As\ x \to 1^+, y \to \infty$

$x \to 1^-, y \to -\infty$

$x \to 1^+, y \to \infty$

$x \to -1-, y \to -\infty$

As $\to x^+$, $x \to -\infty$



c. Given 9,99,999,9999… for a sequence to be geometric.

$r = \frac{U_2}{U_1} = \frac{U_3}{U_2} = \frac{U_4}{U_3}$

$\therefore \frac{U_2}{U_1} = \frac{99}{9} = \frac{111}{11}$

$\frac{U_y}{U_z} = \frac{9999}{9999} = \frac{1111}{111}$

$\therefore$ since $\frac{U_2}{U_1} \neq \frac{U_3}{U_2} \neq \frac{U_4}{U_3}$

= the sequence is not geometric

3. Given

$(1/3 - 2b^3)^5$

From binomial theorem

$(a + b)^n\ a^n + {}^nC_2 a^{n-x} b^2 + {}^n(a^{n-3}b^3 + \cdots$

$\therefore$

$\left(\frac{a}{3} - 2b^3\right)^5 =$

$(a/3)^5 + {}^5C_1\left(\frac{a}{3}\right)^4(-2b^3)^3 + {}^5C_2\left(\frac{a}{3}\right)^3(-2b^3)^6 + {}^5C_1\left(\frac{a}{3}\right)^4(-2b^3)^4 + + {}^5C_q\left(\frac{a}{3}\right)(-2b^3)^4 + (-2b^3)^5$

$= a^5/243 + 5\left(\frac{a4}{31}\right)(-2b^3) + 10\left(\frac{a^3}{27}\right)(4b^6) + 10\left(\frac{a^2}{9}\right)(-8b^2) + 5\left(\frac{a}{3}\right)(16b^{12}) + (-32b^{15})$

$= a^5/243 - \frac{10a^4b^3}{81} + \frac{40a^3b^6}{27} - \frac{80a^2b^a}{9} + \frac{80ab^{12}}{3} - 32b^{15}$

$\therefore (a/12 - 2b^3)^5 = \frac{a5}{243} - \frac{1014b3}{81} + \frac{40a^3b^6}{27} - \frac{80a^2b^a}{9} + \frac{80ab^{12}}{3} = 32b^{15}$

b. Given $\frac{x^n + x^{n+2}}{x + x^3} = x^{n-1}$

$\therefore \frac{x^n + x^{n+2}}{x+x^3} = \frac{x^n(1+x^2)}{x(1+x^2)}$

$= \frac{x^n}{x}$

$\frac{x^n + x^{n+2}}{x+x^3} - x^{n-1}$

c. Given

3Cos 20+Sinθ=1

but Cos20 = $\cos^2\theta - \sin^2\theta$

= (-1 $\sin^2\theta$) – $\sin^2\theta$

Cos2 $\theta = 1 - 2\sin^2\theta$

$\therefore 3(1 - 2\sin^2\theta) + \sin\theta = 1$

$3 - 6\sin^2\theta + \sin\theta = 1$

$\therefore 6\sin^2\theta - \sin\theta\ 1 - 3 = 0$

Let y = sin θ

$6y^2 - y - 2 = 0$

$y = 2/3, -1/2$

but $y = 2/3$

$= \sin\theta = 2/3$

$\theta = \sin^{-1}(2/3)$

$\theta_1 = 41.8103^O$

Since Sin θ is positive $(2/3)$

$\theta_2 = 180.0$

$= 180 - 41.8103$

$= 138.1897^O$

If = - ½

= Sin θ = - ½

$\theta = \sin^{-1}(½)$

= 30

Since is negative

$= \theta 3 = 180 + \theta$

$= 180 + 30$

$Q_3 = 210$

$= \theta\ 4 = 360 - \theta$

= 360 - .030

$Q4 = 330^O$

$\therefore$ the solution of the equation are 41.8103^O,

138.187, 210 and 330°.

4.

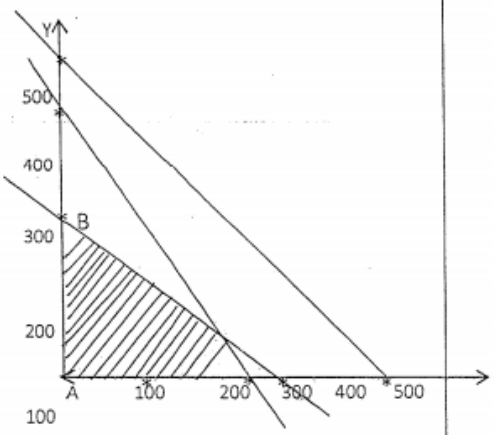
Package	Notebook	Folder	Pens
Package 1	2	1	2
Package 2	3	1	1
Total	≤ 60	≤ 50	≤ 5040

Let x = no. of first package

y no. of second package

∴ for ma x imum benefit,
P = 6.5 x + 7y
With constraints
2 x + 3y ≤ 600 – note book
x + y ≤ 500 – folder
x + y ≤ 400 – pens
x ≤ 0, y ≥ = 0
now, let 2 x + 3y = 600 - (0)
x +y = 500 - (2)
2 x + y = 40 - (3)
We plot the following lines on an x – y plane.
For (1), intercepts are (0,200) and (300,0)

For (2), (9,500) and (500,0)
For (3), (0400) and (200,0)



From the graph the points A(0,0), B(0,200) and ((300,0) are feasible points.
To test for the ma x imum
P = 6.5 x + 7t

If D(2000,0)
P = 6.5(200) + 7 (0)
= 1300
If A(0,0)
P = 6.5(0) + 7 (0) = 0
If B = (0,200)
P = 65(0) + 7 (0) = 0
If B (0,200)
P = 6.3(0) + 7 (200) = 1,400
If C(150, 100)
p (6.5) (150) + 7 (100) = 1675

since p = GH¢1950 is the greatest profit, it implies the point C(150,100) gives the ma x imum benefit.
Hence Akoto and co. should put together 150 of first package and 100 and second package in order to obtain the ma x imum benefit.

b. Given $3|x+1| \geq 1-x$

$\frac{3|x+1|}{3} \geq \frac{1-x}{3}$

$|x+1| \geq \frac{1}{3} - \frac{1}{3}x$

$= x + 1 \geq \frac{1}{3} - \frac{1}{3}x$ and $x + 1 \leq -\left(\frac{1}{3} - \frac{1}{3}x\right)$

$\therefore x + 1 \geq \frac{1}{3} - \frac{1}{3}x$ $\quad x + 1 \leq \frac{1}{3} - \frac{1}{3}x$

$3x + 3 \geq 1 - x$ $\quad 3x + 3 \leq -1 + x$

$3x + \geq 1 + 3$ $\quad 3x - x \leq -1 - 3$

$\frac{4x}{4} \geq \frac{4}{4}$ $\quad \frac{2x}{2} \leq \frac{-4}{2}$

$x \geq 1$ $\quad x \leq -2$

c. Given $A - (\cap C) = (A - B) \cup (A - C)$
Let $x \in A$ and $x \in (B \cap C)^1$
$= x \in A$ and $x \in (B \cap C)$

= x ∈ A and x ∈ Ba nd x ∈ C
= x ∈ A and x ∈ B or x ∈ A
and x ∈C,
= x ∈ A and x ∈ B or x ∈ A
and x ∈ C
= x ∈ A∩ B^1 or x ∈ A∩ C^1
= x ∈ A-B or ∈ A – C
= x ∈ (A-B) ∪ (A – C)

∴ A (A∩ B) ⊆ (A-C)
Conversely
Let x ∈ (A – B) ∪ (A – C)
= x ∈ A and x ∈ B1 or x ∈ A
and x ∈ C^1
= x ∈ A and x ∈ B or x ∈ A
and x ∈ C
= x ∈ A and x ∈ B and x ∈ C
= x ∈ A and x ∈ A∩ B
= x ∩ A *and* K ∈ (A∩ B)^1
= x ∈ A ∩ (B∩ C)^1
= x ∈ A - (B∩ C)
∴ (B – C) ∪ (A – C) ≤ A –
(B ∩ C)
Hence
A - (B ∩ C) = (A – B) ∪ (A – C)

2010 / 11
SECTION B

1. a. Let A = first term of G.P
⇒ A^1 = 13th term of G.P since it is 12 note higher than A
∴ A = U_1 = 400
$A_1 = U_{13} = 800$
but $Un = ar^{n-1}$
⇒ U_1 = a = 400
$U_{13} = ar^{13+1} = 800$
⇒ $ar^{12} = 800$
but a = 400
$= 400r^{12} = 800$
$r^{12} = 2$
$r = \sqrt[12]{2}$
r = 1.0595
∴ the constant ratio is 1.0595 cycle per seconds

ii. C = 4th term of G.P since it's three notice higher than A.
∴ C= $U_4 = ar^{4-1} = ar^3$
$400(1.0595)^3$
= 475.7326
∴ C is 475.7326 cycle per second

b. Log (x +3) + log x = 1
log x (x +3) = 1
10 = x (x +3)
$10 = x^2 + 3x$
⇒ $x^2 + 2x - 10 = 0$
$\alpha^2 + 5\alpha - 2\alpha - 10 = 0$
$\alpha(\alpha + 5) - 2(\alpha + \alpha 5) = 0$
$(\alpha - 2)(\alpha + 5) = 0$
$\alpha = 2, \alpha = -5$
but log of a negative number is undefined
∴ $\alpha = 2$

c. Given $\frac{30Cos^2\theta Sin\theta}{6Sin^2\theta Cos\theta}$

$\therefore \frac{30Cos^3\theta Sin\theta}{6Sin^2\theta Cos\theta} = \frac{5Cos\theta}{Sin\theta}$
$= 5Cot\theta$

2. Given
$f(x) \frac{x^3-7x^2+17x-17}{x^2-6x+9}$

$x^2 - 6x + 9 \overline{)\ x^3 - 7x^2 + 17x - 17}$ (x - 1)
$-x^2 + 8x - 17$
$= (2x - 6x - 9)$
$2x - 8$

⇒ $\frac{x^3-7x^2+17x-17)}{x^2-6x+9} = x - 1 + \frac{2x-8}{x^2-6x+9}$

$\therefore \frac{2x-8}{x^2-6x+9} = \frac{2x-8}{(x-3)^2} = \frac{A}{x-3} + \frac{B}{(x-3)^2}$
⇒ $2x - 8 = A(x-3)^2 + B(x - 3)$

If x = 0
$-8 = 9A - 3B$ --- (1)
If x = 1
$2(1) - 8 = A(1-3)^2 + B(1-3)$
$2 - 8 = A(-2)^2 + B(-2)$
$-6 = 4A - 2B$ ------- (2)

From (2), $2B - 4A + 6$
$B = 2A + 3$ ------- (3)
Put (3) into (1)
$-8 = 9A - 3(2A + 3)$
⇒ $-8 = 9A - 6A - 9$
$\frac{3A}{3} = \frac{1}{3}$
$A = {}^1/_3$
Put $A = {}^1/_3$ into (4)
$B = 2({}^1/_3) + 3$
$= {}^{11}/_3$
$\therefore \frac{2x-8}{(x-3)^2} = \frac{\frac{1}{3}}{x-3} + \frac{{}^{11}/_3}{(x-3)^2}$
⇒ $\frac{1}{3(x-3)} + \frac{11}{3(x-3)^3}$
⇒ $\frac{x^3-7x^2+17-17}{x^2-6x+9} = (x - 1) + \frac{1}{3(x-3)} + \frac{11}{3(x-3)^2}$

b. Refer to 2011/12 question 4c.

c. Given Sin x = sin2 x
but sin2 x =2Sin x Cos x
$\therefore \frac{Sinx}{Sinx} = \frac{2SinxCosx}{Sinx}$
1 = 2Cos x
Cos x = ½
$\therefore$ x =Cos $^{-1}$(½)
$X_1 = 60^0$
Since cosine is also positive in the 4th quadrant, $X_2 = 360 - X_1$
$= 360 - 60$
$X_2 = 300^0$
$\therefore$ the values of x are 60^0 and 300^0

3. Given 3 x – 7y + 11z = 4
x – 2y + 3z =1
-2 x + 8y – 16z = 4

$$\therefore \begin{vmatrix} 2 & -7 & 11 \\ 1 & -2 & 3 \\ -2 & 8 & -16 \end{vmatrix} \begin{vmatrix} x \\ y \\ z \end{vmatrix} = \begin{vmatrix} 4 \\ 1 \\ 4 \end{vmatrix}$$

Using cramer's rules

$$\text{Let A} = \begin{vmatrix} 3 & -7 & 11 \\ 1 & -2 & 3 \\ -2 & 8 & -16 \end{vmatrix}$$

$x = \frac{|Ax|}{|A|}, y] \frac{|Ay|}{|A|}, Z \frac{|Az|}{|A|}$

$$\therefore |A| = \begin{vmatrix} 3 & -7 & 11 \\ 1 & -2 & 3 \\ -2 & 8 & -16 \end{vmatrix}$$

$$= 3\begin{vmatrix} -2 & 3 \\ 8 & -16 \end{vmatrix} + 7\begin{vmatrix} 1 & 3 \\ -2 & -16 \end{vmatrix} + 11\begin{vmatrix} 1 & -2 \\ -2 & 8 \end{vmatrix}$$

= 3 (32 – 24) + 7(-16 + 6) + 11(8 – 4)
= 3(8) + 7 (-10) + 11 (4)
=24-70+44
|A|=-2

$$|Ax| = \begin{vmatrix} 4 & -7 & 11 \\ 1 & -2 & 3 \\ 4 & 8 & -16 \end{vmatrix}$$

$$= 4\begin{vmatrix} -2 & 3 \\ 8 & -16 \end{vmatrix} + 7\begin{vmatrix} 1 & 3 \\ 4 & -16 \end{vmatrix} + 11\begin{vmatrix} 1 & -2 \\ 4 & 8 \end{vmatrix}$$

=4(32-24)+7(-16-12)+11(8+8)
$= 32 - 196 + 176$
$|Ax| = 12$
$\therefore x = \frac{|Ax|}{|A|} = \frac{12}{-2} = -6$

$$|Ay| = \begin{vmatrix} 3 & 4 & 11 \\ 1 & 1 & 3 \\ -2 & 4 & -16 \end{vmatrix}$$

$$= 3\begin{vmatrix} 1 & 3 \\ 4 & -16 \end{vmatrix} - 4\begin{vmatrix} 1 & 3 \\ -2 & -6 \end{vmatrix} + 11\begin{vmatrix} 1 & 1 \\ -2 & 4 \end{vmatrix}$$

=3 (-16 – 12) – 4(-16 + 6) + 11 (4 + 2)
= 3(-28) + 4 (-10) + 11(6)
= -84 + 40 + 66
$|Ay| = 22$
$\therefore y = \frac{|Ay|}{|A|} = \frac{22}{-2} = -11$

$$|Az| = \begin{vmatrix} 3 & -7 & 4 \\ 1 & -2 & 1 \\ -2 & 8 & 4 \end{vmatrix}$$

$$=3\begin{vmatrix} -2 & 1 \\ 8 & 4 \end{vmatrix} + 7\begin{vmatrix} 1 & 1 \\ -2 & 4 \end{vmatrix} + 4\begin{vmatrix} 1 & -2 \\ -2 & 8 \end{vmatrix}$$

= 3(-8 – 8) + 7 (4 + 2) + 4(8 – 4)
= 3(-16)+7(6) + 4(4)
= - 48 + 42 + 16
$|Az| = 10$
$\therefore z = \frac{|Az|}{|A|} = \frac{10}{-2} = -5$

b. Given
$4\tan^2\theta + Sec\theta - 1 = 0$
$4\left(\frac{Sin^2\theta}{Cos^2\theta}\right) + \frac{1}{Cos\theta} - 1 = 0$

$4Sin^2\theta + Cos\theta - Cos^2\theta = 0$
but $Sin^2\theta = 1 - Cos^2\theta$
$\Rightarrow 4(1 - Cos^2\theta) + Cos\theta - Cos^2\theta = 0$

$4 - 4\cos^2\theta + \cos\theta - \cos^2\theta = 0$

$5\cos^2\theta + \cos\theta - 4 = 0$

Let $y = \cos\theta$

$\Rightarrow 5y^2 - y - 4 = 0$

$y = 1, \ ^{-4}/_{5}$

but $y = \cos\theta$

$\therefore$ if $y = 1$

$\cos\theta = 1$

$\theta = \cos^{-1}(1)$

$\theta = 0$

If $y = ^{-4}/_{5}$

$\cos\theta = ^{-4}/_{5}$

$\theta = \cos -1(^{-4}/_{5})$

$\theta = 143.13$

Hence, 0° and 143.13° are the solution.

c. Refer to 2011/12 question 3b.

4. a Given $f(x) = \frac{x}{x^2-x-2}$

Let $y = \frac{x}{x^2-x-2}$

$\therefore$ x – intercept is (0,0)

y – intercept is (0,0)

For vertical asymptotes,

$x^2 - x - 2 = 0$

$x = 2, \ x = -1$

For horizontal asymptote, $y = 0$

For Domain$\{x : x \in R, \ x \neq 2,01\}$

For behavior of curve,

As $x \to 2^+, y \to \infty$

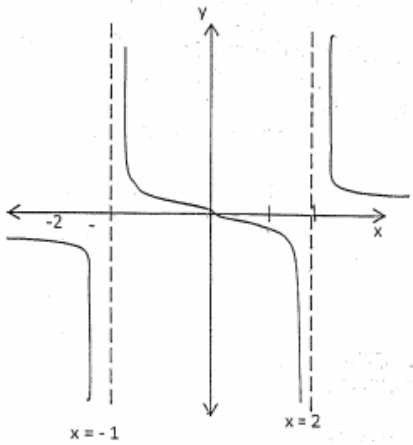
$x \to 2^-, y \to -\infty$

$x \to -1^+, y \to \infty$

$x \to -1^-, y \to -\infty$

As $y \to 0^+, y \to -\infty$

$y \to 0^-, y \to \infty$



b. Given

$log_9(a+3) - log_9 b = c + ½$

and $log_3(a-3) - log_3 b = c -$

1

$\Rightarrow log_9 \frac{(a+3)}{b} = c + ½$

$\therefore \frac{a+3}{b} = 9^{c+\frac{1}{2}}$ ------ (1)

Also, $log_3 \frac{(a-3)}{b} = c - 1$

$\therefore \frac{(a-3)}{b} = 3^{c-1}$

(2)

(1) × (2)

$\therefore \frac{(a+3)}{b}\frac{(a-3)}{b} = 9^{c+\frac{1}{2}} \times 3^{c-1}$

$\frac{a^2-3^2}{b^2} = 3^{2c+1} \times 3^{c-1}$

$\therefore \frac{a^2-9}{b^2} = 3^{2c+1+c-1}$

$\frac{a^2-9}{b^2} = 3^{3c}$

$a^2 - 9 = b^2 3^{3c}$

$a^2 = 9 + b^2 33c$

c. Given $0.\dot{2}\dot{3}=0.232323\ldots$

$= 0.\dot{2}\dot{3} =$

$0.23+0.0023+0.000023+\ldots$

$= 2.3\times10^{-1} + 2.3\times10^{-3} + 2.3\times10^{-5} + \ldots$

$= 2.3\times10^{-1}\,[1+10^{-2}+10^{-4}+10^{-6}+\ldots]$

$= 2.3\times10^{-1}$ (sum to infinity of G.P with $r = 10^{-2}$)

$= 2.3 \times 10^{-1} \quad \left(\frac{1}{1-1/100}\right)$

$= \frac{23}{100}\left(\frac{1}{99/100}\right)$

$= \frac{23}{100} \times \frac{100}{99}$

$= \frac{23}{90}$

$\therefore 0.\dot{2}\dot{3} = \frac{23}{90}$

2009/10
SECTION B

Given IA = A and AI = A

Since A is a 3×3 matri x

Let $A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ and $I = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$

$$\therefore IA = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= \begin{vmatrix} a+0+0 & b+0+0 & c+0+0 \\ 0+d+0 & 0+e+0 & 0+f+0 \\ 0+0+g & o+o+h & 0+0+i \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = A$$

$\therefore AI = A$

1. b. Given $\frac{2x^2-5x+4}{x^3}$

$= \frac{2x^2-5x+4}{x^3} = \frac{2x^2-5x+4}{x^3}$

$= \frac{2x^2-5x+4}{x^3} = \frac{x^2+Bx+C}{X3}$

Since denominators are the same.

$= 2x^2 - 5X + 4 = A\,x^2 + BX + C$

When X= 0

$2(0)^2 - 5(0)+4 = A(0)^2 + B(0+C$

$4 = C$

When X= 1

$2(1)^2 - 5(1) + 4 = A(1)^2 + B(1) + C$

$2 - 5+4=A+B-4$

$2 - 5 + 4 = A + B - 4$

$\therefore A + B = -3 \quad - \quad (1)$

When X= 2

$2(2)^2 - 5(2) + 4 = A(2)^2 + B(2) + C$

$8 - 10+4 = 4A+2B+4$

$2 - 4 = 4A + 2B$

$2A+B = -1 \quad - \quad (2)$

$\therefore (2) - (1)$ gives

$2A - A + B - B = -1 - -3$

$A = -1 + 3$

$A = 2$

Put A = 2 into (1)

$\therefore A+B = -3$

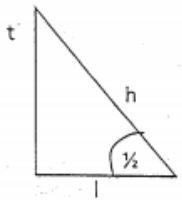
$B = -3 - 2 = -5$

$\therefore \frac{2x^2-5x+4}{x^3} = \frac{2}{x} - \frac{5}{x^2} + \frac{4}{x^3}$

c. Given $\text{Sin}\,\theta = \frac{2tan\frac{1}{2}\theta}{1+1tan^2\frac{1}{2}\theta}$

Let $t = \tan ½\,\theta$

Representing on a right angle triangle



From Pythagoras theorem, $h^2 = t^2 + 1$

$= h = \sqrt{t^2+1}$

From the right angled triangles,

Sin θ ½ $= \frac{t}{h} = \frac{t}{\sqrt{t^2+1}}$

Cos ½ θ $\frac{1}{h} = \frac{t}{\sqrt{t^2+1}}$

But sin θ = sin $\left(\frac{1}{2}\theta + \frac{1}{2}\theta\right)$

= Sin ½ θ Cos ½ θ +Cos ½ θ Sin ½ θ

=2Sim ½ θ Cos ½ θ

$= 2\left(\frac{t}{\sqrt{1+t^2}}\right)\left(\frac{1}{\sqrt{1+t^2}}\right)$

$= \frac{2t}{1+t^2}$, but t = tan½ θ

$\therefore$ Sin $\frac{2\tan\frac{1}{2}\theta}{1+\tan^2\frac{1}{2}\theta}$

2. a. Let Un = $n^2 - 2n + 1$

$Sn = \sum_{k=1}^{n} Uk$

but Uk = $2^k - 2k + 1$

$= 2.2k^{-1} - 2k + 1$

$\therefore Sn = \sum_{k=1}^{n} (2.2^{k-1} - 2k + 1)$

$= \sum_{k=1}^{n} 2.2^{k-1}$ d– $\sum_{k=1}^{n} 2k + \sum_{k=1}^{n} 1$

$= \left(\frac{2^n 1}{2-1}\right) - 2\left|\frac{1}{2}(n+1)\right| + n$

$= 2(2^n - 1) - (n + 1) + n$

$Sn = 2(2^{n-1}) - 1$

b. Given log_8 (p+2)+logs q = r – ⅓

$log_2(p-2) - log_2 q = 2r+1$

$= log_8 (p+2)\ q = r - ⅓$ - (1)

$log2(\frac{p-2}{q}) = 2r + 1$ - (2)

changing (1) and (2) to indices

$= 8\ r - ⅓ = q(p+2)$ - (3)

$2^{2r+1} = \frac{p-2}{q}$ - (4)

Making q of the subject in (3) and (4)

$q = \frac{8^{r-\frac{1}{3}}}{p+2}$ - (5)

$q = \frac{p-2}{2^{2r+1}}$ - (6)

equating (5) and (6) gives

$\frac{8^{r-1/3}}{p+2} = \frac{}{2^{2r+1}}$

$\therefore 2^{2r+1} \times 8^{r-1/2} = (p-2)(p+2)$

n 2r+1×34r1 = p2 – 4

$2^{5r} = p2\ 4$

$P^2 = (5) + 4$

$P^2 = 4 + 32^r$

If r = 1

P2 = 4+32

$\sqrt{p^2} = \sqrt{36}$

p = 6

$\therefore\ q = \frac{6-2}{2^{2+1}} = \frac{4}{2^3} = \frac{4}{8} = \frac{1}{2}$

$\therefore$ the value of p = 6 and q = ½

c. Let X= amount invested in 12 coles stock

y = amount invested in cape coast

metropolitan bond to ma x imize

p = 0.2X+ 0.1y

with costraints;

$\frac{x}{y} = \frac{3}{1} = X = 3y \leq 0$

X+ y ≤ 100,000

X≤ 70,000

y ≥ 20,000

now let X= 3y = 0 - (1)

X+ y =100,00 - (2)

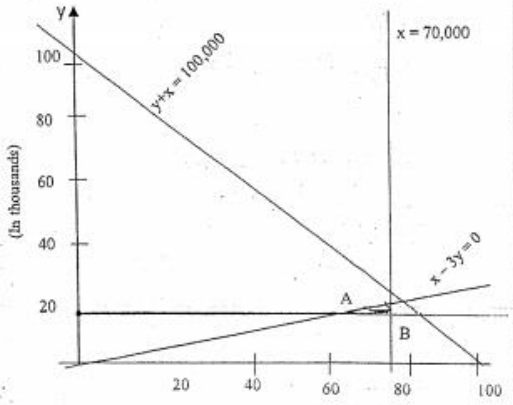
X= 70,000 - (3)

y = 20,000 - (4)

we plot the following pints from te above equations on an x -y plane.

For (1), (0,) and (60,000, 20,000)

For (2), (1000,000) and (0,100,000)



From the graph, the points

A(60,000, 20,000), B(70,000, 20,000)

And C(70,000, 2,333.333) are feasible point in the shaded region ABC

To te x t for the maximum.

P = 0.2 x +0.1y

If A (60000,20000)

P=0.2(60,000) + 0.1(20000)

=12000+2000

P = GH14000

If B(70,000, 2000)

P = 0.2(70000)+ 0.1 (20000)

=14000 + 2000

P = GH¢16000

If C (70000 + 0.1 (23,333.3)

=14000+2333.33

P=GH¢16,333.00

Since P=GH¢16,333.33 is the greatest returns, it implies c is the point that gives the maximum returns.

Hence, GH¢70,000 should be invested in P.coles stock and GH¢3,3333 should be invested in Cape Coast metropolitan bond

3. Given $(2.99)^{\frac{1}{3}} = (3 - 0.01)^{\frac{1}{3}}$

$[3(1 - {}^{0.01}/_{3})]$

$= 3^{\frac{1}{3}}(1-\frac{1}{300})^{\frac{1}{3}}$

Let $X = {}^{1}/_{300}$ and $n = {}^{1}/_{3}$

$\therefore (1- x)n = 1 - nx + \frac{n(n-1)(-x)^2}{2!} + \frac{n(n-n)(n-2)tx^3}{3!} + \frac{n(n-n)(n-2)(n-3)}{4!}(-x)^4 \ldots$

$(1 - x) = 1\frac{-1}{3}x + \frac{\frac{1}{3}(\frac{-2}{5})(\frac{-1}{3})(-x)^3}{3!} + \frac{\frac{1}{3}(\frac{2}{3})(\frac{-5}{3})(\frac{-8}{3})x4\ldots}{4!}$

$(1 - x)^{\frac{1}{3}} = 1 - \frac{1}{3}x - \frac{1}{6}x^2 - \frac{5}{81}x^3 - \frac{10}{243}\times^4 + \cdots$

But $X = {}^{1}/_{300}$

$\therefore (1 - \frac{1}{300})^{\frac{1}{3}} = 1 - \frac{1}{3}(\frac{1}{300}) - \frac{1}{6}(\frac{1}{300})^2 - \frac{5}{81}(\frac{1}{300})^3$

$\frac{10}{243}(\frac{1}{300})^4 + \cdots$

= 0.998887

$\therefore (2.99)^{\frac{1}{3}} = 3^{\frac{1}{3}}(1 - \frac{1}{300})^{\frac{1}{3}}$

$= 3^{1}/_{3}\,(0.998887)$

$= 1.4406$

b. Given $\left|\frac{x+4}{x-5}\right| \leq 3$

$\left(\frac{x+5}{x-5}\right)^2 \leq 3^2$

$(X+4)2 \leq 9)\ x\ -5)^2$

$x^2 + 8X + 16 \leq 9(x^2 - 10x + 25)$

$x^2 + 8x + 16 \leq 9x^2 - 90X + 225$

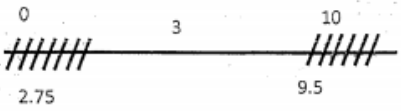
$9x^2 - x^2 - 90x - 8X + 225 - 16 \geq 0$

$8x2 - 98X + 209 \geq 0$ - (1)

Let $8x2 - 98 + 208 = 0$

X= 9.5, 2.75

representing on number line gives,



if x= 0, from (1)

$= 8(0)^2 - 98(0) + 209 = 209 \geq 0$

If x= 3

$8(3)2 - 98(3) + 209 = 13 \geq 0$

If x= 1=

$B(10)^2 - 98\,(10) + 209 - 29 \geq 0$

Since x= 0 and x = 10 satisfies the inequality, it implies

x ≤ 2.75 and x≥ 9.5 are solutions.

∴ e x pressing in interval notation give

$(-\infty, 2.75] \cup [9.5, \infty)$

c. Given $\frac{sec20 + xot20}{sec20} = it\ cosec20 - sin20$

$\frac{sec20 + xot20}{sec20} = it\ cosec20 - sin20$

$1 + \frac{Cos20}{Sin20} \times \frac{Cos20}{1}$

$1 + \frac{Cos20}{Sin20}$

But $Cos^2 20 = 1 - Sin^2 20$

$= 1 + \frac{1 - sin^2 20}{Sin20}$

$= 1 + Cosec20 - \sin 20$

$\therefore \frac{Sec20 + Cot20}{Sec20} = 1 + Cos20 - Sin20$

4. Given $x - 2y - 5x = -12$

$2x - y = 7$

$4x + 5y + 6z = 1$

$$= \begin{vmatrix} 1 & -2 & 5 \\ 2 & -1 & 0 \\ -4 & 5 & 6 \end{vmatrix} \begin{vmatrix} x \\ y \\ z \end{vmatrix} = \begin{vmatrix} -12 \\ 7 \\ 1 \end{vmatrix}$$

Using Cramer's rule

$$\text{Let A} \begin{vmatrix} 1 & -2 & 5 \\ 2 & -1 & 0 \\ -4 & 5 & 6 \end{vmatrix}$$

$\therefore = \frac{|Ax|}{|A|} + y\frac{|Ay|}{|A|}$

$Z = \therefore \frac{|Az|}{|A|}$

$$\therefore |A| = \begin{vmatrix} 1 & -1 & 5 \\ 2 & -1 & 0 \\ -4 & 5 & 6 \end{vmatrix}$$

$$= -2\begin{vmatrix} -2 & 5 \\ 5 & 6 \end{vmatrix} - \begin{vmatrix} 1 & 5 \\ -4 & 6 \end{vmatrix} - 0\begin{vmatrix} 1 & -2 \\ -4 & 5 \end{vmatrix}$$

$= -2(-12 - 25) - 96 + 20)$

$= -2(-37) - (26)$

$= 74 - 26$

$|A| = 48$

$$|Ax| = \begin{vmatrix} 12 & -2 & 5 \\ 7 & -1 & 0 \\ 1 & 5 & 6 \end{vmatrix}$$

$$-7\begin{vmatrix} -2 & 5 \\ 5 & 6 \end{vmatrix} - \begin{vmatrix} 12 & 5 \\ 1 & 6 \end{vmatrix} - 0\begin{vmatrix} 12 & -2 \\ 1 & 5 \end{vmatrix}$$

$= -7(12 - 25) - (75 - 5)$

$= -7(-37) - (-77)$

$259 + 77$

$|Ax| = 336$

$\therefore x = \frac{|Ax|}{|A|} = \frac{336}{48} = 7$

$$|Ay| = \begin{vmatrix} 1 & -12 & 5 \\ 2 & -1 & 0 \\ -4 & 1 & 6 \end{vmatrix}$$

$$-2\begin{vmatrix} -12 & 5 \\ 1 & 6 \end{vmatrix} + 7\begin{vmatrix} 1 & 5 \\ -4 & 6 \end{vmatrix} - 0\begin{vmatrix} 1 & -12 \\ -4 & 1 \end{vmatrix}$$

$= -2(-72 - 5) + 7(6 + 20)$

$= -2(-77) + 7(26)$

$= 154 + 182$

$|Ay| = 336$

$\therefore y = \frac{|Ay|}{|A|} = \frac{336}{48} = 7$

$$|Az| = \begin{vmatrix} 1 & -12 & -12 \\ 2 & -1 & 7 \\ -4 & 5 & 1 \end{vmatrix}$$

$$-2\begin{vmatrix} -12 & -12 \\ 5 & 1 \end{vmatrix} - \begin{vmatrix} 1 & -12 \\ -4 & 1 \end{vmatrix} - 7\begin{vmatrix} 1 & -2 \\ -4 & 5 \end{vmatrix}$$

$= -2(-2-60) - (1 - 48) - 7(5 - 8)$

$-3(-62) - (-47) - 7(-3)$

$= 124 + 47 + 21$

$|Az| = 192$

$\therefore z = \frac{|Az|}{|A|} = \frac{192}{48} = 4$

b. Given $y+1 = \frac{-3}{x-2}$

$y = \frac{-3}{x-2} - 1$

$= \frac{-3-(x-2)}{x-2}$

$y = \frac{-1-x}{x-2}$

now x– intercept is (1,0)

y – intercept is (0,1/2)

for vertical asymptote, $y = -1$

Domain: $\{ x : x \in R, x \neq 2 \}$

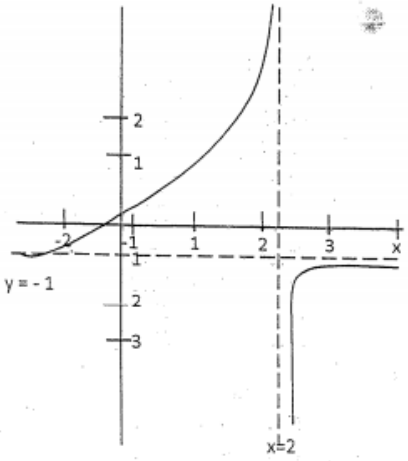
For behavior of curvey

As $x \to 2^+, y \to -\infty$

$x \to 2^-, y \to \infty$

As $y \to -1^+, y \to -\infty$

$y \to -1^-, x \to \infty$



4. Given

$$\frac{Sec^2-tan^220}{tan20+Sec20} = \frac{Cos20}{1-Sin20}$$

$$\frac{Sec^2-tan^220}{tan20+Sec20} = \frac{1}{Cos^220} - \frac{Sin^220}{Cos^220}$$

$$\frac{Sin20}{COs20} - \frac{1}{Cos20}$$

$$= -\frac{1-Sin^220}{Cos^220}$$

$$\frac{Cos20+1}{Sin20}$$

But $Cos^220 = 1-Sin^220$

$$\frac{Sec^220 - tan^220}{tan20 + Sec20} = \frac{Cos^220}{Cos^220} - \frac{Cos20}{1+Sin20}$$

$$1 \times -\frac{Cos20}{1+Sin20}$$

$$\therefore \frac{Sin^220-tan^220}{tan20+Sec20} = \frac{Cos20}{1+Sin20}$$

OBJECTIVE ANSWERS

2017/18

1. D	11. C	21.Bonus
2. A	12. C	22.C
3. C	13. C	23.B / A
4. C	14.B	24.Bonus
5. D	15.C & D	25.Bonus
6. B	16. D	26.B
7. A	17.C	27.C
8. C	18.A	28.A
9. C	19.B	29.C
10. A	20.C	30.C

2016/17

1. D	11. Bonus	21.Bonus
2. A	12. C	22.C
3. C	13. C	23.B
4. C	14.	24.Bonus
5. A	15.B	25.Bonus
6. B	16. D	26.B
7. B	17.C	27.C
8. C	18.A	28.A
9. A	19.B	29.C
10. A	20.C	30.C

2015/16

1.C	11. C	21.A
2.A	12. A	22.C
3.B	13. D	23.A
4. B	14. D	24.D
5. C	15. A	25.C
6.C	16. C	26.A
7.D	17. A	27. B
8.B	18. C	28.A
9.D	19.B	29.C
10. A	20. B	30.D

2014/15

11. A	11. A	21.B
12. A	12. C	22.B
13. B	13. A	23.C
14. D	14.D	24.A
15. D	15.C	25.A
16. C	16. D	26.A
17. B	17.C	27.C
18. A	18.A	28.C
19. B	19.C	29.C / A
20. A	20.C	30.D

2013/ 14

1. C	11. C	21. A	31. D	41.D
2. B	12. A	22.A	32.A	42.C
3. C	13.D	23.A	33.C	43.A
4. A	14.C	24.B	34.A	44.B
5. B	15B/C	25.D	35.C	45.A
6. C	16.B	26.A	36.B	46.C
7. D	17.D	27.A	37.B	47.B
8. C/B	18.C	28.D	38.B	48.B
9. D	19.B	29.C	39.C	49.A
10. B/C	20. C	30.B	40.A	50.D

2012/13

1. C	11. A	21.C
2. B	12.C	22.A
3. C	13.bonus	23.C
4. A	14.C	24.B
5. B	15.A	25.D
6. C	16.A	26.C
7. D	17.C	27.A
8. C	18.D	28.D
9. A	19.B	29.C
10. A	20.B	30.B

ISRAEL SERIES-0546751655

2011/12

1. D	11. C	21. D
2. C	12. A	22. B
3. B	13. D	23. B
4. D	14. D	24. D
5. B	15. C	25. B
6. D	16. Bonus	26. C
7. C	17. B	27. Bonus
8. B	18. A	28. A
9. A	19. C	29. Bonus
10. A	20.D	30. C

2010/11

1. C	11. A	21. A
2. C	12. B	22. A
3. B	13. A	23. C
4. B	14. D	24. B
5. D	15. D	25. C
6. D	16. C	26. B
7. Bonus	17. D	27. B
8. B	18. C	28. A
9. Bonus	19. D	29.
10. C	20. Bonus	30. A

2009/10

11. B	11. C	21. C
12. D	12. B	22. A
13. B	13. B	23. D
14. C	14. C	24. A
15. D	15. A	25. C
16. B	16. D	26. A
17. A	17. D	27. D
18. D	18. B	28. C
19. A	19. A	29. C
20. C	20. A	30. B